

RUHR-UNIVERSITÄT BOCHUM

CHEMICAL ANALYSIS OF ORGANIC COMPOUNDS

Dr. Jens Prothmann

Content of the lecture

- Introduction to environmental analysis
- Basic concepts of instrumental analysis
- Sampling strategies and techniques
- Sample storage and sample preparation
- Qualitative and quantitative analysis
- Quality assurance and quality control
- Examples of research studies in environmental analysis

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Main literature used for this lecture:

Fundamentals of Environmental Sampling and Analysis, Chunlong Zhang, 2nd Edition, 2024

Fundamentals of Analytical Chemistry, Skoog, West, Holler, Crouch, 8th Edition, 2004

Some objectives of environmental analysis

- Determine how much of a pollutant enters the environment
- Measure environmental concentrations of pollutants
- Asses the degree of pollution
- Evaluate the risk and toxicity to humans and biota
- Study the fate and transport of pollutants



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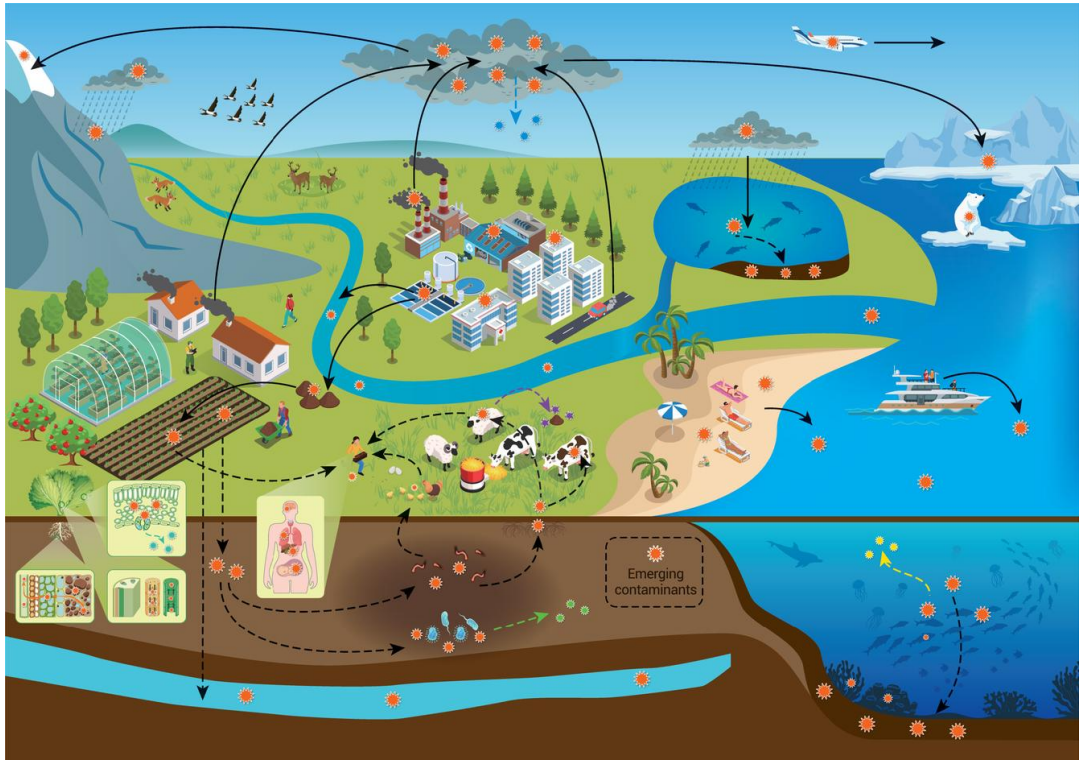
- Definition of pollution by the Organisation for Economic Co-operation and Development (OECD):

  Pollution means any introduction by man, directly or indirectly, of substance or energy into the environment resulting in deleterious effects of such a nature as to endanger human health, harm living resources and ecosystems, impair amenities or interfere with other legitimate uses of the environment.

Environmental pollution

- Emission of pollutants from various anthropogenic sources:
  - Municipal and industrial wastewater treatment plants
  - Stack emission
  - Agriculture
  - Traffic
  - Personal care products
  - …

Figure taken from Wang et al., *Emerging contaminants: A One Health perspective*, The Innovation, **2024**.



Transport of pollutants

- Water
  - Pollutants soluble in water
- Atmosphere
  - Volatile pollutants
- Solid particulate matter
  - Non-water-soluble and non-volatile pollutants





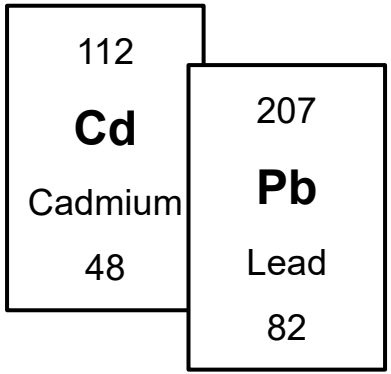


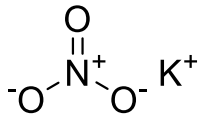


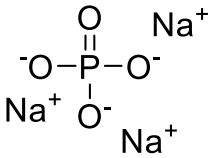
Pictures generated with AI

Inorganic and organic pollutants

- Inorganic pollutants
  - Metals and inorganic compounds

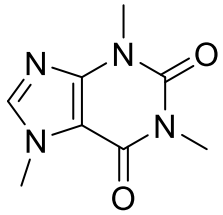


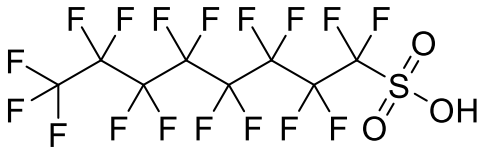


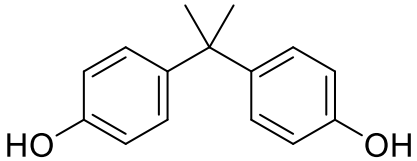


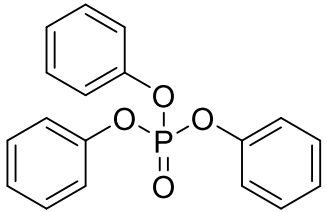
This lecture

- Organic pollutants
  - Natural and synthetic organic compounds









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Environmental concentrations of pollutants

- All compounds are toxic at high enough concentrations
- (Eco)toxicological effects of pollutants are connected to their concentrations
  - ➢ Environmental analysis is needed to monitor concentrations of pollutants in various environmental compartments

- LD50 of NaCl (oral rat): 3000 mg/kg (LD50 = lethal dose 50%)
  - If transferred to a human with 80 kg → 240 g NaCl



Picture from jakob5200 from Pixabay

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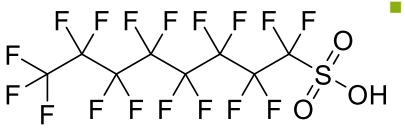
Environmental risk assessment

- Predicted no effect concentration (PNEC)
  - The concentration below which, according to ecotoxicological data, no adverse effects on organisms are expected
- Environmental quality standards (EQS)
  - A legally binding environmental quality standard that takes into account not only ecotoxicity but also bioaccumulation, food chains, and long-term protection



Picture generated with AI

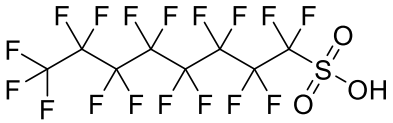
Environmental concentrations of pollutants

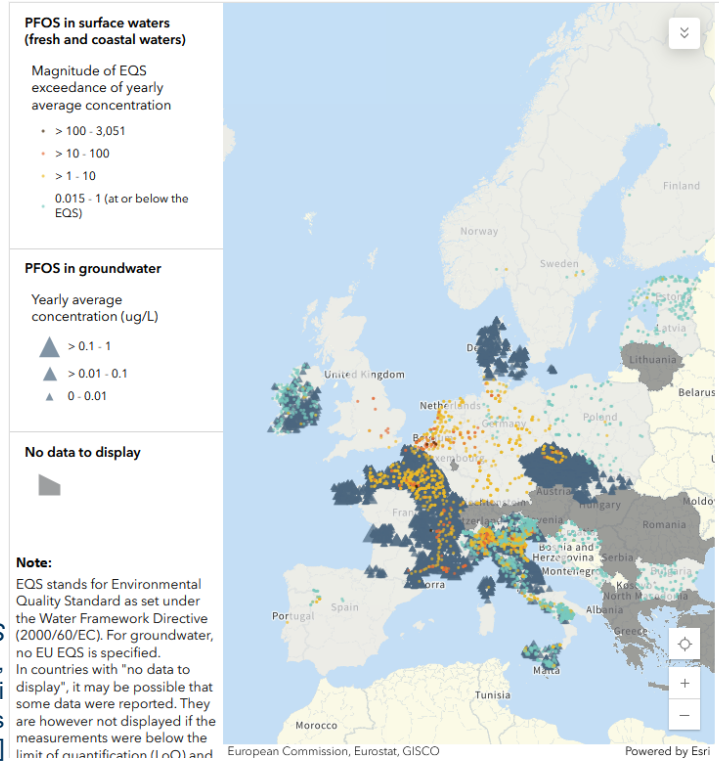


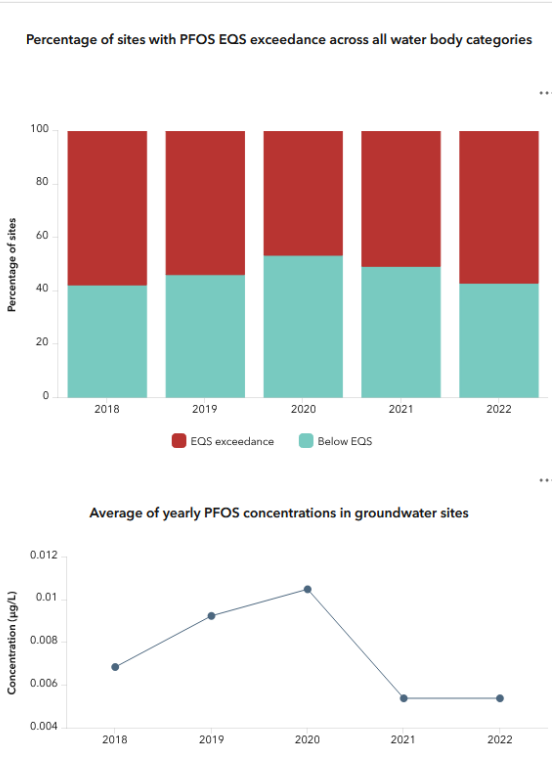
- Perfluorooctane sulfonate (PFOS)
  - Chemical class: per- and polyfluoroalkyl substances (PFAS)
    - High persistence in the environment → *“forever chemicals”*
    - (Eco)toxicological effects that impact humans and the environment
  - PFOS is widespread in European waters
  - Often exceeding the annual average environmental quality standards (EQS)
    - From 2018 to 2022 in 51-60% of rivers and 11-35% of lakes
  - EQS for PFOS in inland surface waters: 0.65 ng/L

European Environment Agency (EEA), *PFAS pollution in European waters*, Dec 2024, https://www.eea.europa.eu/en/analysis/publications/pfas-pollution-in-european-waters [Accessed 25.06.2026]

Environmental concentrations of pollutants

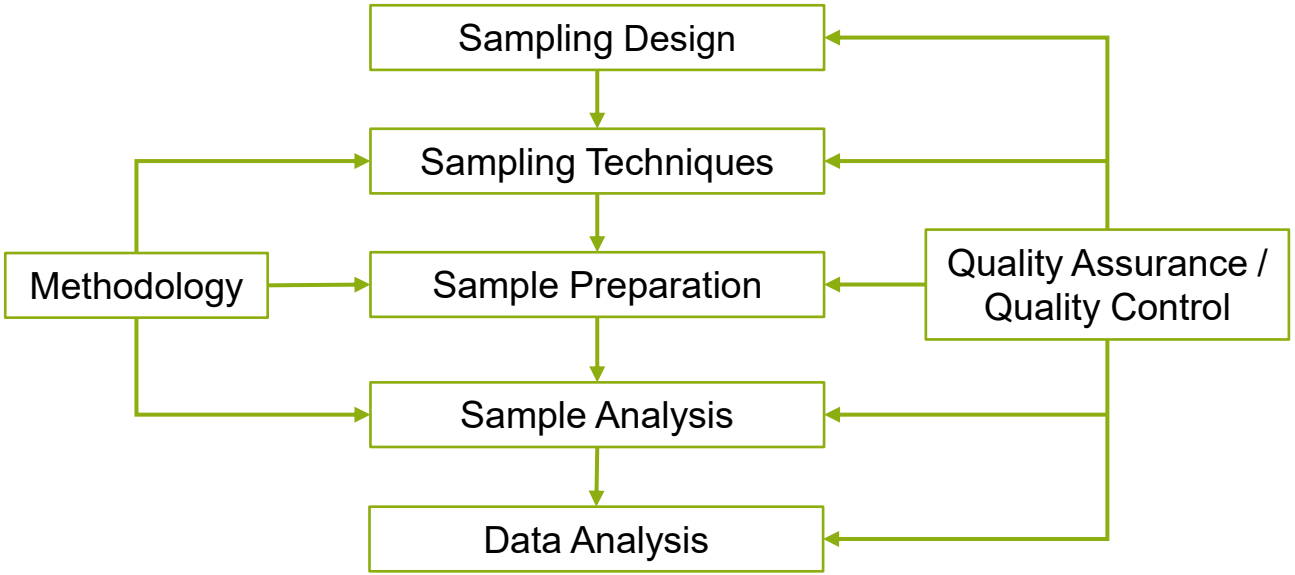






European Environment Agency (EEA), *PFAS pollution in European waters*, Dec 2024, https://www.eea.europa.eu/en/analysis/publications/pfas-pollution-in-european-waters [Accessed 25.06.2026]

Workflow of environmental data acquisition

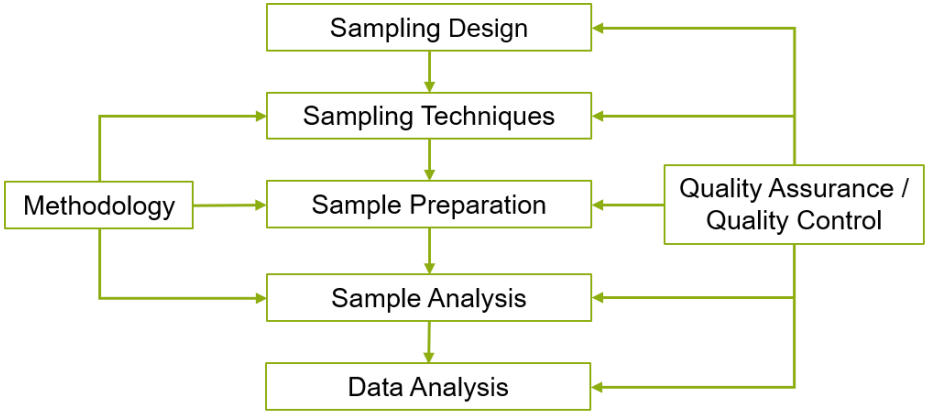


Workflow of environmental data acquisition

- Environmental data need to be scientifically reliable
- Appropriate procedures for sampling and analysis are important
- Environmental data should accurately reflect the content of the sample

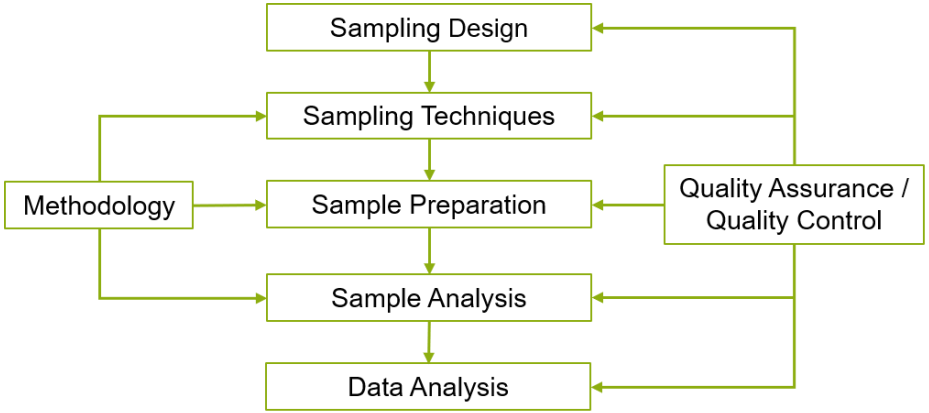
➢ It's good to have a plan!





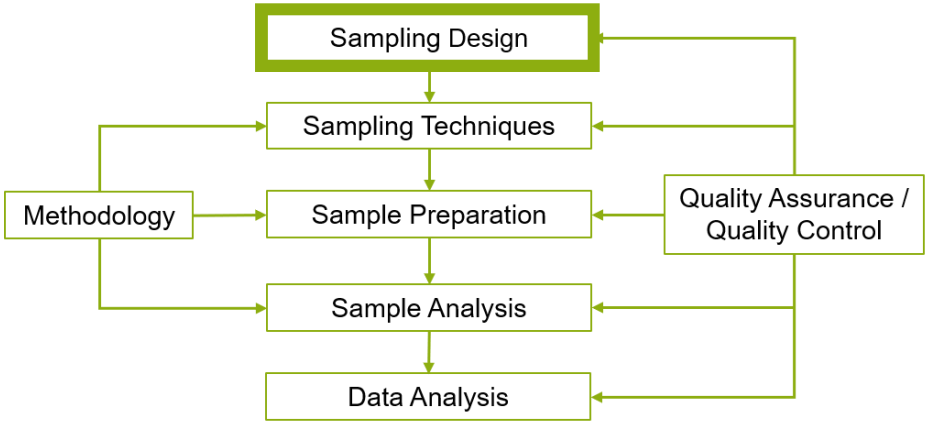
Workflow of environmental data acquisition

- General questions before starting sampling:
  - Where to take samples?
  - When to take samples?
  - How to take samples?
  - Who will take the samples?
  - How many samples should be taken?
  - How often samples should be taken?
  - How much of a sample is needed?
  - How to preserve taken samples?
  - How long will the samples be stable?
  - What to take? (e.g. air, water, soil)
  - What to analyse? (e.g. metals, PFAS)



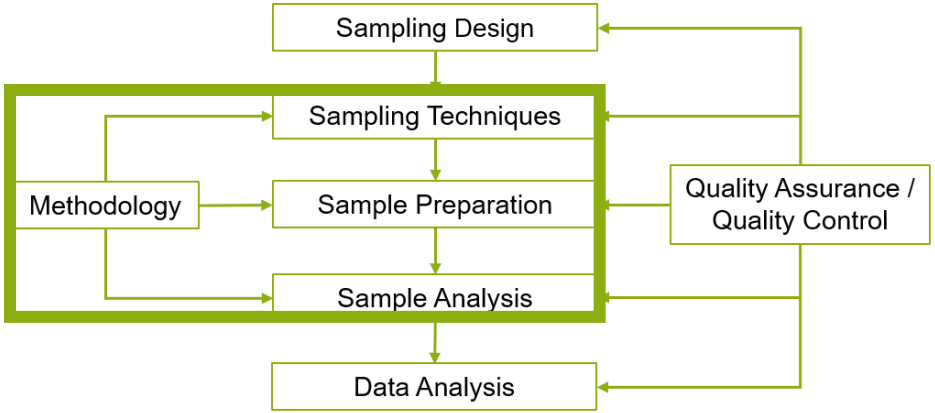
Sampling Design

- Randomly going out in the field, collecting some samples and bring it back to the lab is usually not the way to go!
- Sampling design must be consistent with the research question
- Data analysis often based on spatially (where) and temporally (when) sample parameters
- Frequency of sampling can be limited by available resources, like:
  - Time for sampling campaign
  - Time for sample preparation and analysis
  - Costs of human resources



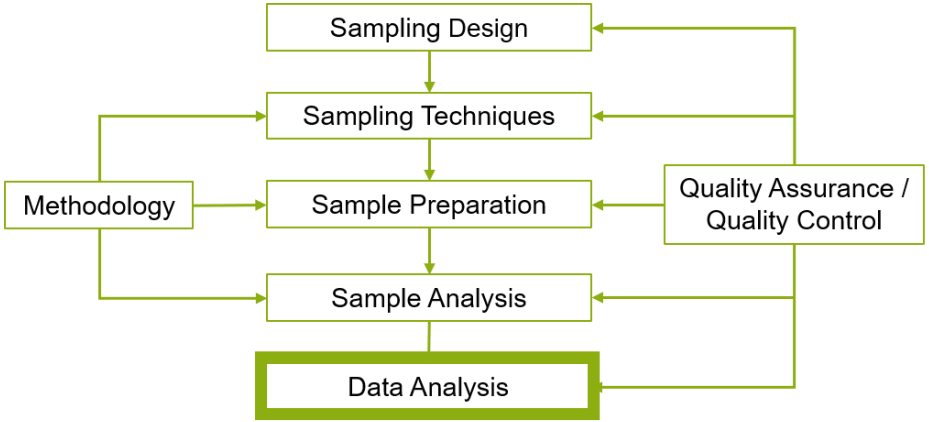
Methodology

- Sampling technique must fit for the sample type (water, sediment, soil, air, biomass, …)
- Sampling technique must be able to produce a representative sample
- Sampling technique, sample preparation and sample analysis must be coordinated for the analyte of interest



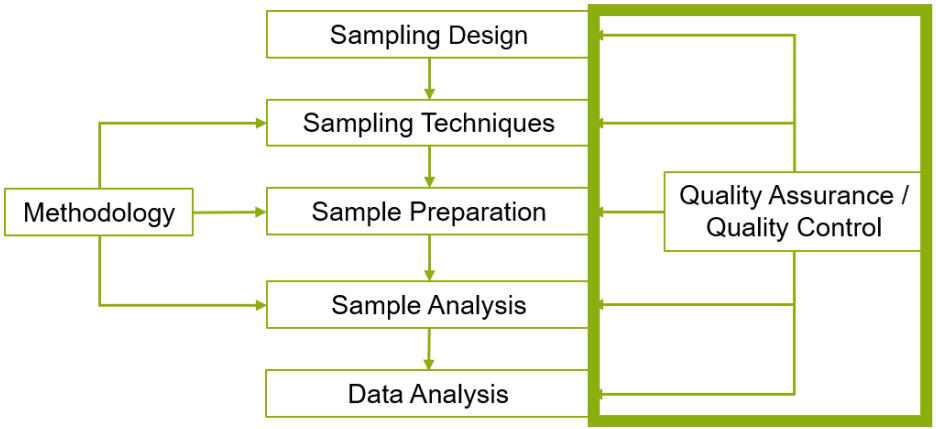
Data analysis

- The validity of the data analysis depends strongly on the sampling design and methodology
- Data analysis answers research question
- How the data analysis is performed and which data analysis tools will be used should be decided from the beginning
- Often most time-consuming working step



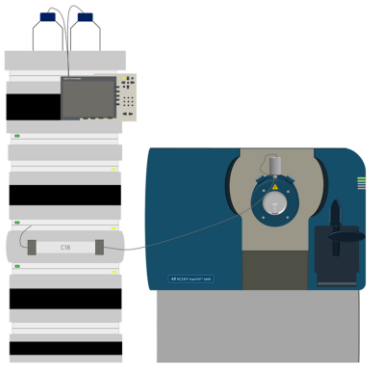
Quality assurance / Quality control (QA/QC)

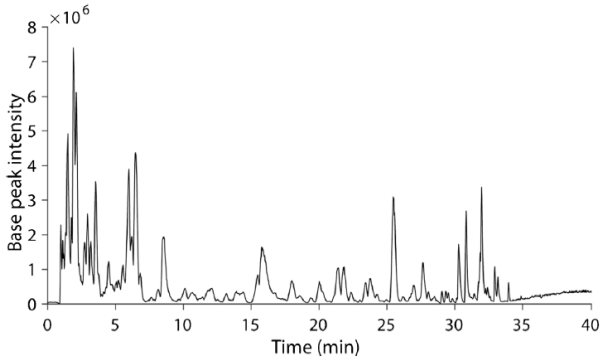
- QA and QC secures the scientific reliability of sampling design, methodology and data analysis
- Determines error rates associated with sampling and analysis
- Describes the accuracy of the analyte concentration in a sample



Basic concepts of instrumental analysis

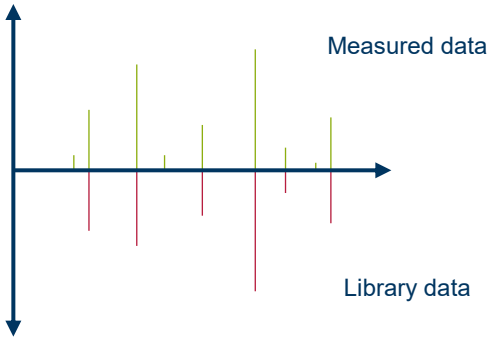
- Qualitative and Quantitative analysis
- Concentration units in environmental samples
- Analytical accuracy and precision
- Standard solution and standard calibration curve
- Limit of detection (LOD) and limit of quantification (LOQ)

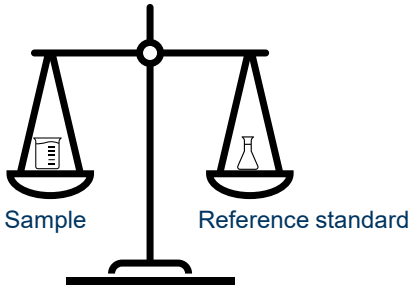




Qualitative and Quantitative Analysis

- Qualitative analysis
  - What is in the sample?
  - Identification of compounds within a sample
  - Identification for example of comparison of measured data with library data
- Quantitative analysis
  - How much is in the sample?
  - Determination of concentrations of compounds of interest
  - Comparison of measured signal intensities from compounds of interest with reference standards with known concentrations





Concentration units in environmental samples

- Concentration units in liquid samples (with density = $1.0\,\frac{g}{mL}$)
  - Freshwater, groundwater, urine, blood, …
  - Most common are mass/volume (m/v) units
  - $1\,\frac{mg}{L}$ = 1 ppm $1\,\frac{\mu g}{L}$ = 1 ppb $1\,\frac{ng}{L}$ = 1 ppt



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Concentration units in environmental samples

- Concentration units in solid samples
  - Soil, sediment, particulate solid matter, biomass, …
  - Most common are mass/mass (m/m) units
  - $1\,\frac{mg}{kg}$ = 1 ppm $1\,\frac{\mu g}{kg}$ = 1 ppb $1\,\frac{ng}{kg}$ = 1 ppt
  - Differentiation between concentration on dry basis (c_d) and wet basis (c_w)
  - c_d better for comparison of samples with different moisture content
  - Drying of wet sample to determine moisture content (% MC)



$$\%\ MC_{wet\ basis} = \frac{m_{wet\ sample} - m_{dry\ sample}}{m_{wet\ sample}} \times 100$$

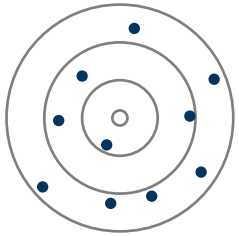
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Analytical accuracy and precision

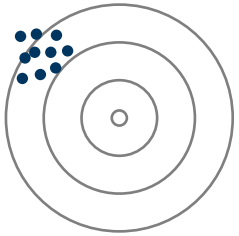
- Accuracy describes how closely the measured value matches the true or expected value
- Precision is a measure of the variation in measurement results

- Be careful:

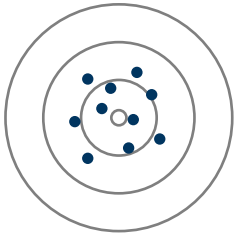
  A good precision does not guarantee an accurate analysis!



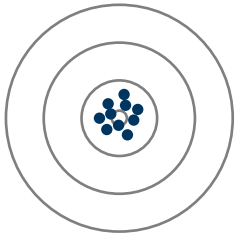
Low accuracy, low precision



Low accuracy, high precision



High accuracy, low precision



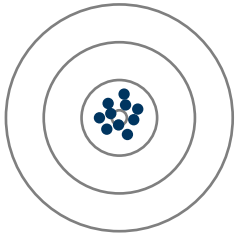
High accuracy, high precision

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Analytical accuracy and precision

- Accuracy determined and reported as Recovery (Rec) in % and an accurate method should be close to 100%

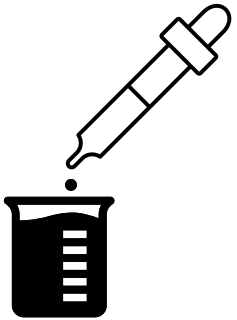
$$Rec\ (\%) = \frac{Measured\ value}{True\ value} \times 100$$



High accuracy, high precision

- Accuracy usually determined using spiked samples, because true value is often not known

$$Rec\ (\%)_{on\ spike} = \frac{Spiked\ sample\ value - Sample\ value}{Spiked\ value} \times 100$$



Analytical accuracy and precision

- Precision is determined by repeated measurements (x_1, x_2, … x_n) and reported by the absolute standard deviation (s) or the relative standard deviation (RSD)

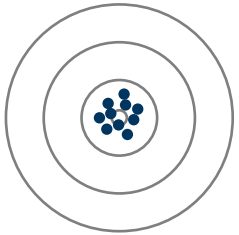
$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

n = Number of measurements

$\bar{x}$ = Mean value

$$RSD = \frac{s}{|\bar{x}|} \times 100$$

$|\bar{x}|$ = Absolut value of the mean



High accuracy, high precision

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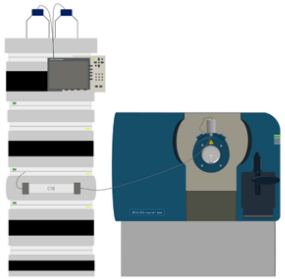
External standard calibration curve

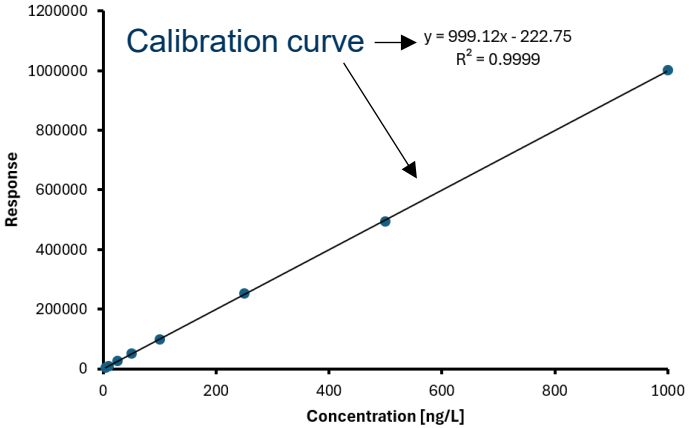
- Preparation of reference standards with different concentrations in pure solvent

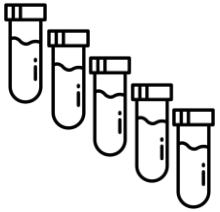
PFOS 

Concentration [ng/L]	Response
5	4970
10	10140
25	25110
50	50190
100	99540
250	251980
500	493290
1000	1001300









Tube-Icon from Flaticon.com

External standard calibration curve

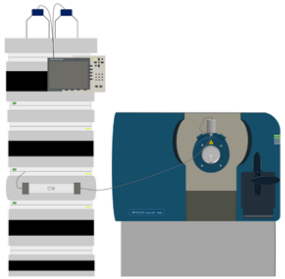
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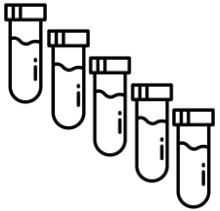
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	179619

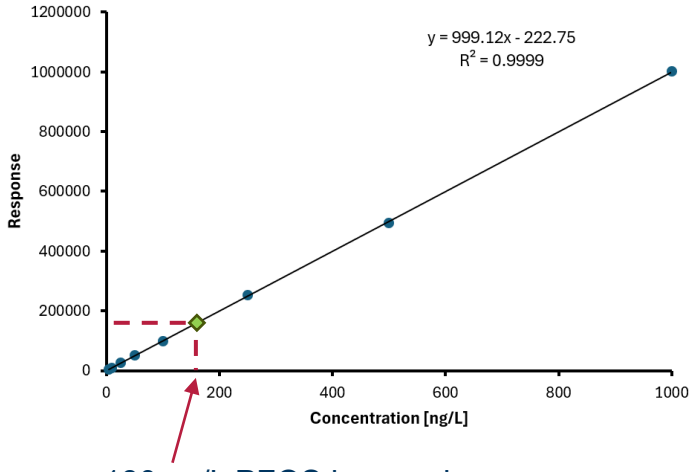
Sample







Sample

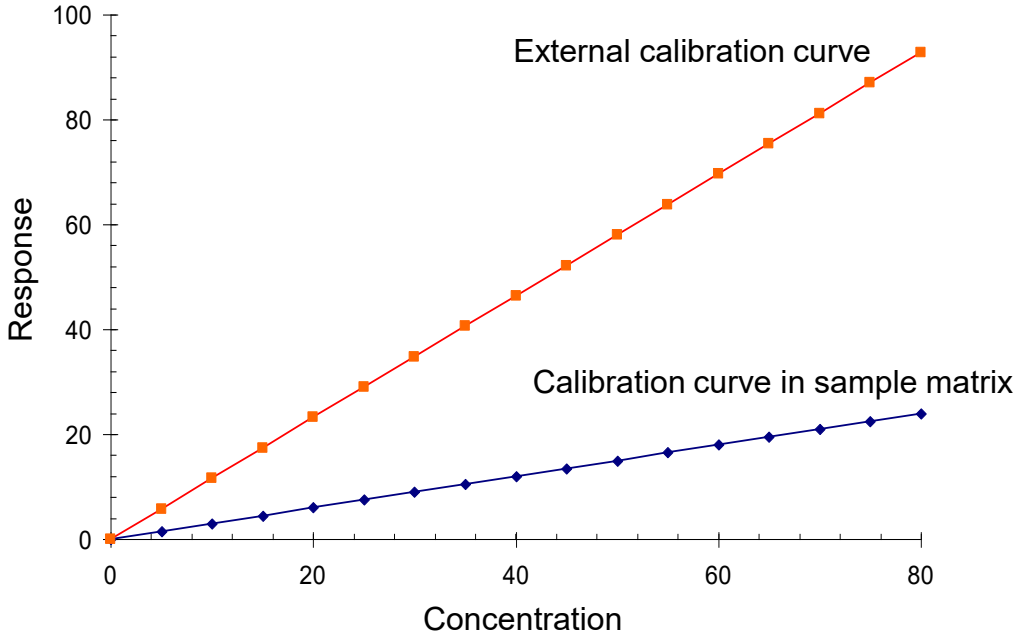


180 ng/L PFOS in sample

Tube-Icon from Flaticon.com

Matrix effects

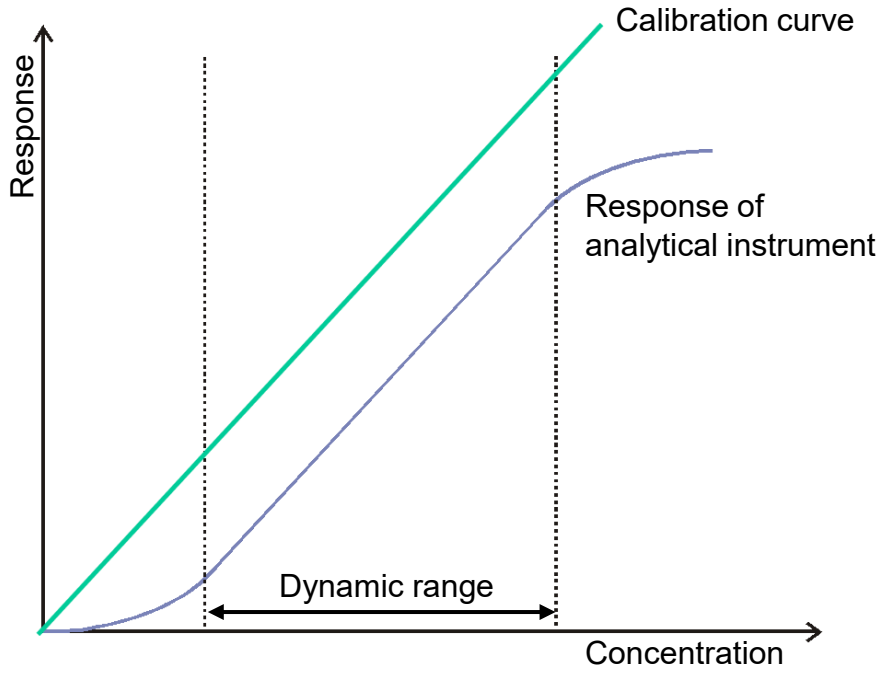
- Sample matrix = everything in the sample except the analytes of interest
- Sample matrix can interfere in the detection of an analyte
  - ➢ Signal suppression
  - ➢ Signal enhancement
- Comparison of calibration curves of external calibration and calibration in sample matrix



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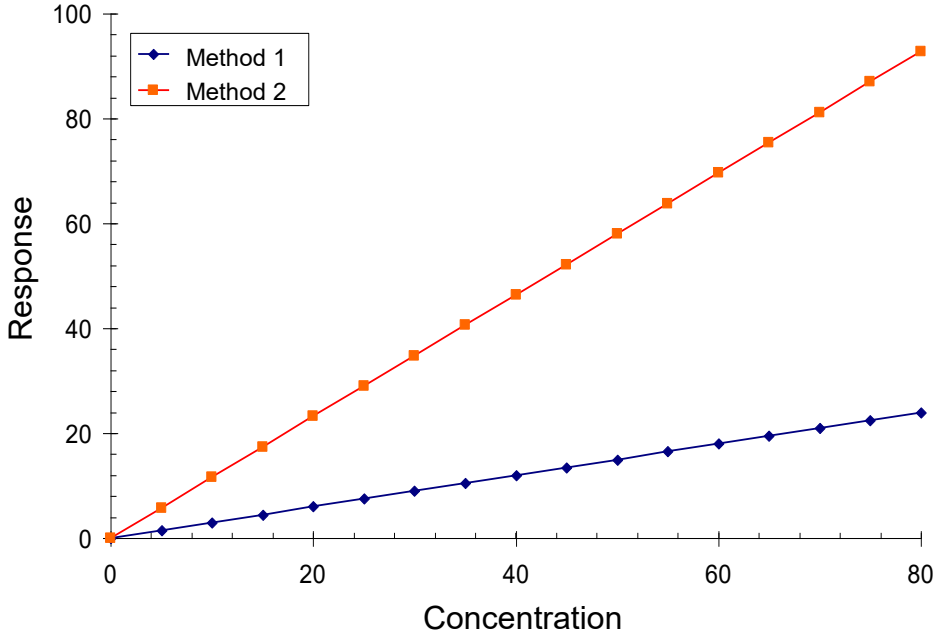
Dynamic range of an analytical method

- Linear range of detector response
- Dynamic range varies between detector types



Sensitivity of an analytical method

- The ability of an analytical method to detect the smallest possible changes in the concentration of the analyte
- Slope of calibration curve describes sensitivity
- Sensitivity depends e.g. on detector abilities, background noise, sample matrix, …



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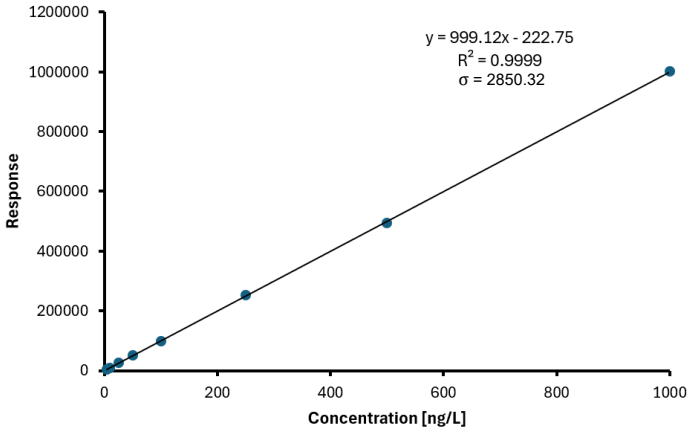
Limit of detection (LOD)

- Definition: smallest amount of a substance that can be **qualitatively** detected with a 95% probability
- Calculation based on calibration curve according to the International Committee of Harmonization (ICH)

$$LOD = 3.3 \times \frac{\sigma}{s}$$

σ = standard error of calibration curve

s = slope of calibration curve



$$LOD = 3.3 \times \frac{2850.32}{999.12} = 9.4 \frac{ng}{L}$$

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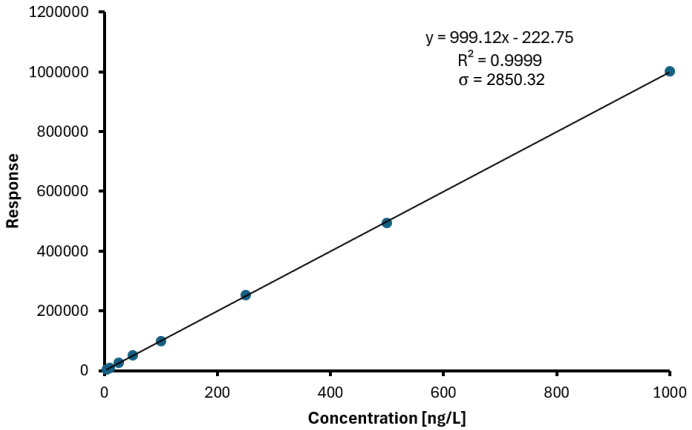
Limit of quantification (LOQ)

- Definition: smallest amount of a substance that can be **quantitively** detected with a 95% probability
- Calculation based on calibration curve according to the International Committee of Harmonization (ICH)

$$LOQ = 10 \times \frac{\sigma}{s}$$

σ = standard error of calibration curve

s = slope of calibration curve

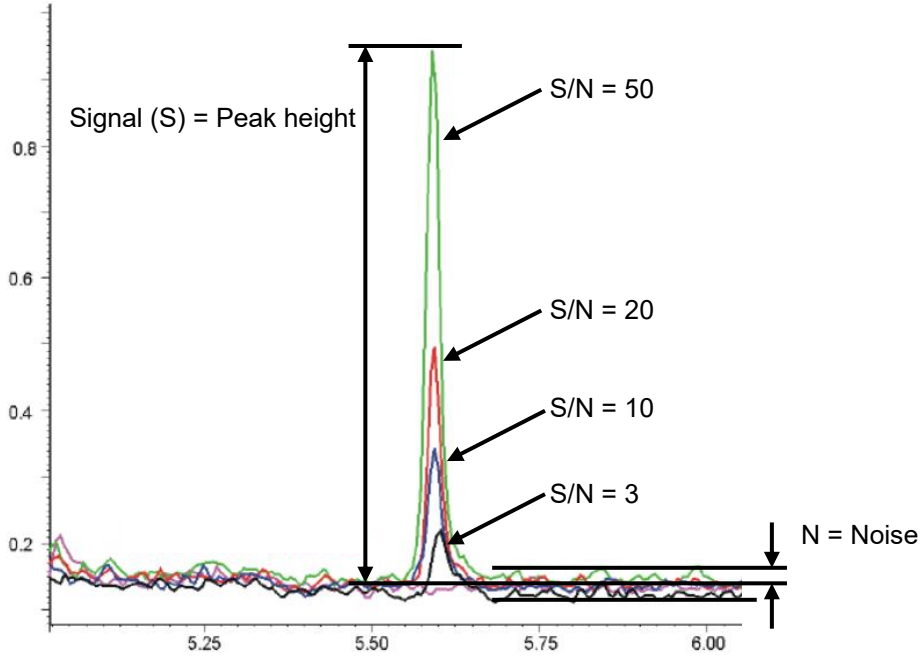


$$LOQ = 10 \times \frac{2850.32}{999.12} = 28.5 \frac{ng}{L}$$

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LOD and LOQ using Signal-to-noise ratio (S/N)

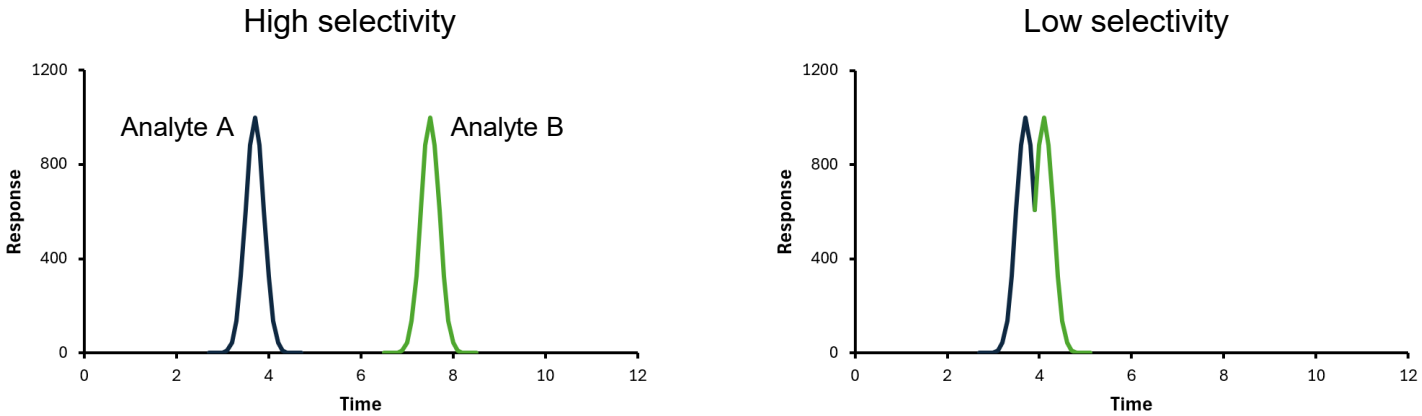
- Determination of LOD and LOQ by S/N using measurement data
  - LOD = S/N = 3
  - LOQ = S/N = 10
- Measurements of different concentration values around LOD and LOQ



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Selectivity of an analytical method

- The ability of a method to accurately and specifically determine all analytes of interest simultaneously and in the presence of other substances in the sample matrix without mutual interference
- Example of selectivity in chromatography:



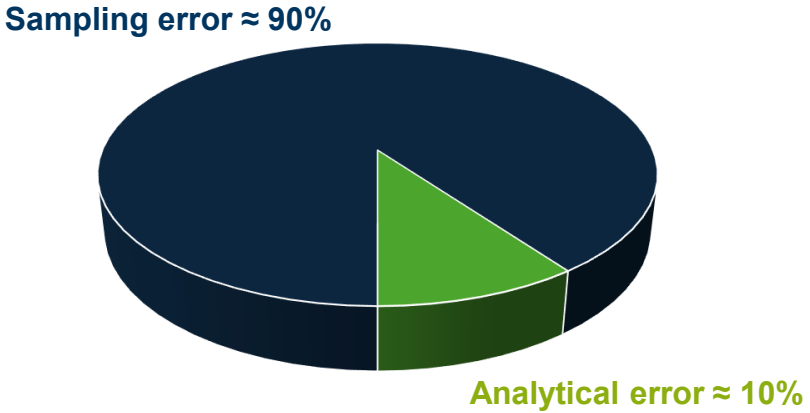
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Sampling error (S_s) and analytical error (S_a)

- Total error (S_t) of environmental data are expressed as variances:

$$S_t^2 = S_s^2 + S_a^2$$

- Sampling error mainly dominated by sampling variability
  - Heterogeneity of environmental matrices
  - Representative samples are most important
    - Critical elements e.g. sample's physical dimensions, location and timing of collection



➢ If representativeness cannot be established, the quality of the analysis is not relevant

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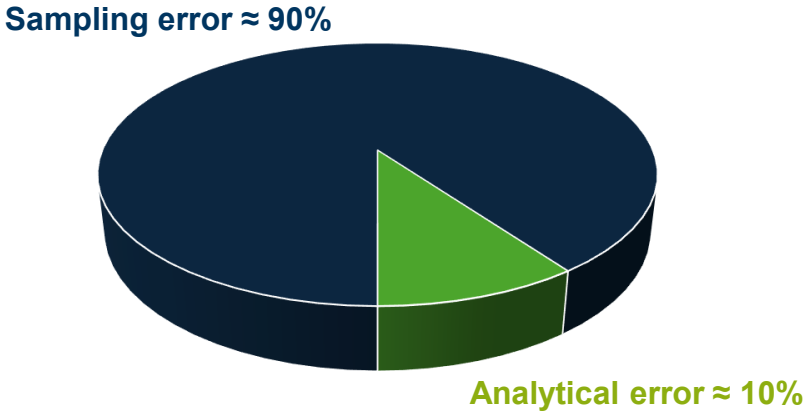
Sampling error (S_s) and analytical error (S_a)

- Total error (S_t) of environmental data are expressed as variances:

$$S_t^2 = S_s^2 + S_a^2$$

- Analytical error mainly dominated by analytical method
  - Can be determined e.g. with accuracy and precision

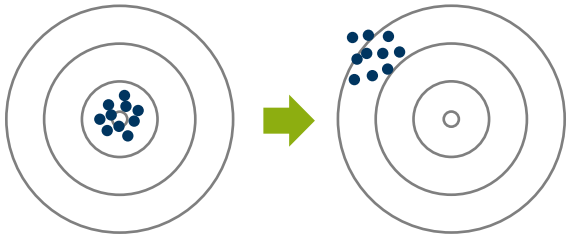
- Both S_s and S_a can be minimized using tools of quality control and quality assurance (QC/QA)



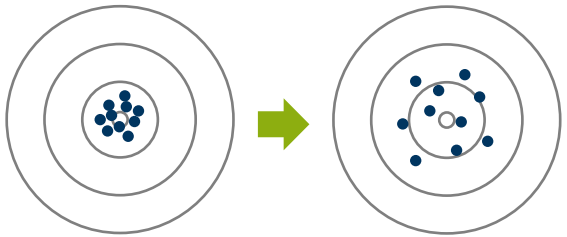
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Systematic error vs. random error

- Systematic errors
  - Shifts all measured values in one direction
    - ➢ Affects accuracy
- Random errors
  - Causes variation in the measured values
    - ➢ Affects precision



High accuracy Low accuracy



High precision Low precision

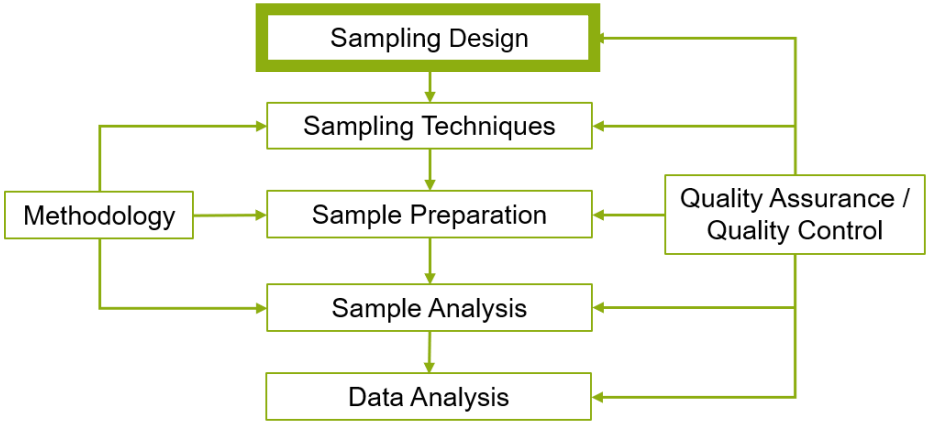
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Systematic error vs. random error

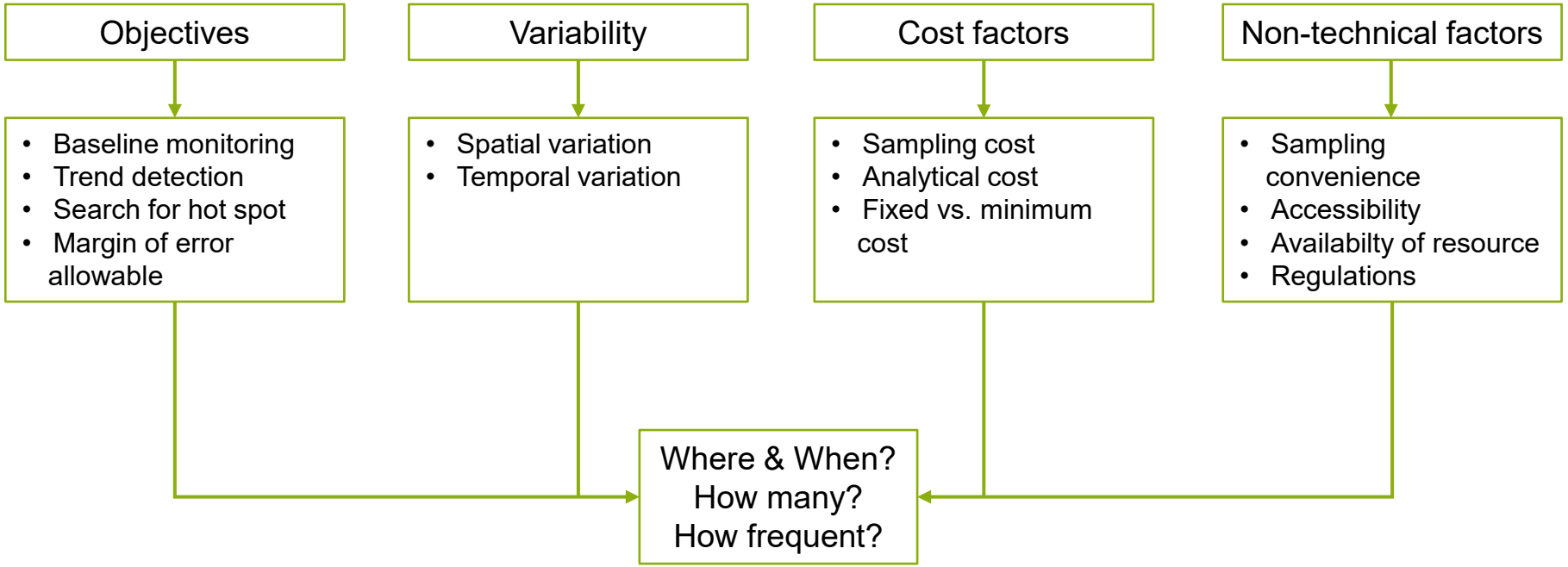
- Systematic errors
  - Shifts all measured values in one direction
    - ➢ Affects accuracy
  - Caused for example by
    - False calibration of instruments (e.g. lab scales, pH-electrode)
    - Matrix effects
    - False sample storage (e.g. evaporation or adsorption of analyte)
- Random errors
  - Causes variation in the measured values
    - ➢ Affects precision
  - Caused for example by
    - Electronical noise of analytical instruments
    - Inhomogeneous samples (uneven distribution of analyte)
    - Small variation of temperature and pressure during measurements

Sampling strategies

- Where and when to take samples?
- How many samples should be taken?
- How often samples should be taken?



Basic criteria for selecting a sampling plan



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Where (Space) and When (Time) to sample

- Only one dimension (1-D) in time domain:
  - Sampling (t_1, t_2, t_3, …, t_N) in a specific time period (e.g. hours, days, weeks, months, years)

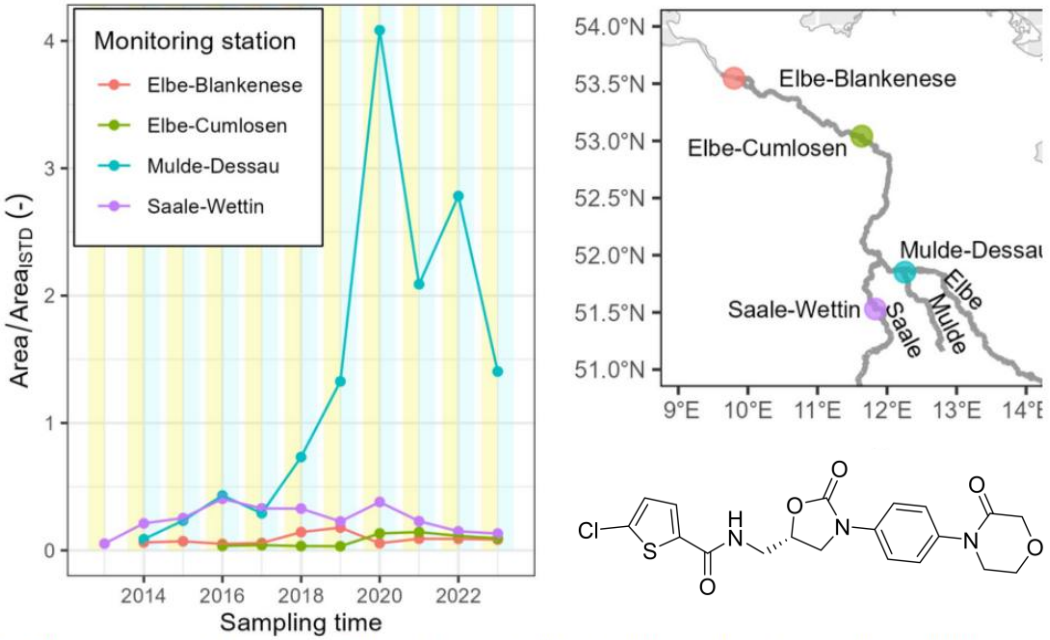


Fig. 4 Left: Peak area-time series of rivaroxaban (RIV) in annual, composite SPM samples. Peak area is relative to the internal standard (ISTD) peak area, bezafibrate-*d*4 at 1 µg/L. Data is reproduced as it is displayed on the Time Series SPM dashboard. Yellow bars indicate a measurement file is available for at least one station for that year, blue bars indicate at least one *MS2-RT* level annotation for that year. The points indicate a detection, showing the median peak area of triplicate measurements. Lines are the linear interpolation between points. Right-top: Map of the Elbe catchment in Germany showing the locations where RIV was detected (map does not show non-detects). Right-bottom: Structural formula of RIV

Figure taken from Jewell et al., Environmental Sciences Europe, **2026**.

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Where (Space) and When (Time) to sample

- Up to three dimensions (1-D, 2-D, 3-D) in space domain:
  - Sampling points can be designated by the coordinates on 1-D (x), 2-D (x, y) or 3-D (x, y, z)

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Where (Space) and When (Time) to sample

- Up to three dimensions (1-D, 2-D, 3-D) in space domain:
  - Sampling points can be designated by the coordinates on **1-D (x)**, 2-D (x, y) or 3-D (x, y, z)

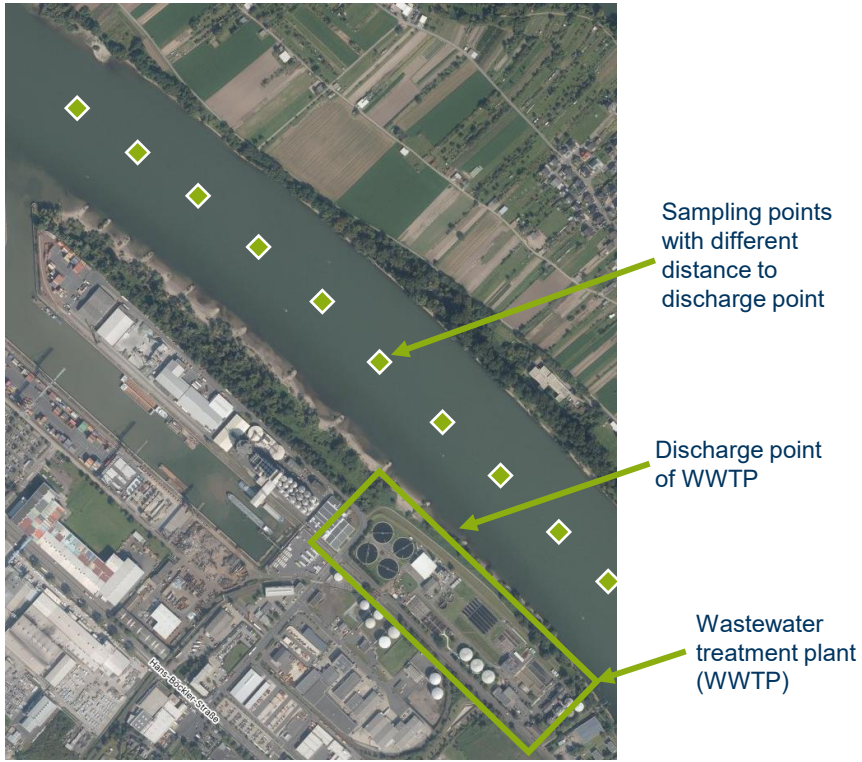


Figure taken from mapy.com

Where (Space) and When (Time) to sample

- Up to three dimensions (1-D, 2-D, 3-D) in space domain:
  - Sampling points can be designated by the coordinates on 1-D (x), **2-D (x, y)** or 3-D (x, y, z)

Fig. 1 Map of the sampling area including each sampling location and showing the position of the incineration site (1), the Ilulissat airport (2) and the city of Ilulissat (3). The exact coordinates of each sampling location are shown in Table S1

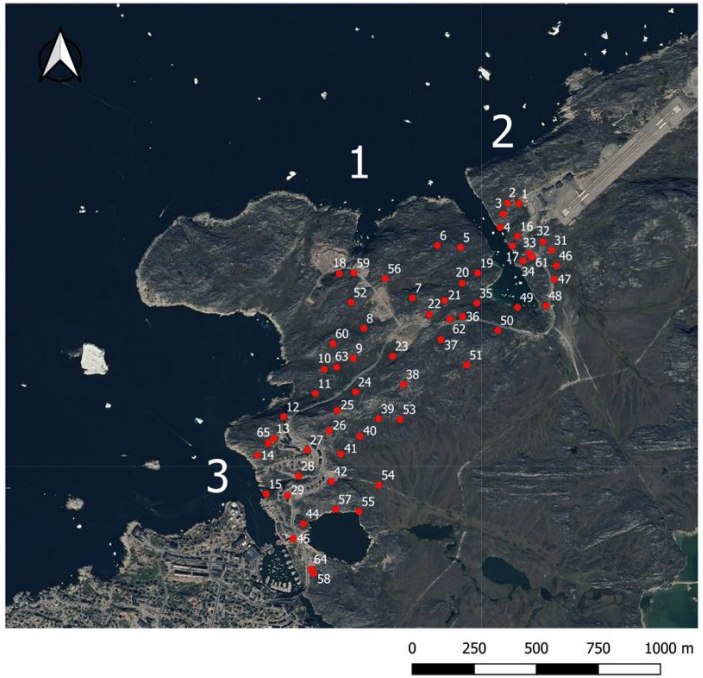
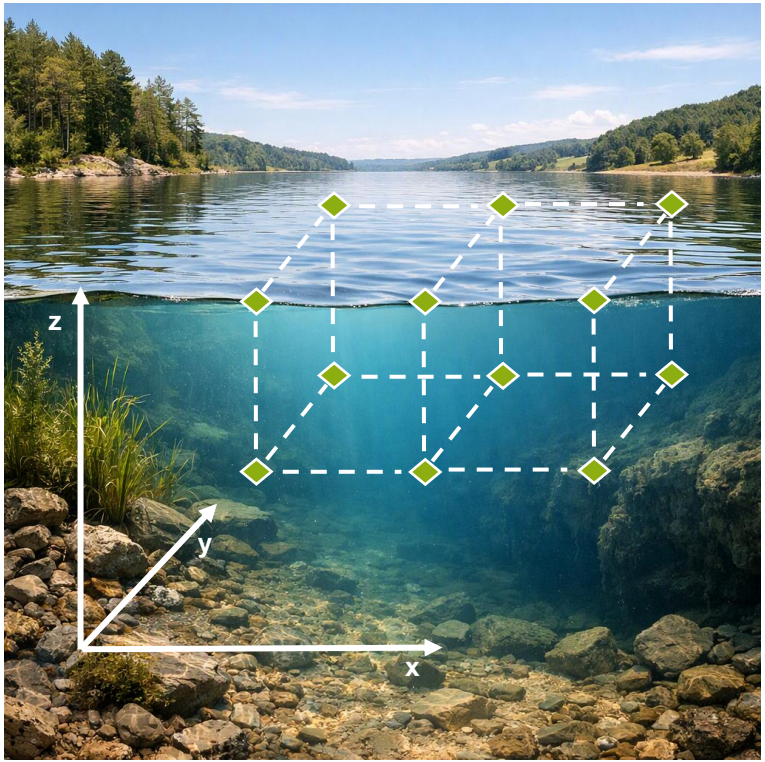


Figure taken from Kronik, Prothmann and Troché et al., Environmental Science and Pollution Research, **2021**.

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Where (Space) and When (Time) to sample

- Up to three dimensions (1-D, 2-D, 3-D) in space domain:
  - Sampling points can be designated by the coordinates on 1-D (x), 2-D (x, y) or **3-D (x, y, z)**



Lake picture generated with AI

Environmental sampling approaches

- Non-statistically based approaches
  - Haphazard sampling
  - Judgmental sampling
- Probabilistic sampling
  - Simple random sampling
  - Stratified random sampling
  - Systematic grid sampling
  - Systematic random sampling
- Composite sampling

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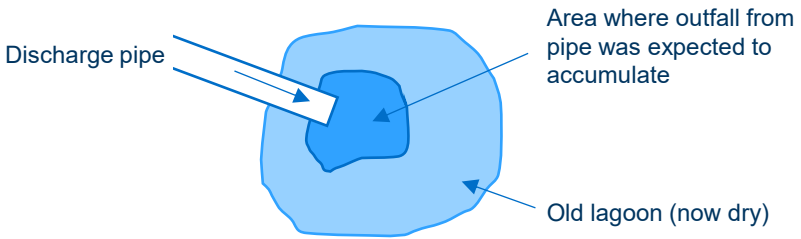
Non-statistically based approaches

- Haphazard sampling
  - Sampling in a haphazard manner, in which the sampling person may consciously or subconsciously favor the selection of certain units of the population
  - “Any sampling location will do”

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Non-statistically based approaches

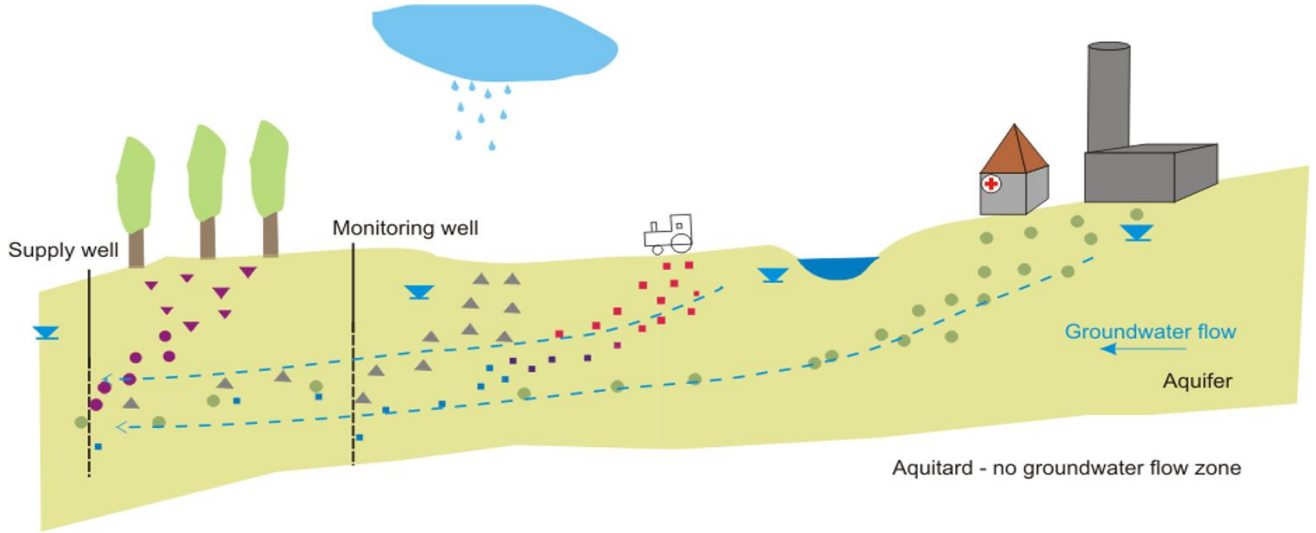
- Judgmental sampling
  - Subjective selection of sampling locations based on professional judgment using prior information on the sampling site and/or personal knowledge and experience
  - Used when schedule and budget are tight or for a first screening of a possible contamination side



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Non-statistically based approaches

- Judgmental sampling
  - Primary representative sampling approach for groundwater assessment in the selection of monitoring wells



Probabilistic sampling

- Simple random sampling
- Stratified random sampling
- Systematic grid sampling
- Systematic random sampling

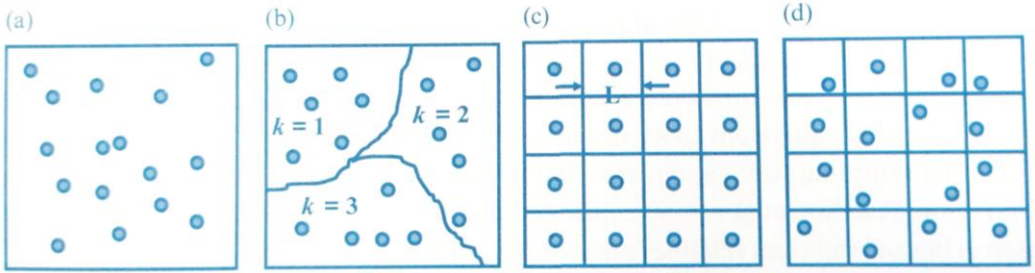


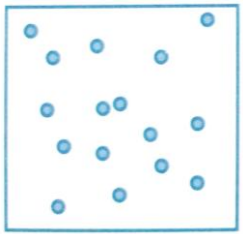
Figure 4.7 Four common probability-based sampling designs for sampling in two-dimensional space: (a) simple random sampling; (b) stratified sampling (three strata); (c) systematic grid sampling; and (d) systematic random sampling (L = spacing).

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Probabilistic sampling

- Simple random sampling
  - Arbitrary collection of samples with each sample having the same probability of being chosen
  - Simplest and most fundamental probability-based sampling design
  - Assumes that variability of sampled medium is insignificant
    → used for homogenous populations
  - Statistical analysis using arithmetic mean ($\bar{x}$) and standard deviation (s)
  - Not applicable for heterogenous populations



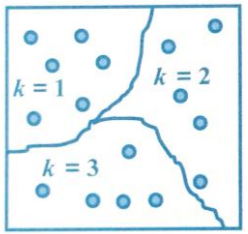
$$\bar{x} = \frac{1}{n}\sum_{n}^{i=1} x_i$$

$$s = \sqrt{\frac{\sum_{n=1}^{n=n}(x-\bar{x})^2}{n-1}}$$

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Probabilistic sampling

- Stratified random sampling
  - Dividing of sampling population into several nonoverlapping strata
  - Simple random sampling in each stratum
  - Each stratum should be relatively more homogenous than whole population
  - Temporal and spatial strata



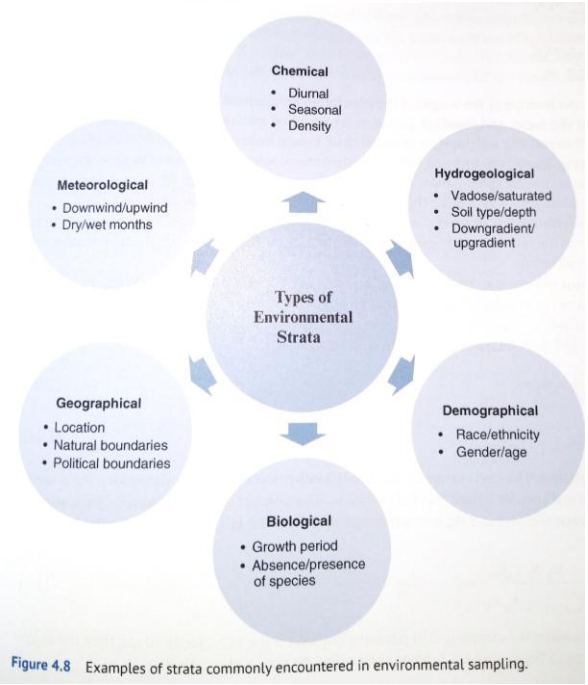
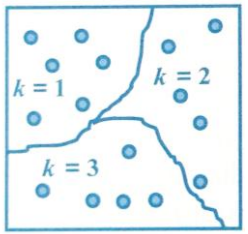


Figure taken from C. Zhang,
Fundamentals of Environmental Sampling and Analysis,
2nd Edition, **2024**.

Probabilistic sampling



- Stratified random sampling
  - Sampling size can be adjusted depending on the variations or the cost of sampling in various strata
  - Smaller standard deviation compared to simple random sampling
  - Equal allocation: each stratum has the same number of samples
  - Proportional allocation: number of samples in each stratum is proportional to the size or volume of the stratum

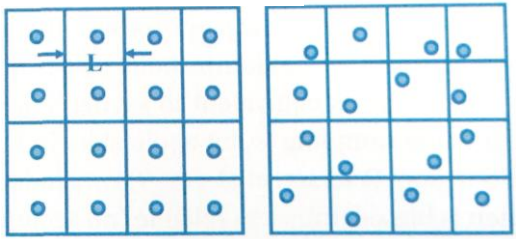
$$\bar{x} = \sum_{k=1}^{r} w_k \bar{x}_k \qquad s = \sqrt{\frac{\sum_{k=1}^{r} w_k^2 s_k^2}{n_k}} \qquad w_k = \frac{n_k}{n}$$

r = total number of strata (k = 1, 2, … r)
$\bar{x}_k$ = stratum mean
s_k = stratum standard deviation
$\bar{x}$ = population mean

s = population standard deviation
w_k = weight of the population represented by stratum k
n = total sample units
n_k = total sample units in stratum k

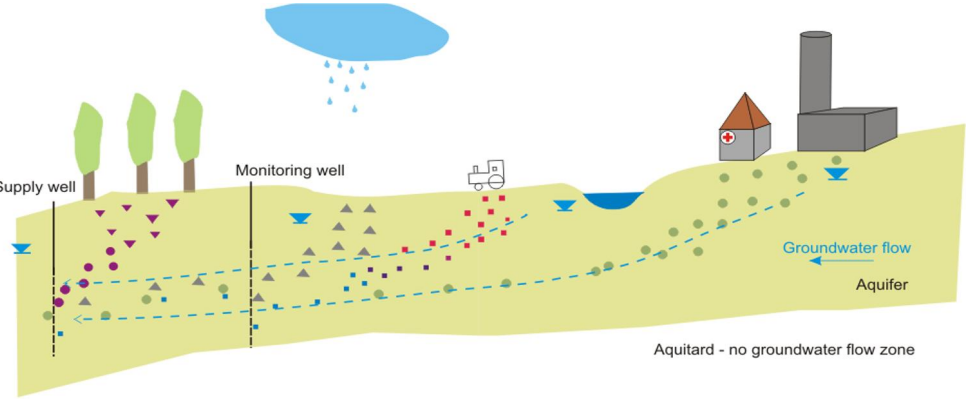
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Probabilistic sampling



- Systematic grid sampling and systematic random sampling
  - Selecting sample units according to specific pattern in time or space
  - Sampling area can be subdivided using square or triangular grids
  - Sample taken a specific point in each grid (e.g. center) or at random point within grid
  - Easy to implement and convenient in practice

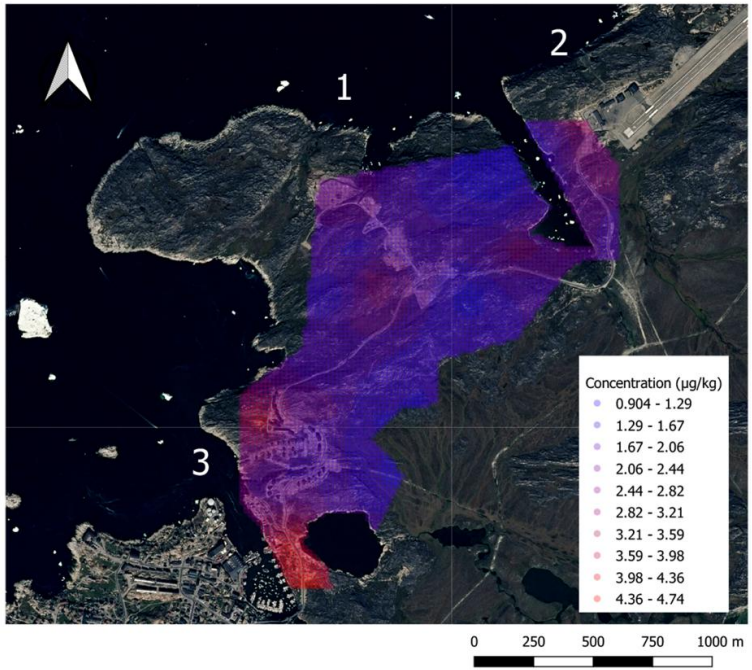
Example: Groundwater monitoring:
-Sampling at monitoring well over time with fixed sampling schedule (e.g. every 12 hours, every Monday)



Probabilistic sampling

- Systematic grid sampling and systematic random sampling
  - Helps to find hot spots in sampling area or in time domain
  - Possibility to define concentration gradients in sampling areas

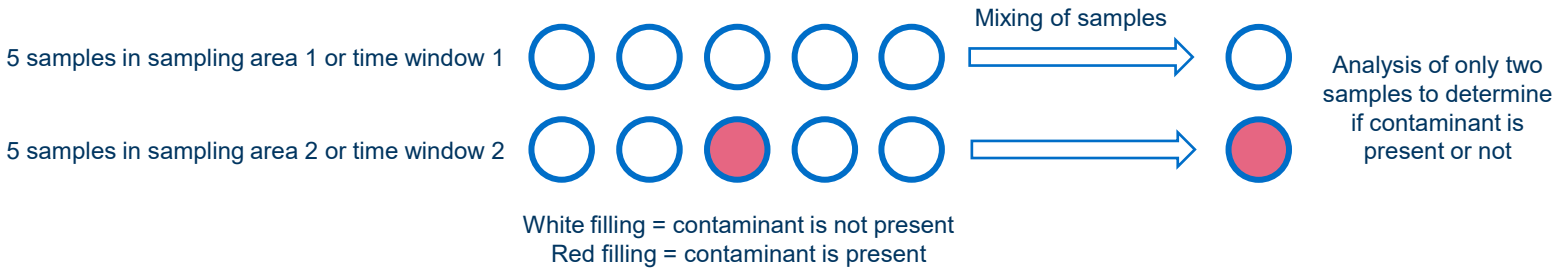
Fig. 4 Kriging plot obtained for pyrene covering the sampling area between the incineration site (1), the Ilulissat airport (2) and the city of Ilulissat (3) showing the distribution of the predicted pyrene concentrations



Figures taken from Kronik, Prothmann and Troché et al., Environmental Science and Pollution Research, **2021**.

Composite sampling

- Mixing discrete samples collected at different times (temporal) or different locations (spatial)
- Cost-saving → less samples need to be analysed
- Common sampling approach for sediment samples due to suspected heterogeneity



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Composite sampling

- Mixing discrete samples collected at different times (temporal) or different locations (spatial)
- Cost-saving → less samples need to be analysed
- Also mixing of continuously collected samples on the flow, volume, or time proportional basis possible
  - Automated samplers

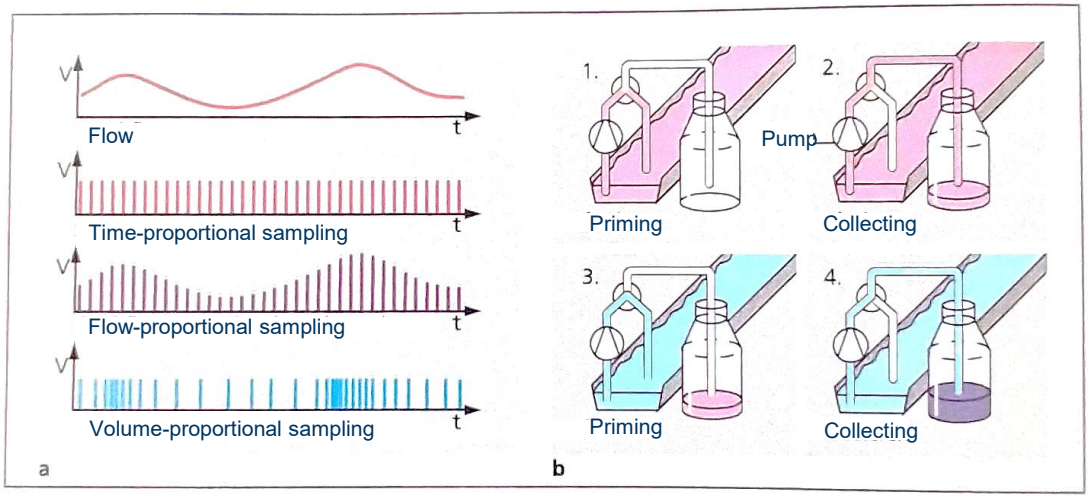
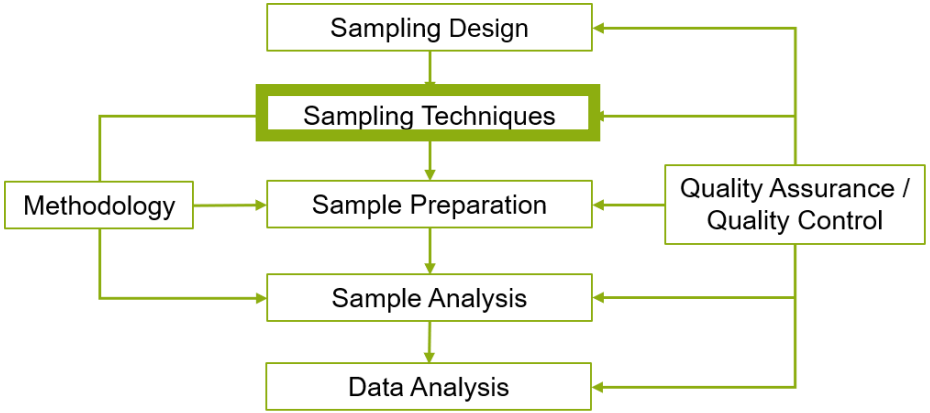


Figure taken and adapted from Schwedt, Schmidt, Schmitz, Analytische Chemie, 3rd Edition, **2016**.

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Environmental sampling techniques

- How to take representative samples?
- How much sample should be taken?
- How should the sample be preserved prior analysis?

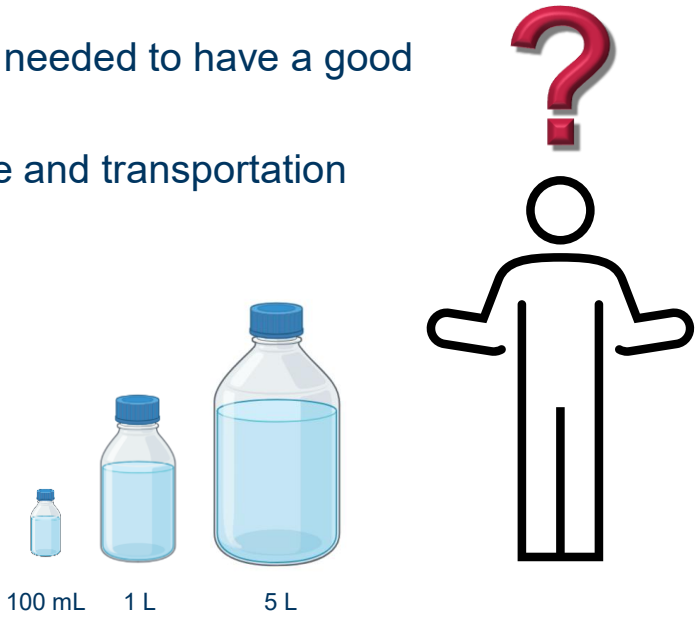


Sequence of sampling

- Try not to contaminate sampling locations while sampling
- Examples:
  - Collect from the least to the most contaminated sampling location whenever possible to reduce potential cross-contamination between samples
  - If water and sediment is collected at the same side → collect water first to minimize effects from suspended bed material
  - Stay downstream if you collect a sediment sample while standing in the water
  - If sampling at different water depths → collect surface water first and then deeper samples for minimal interference with the water column

Sample amount

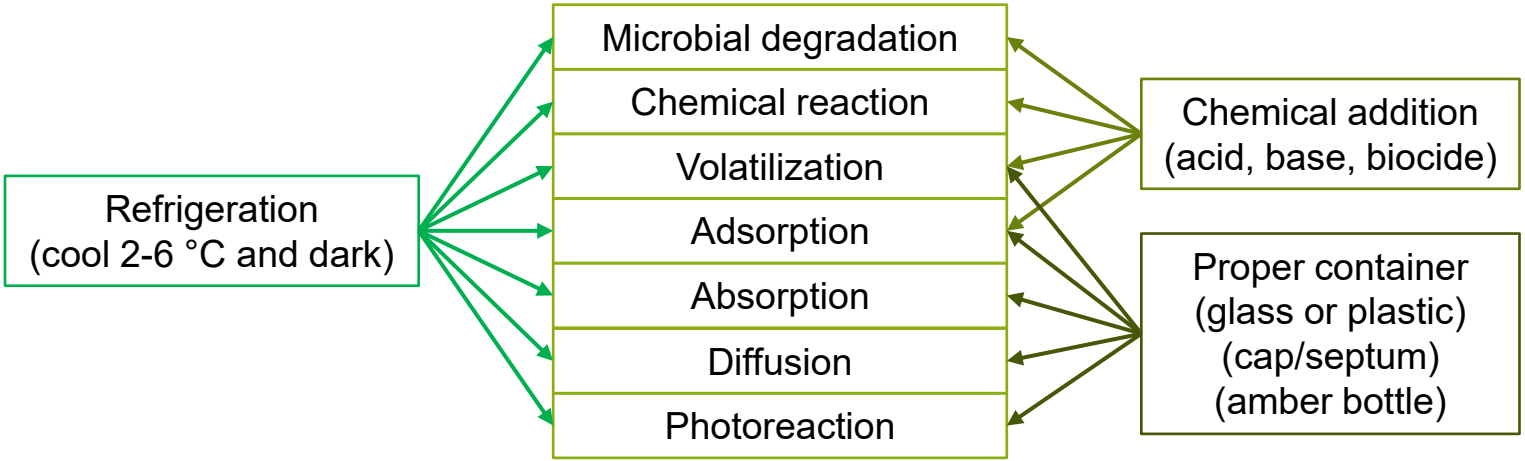
- Depends primarily on the concentration of the contaminant in the sample matrix
- Sufficient sample volume for analysis including QA/QC
- For heterogenous samples generally larger amounts are needed to have a good representativity of the sample variations
- Too large samples can be problematic for sample storage and transportation



Bottles from BioRender.com

Sample preservation and storage

- Minimizing any physical, chemical, and/or biological changes in a sample from the time of sample collection until sample analysis



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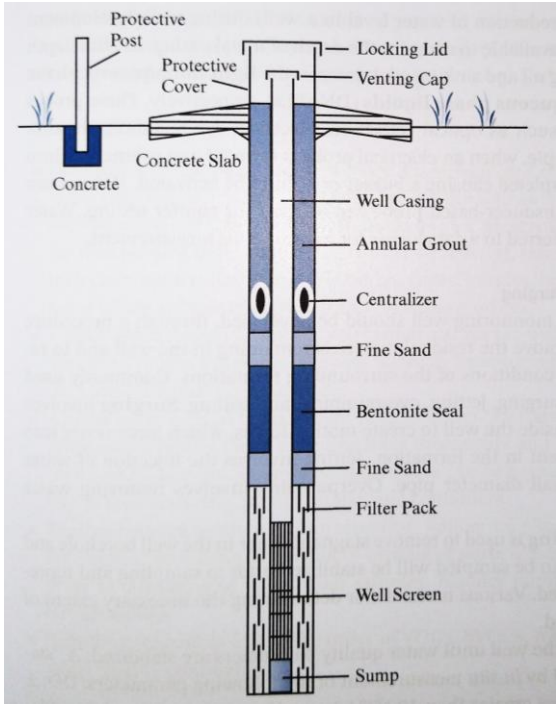
Sample preservation and storage

- Minimizing any physical, chemical, and/or biological changes in a sample from the time of sample collection until sample analysis

Examples of recommended preservation for some classes of organic chemicals

Analytes	Change during storage	Preservation
Phthalate ester	Diffusion from plastics	Use Teflon or glass bottles
Volatile organic compounds	Volatilisation	Avoid headspace
Polyaromatic hydrocarbons	Photochemical degradation	Use amber glass bottles

Sample equipment for groundwater samples



Scheme of a monitoring well

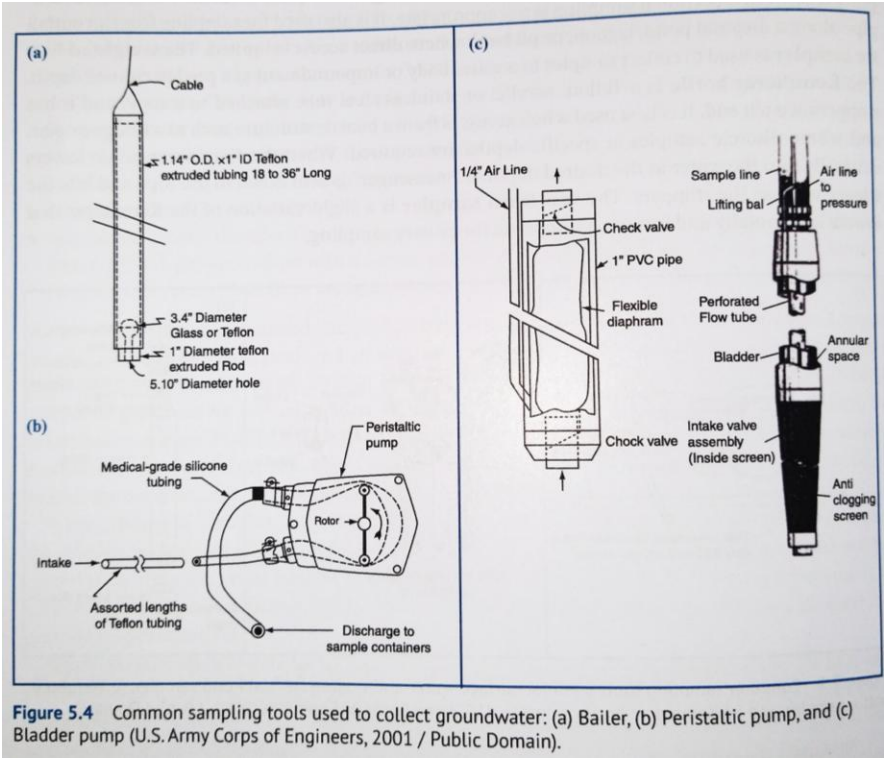


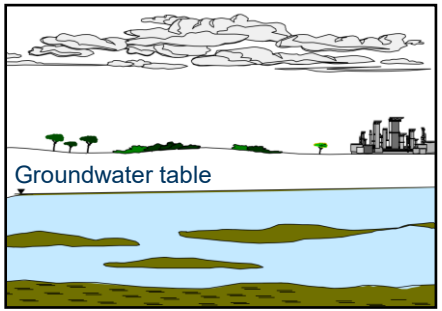
Figure 5.4 Common sampling tools used to collect groundwater: (a) Bailer, (b) Peristaltic pump, and (c) Bladder pump (U.S. Army Corps of Engineers, 2001 / Public Domain).

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Sampling of groundwater samples

- Groundwater flow direction
  - Measuring of water table in three monitoring wells
  - Determination of groundwater table using a water level meter





Water level meter



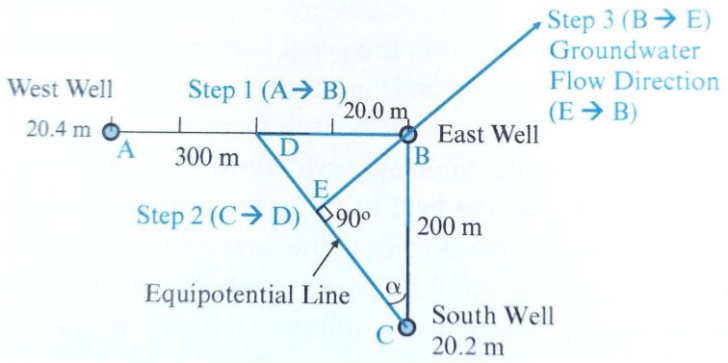
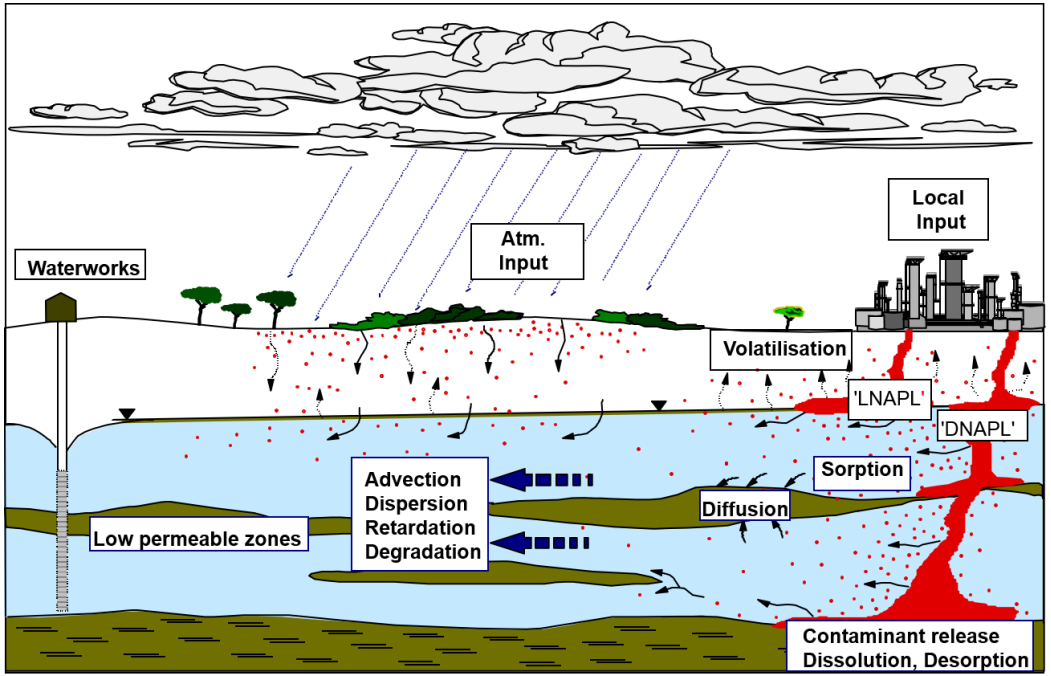


Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

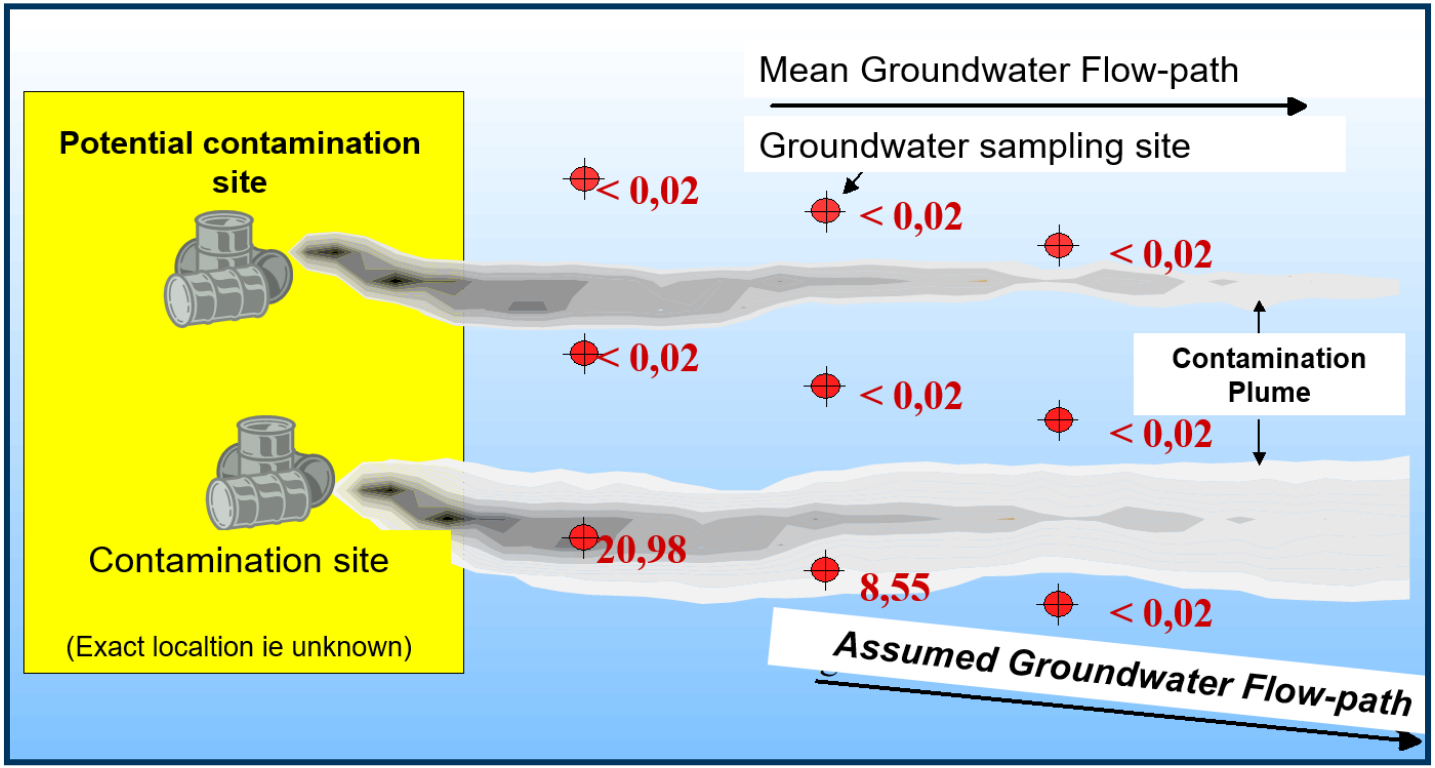
Sampling of groundwater samples

- Contaminants are transported with the groundwater flow direction
- Light nonaqueous phase liquids (LNAPLs) accumulate above groundwater table
- Dense nonaqueous phase liquids (DNAPLs) sink downward and accumulate at non permeable zones



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Sampling of groundwater samples



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Sample equipment for water samples

- Surface water
  - Pond sampler used at shore sampling
  - Weighted bottle sampler and Kemmerer bottle sample used for sampling at specific depth

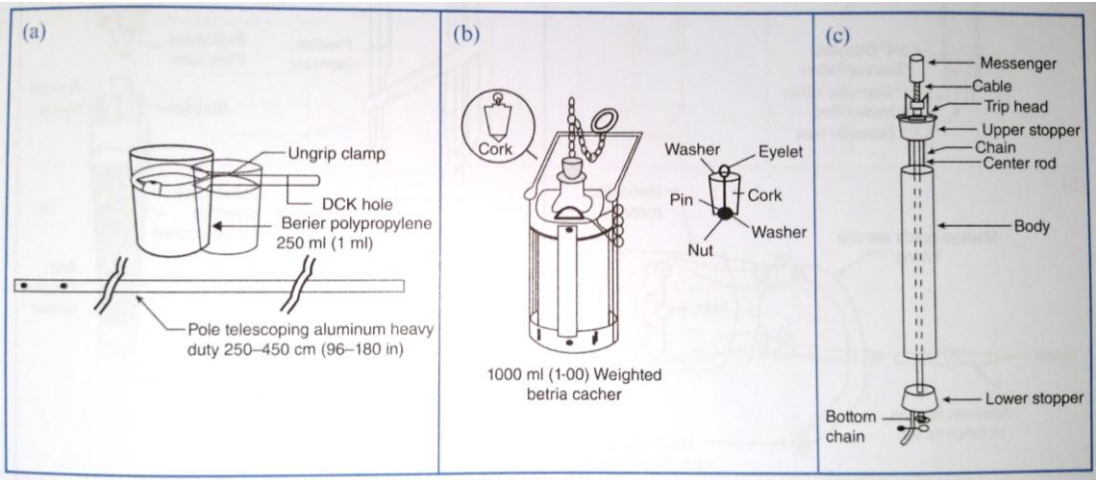


Figure 5.3 Common sampling tools used for surface water and wastewater: (a) Pond sampler, (b)Weighted bottle sampler, and (c) Kemmerer bottle sampler (U.S. Army Corps of Engineers, 2001 / Public Domain).

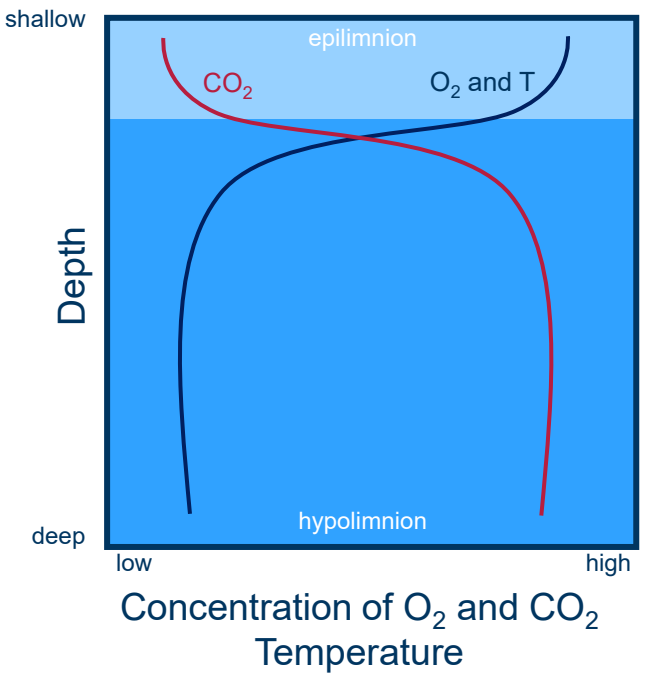
Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Sampling in stagnant water bodies

- Stratification (vertical) and inhomogeneity (horizontal)

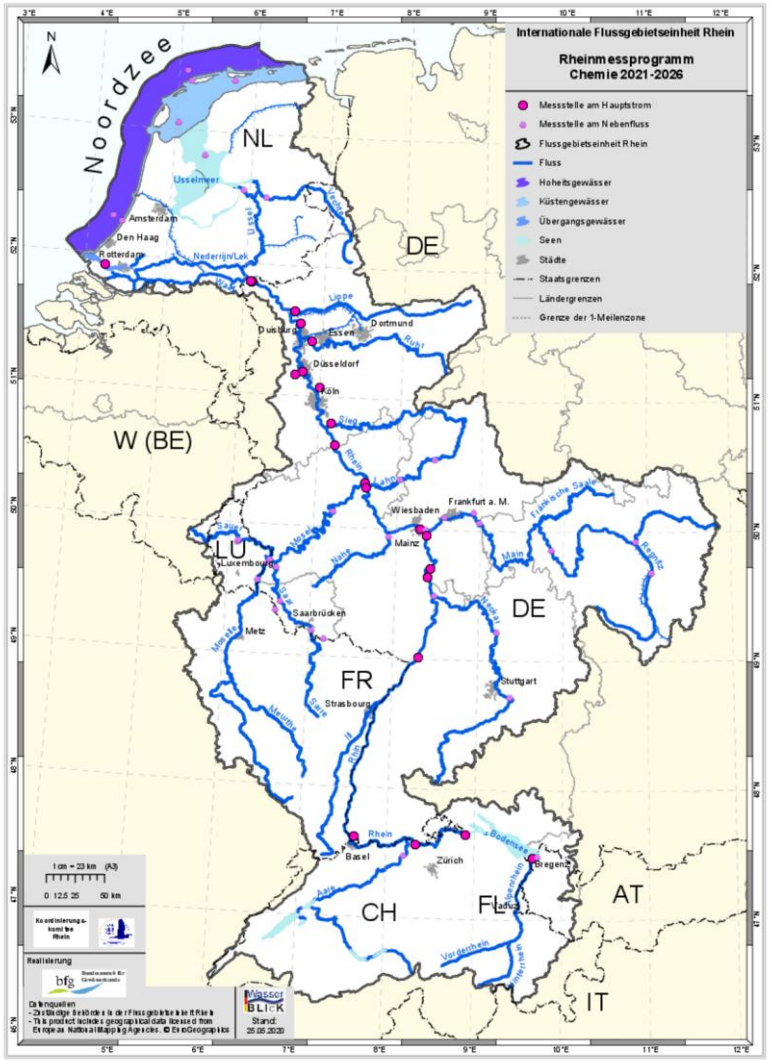




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Sampling in rivers

- High temporal variability
- Installations of monitoring stations with automated sampling along large rivers



Net of monitoring stations of the Rhine Monitoring Programme Chemistry 2021-2026
Figure taken from International Commission for the Protection of the Rhine (ICPR, www.iksr.org)

Sampling in rivers

- Mixing zones
- Dissolved and particulate phase
- Flow profile
  - Higher flowrate at surface and lower flowrate at riverbed
- Profile of concentration of particulate matter
  - Lower concentration at surface and higher concentration at riverbed



Confluence of Rhine and Moselle Rivers in Koblenz, Germany
(Picture taken from Wikipedia article “Moselle”, 20.08.2026)

Sample equipment for sediment samples

- Sampling around shorelines of shallow and slow-moving waters
- Sediment samples are often collected as composite samples due to suspected heterogeneity

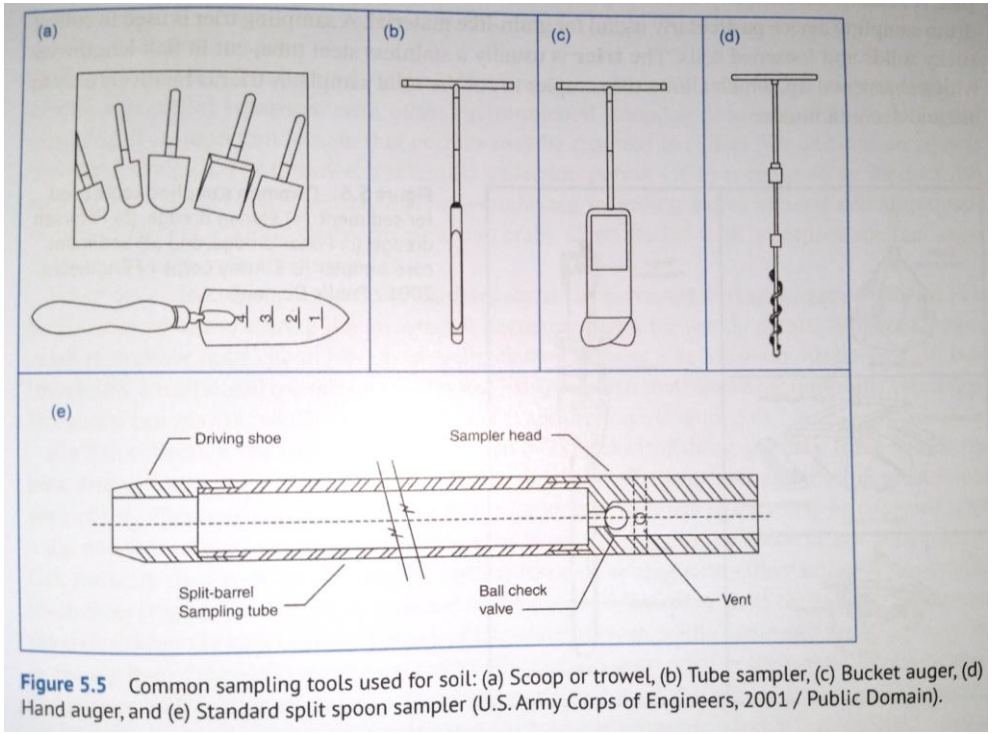


Figure 5.5 Common sampling tools used for soil: (a) Scoop or trowel, (b) Tube sampler, (c) Bucket auger, (d) Hand auger, and (e) Standard split spoon sampler (U.S. Army Corps of Engineers, 2001 / Public Domain).

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Sample equipment for sediment samples

- Grab samplers are used from a boat, bridge or pier
- Core sampler is used to compare recent surficial vs. historical deeper sediments

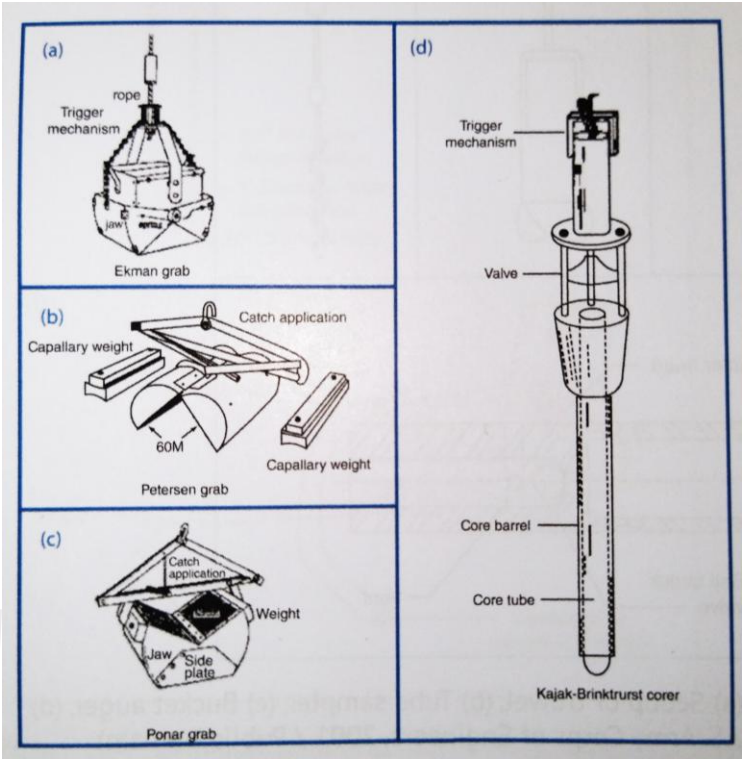


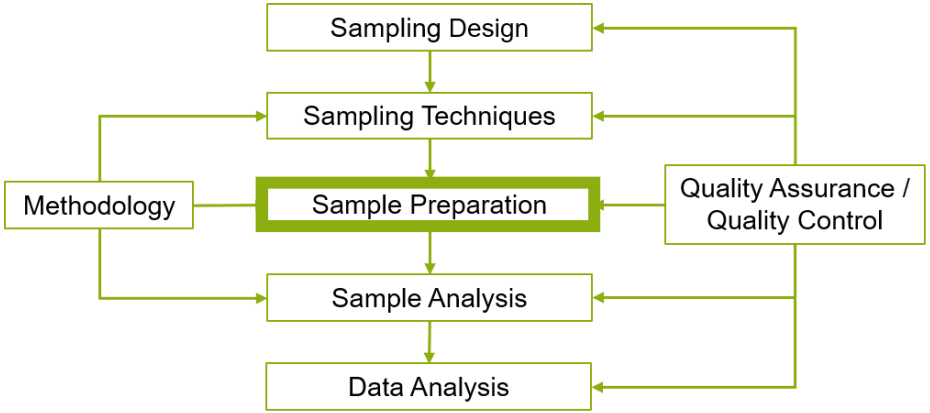
Figure 5.6 Common sampling tools used for sediment: (a) Ekman dredge, (b) Petersen dredge, (c) Ponar dredge, and (d) Sediment core sampler (U.S. Army Corps of Engineers, 2001 / Public Domain).

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Sample preparation

- How to prepare the sample for analysis?
- How to remove disturbing matrix?
- How to increase analyte concentration?



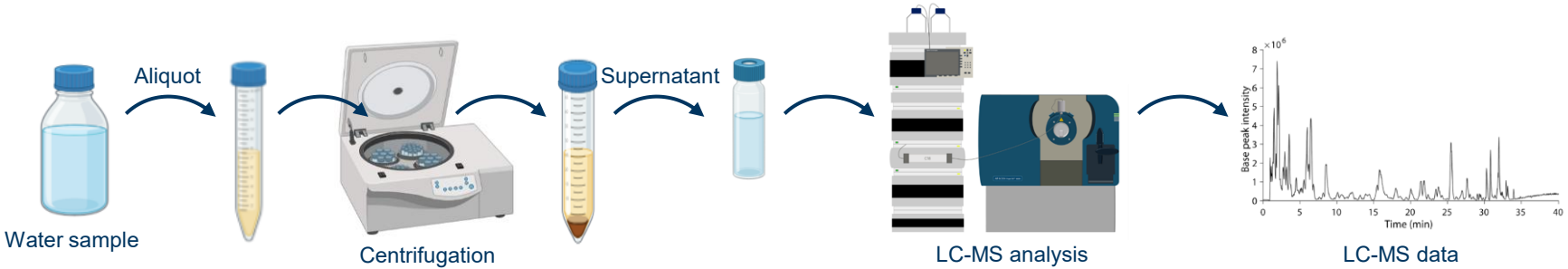
Sample preparation

- Important part of analytical process
  - Can be labor-intensive and time-consuming
  - Environmental samples can rarely be directly injected into an analytical instrument without pretreatment
    - Change of sample phase → most instruments work with aqueous samples
    - Only a few exception → for example drinking water
  - Increasing analyte concentration → trace analysis
  - Removing disturbing matrix → reducing matrix effects

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Sample preparation of water samples

- Depending on the aim of the project and sensitivity of analytical instrument centrifugation of surface water or groundwater samples can be an easy and sufficient sample preparation



- Possible drawbacks:
  - Compounds at low concentrations may not be detected → below LOD
  - Matrix effects may disturb measurements

Bottles and centrifuge from BioRender.com

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Sample preparation techniques for aqueous samples

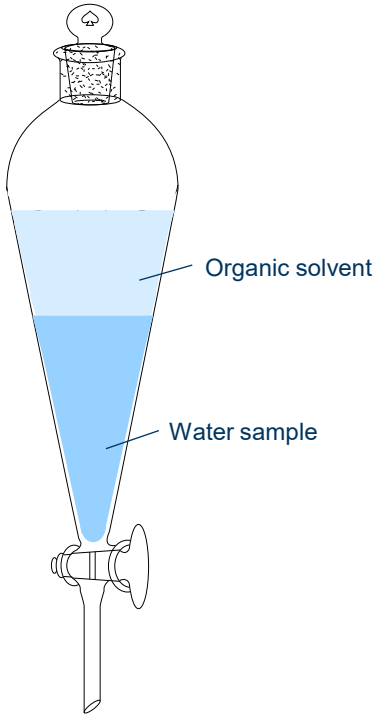
- Liquid-liquid extraction (LLE)
- Solid phase extraction (SPE)
- Static headspace extraction (SHE)
- Solid phase microextraction (SPME)
- Headspace solid phase microextraction (HS SPME)
- Magnetic solid phase extraction (MSPE)
- Stir-bar sorptive extraction (SBSE)
- Single-drop microextraction (SDME)
- …

Liquid-liquid extraction (LLE)

- Extraction of semi-volatile organic compounds (NVOCs) and non-volatile organic compounds (NVOCs)
- Relies on partitioning of analytes between two immiscible liquid phases with different densities
  - Normally a water phase and an organic solvent phase
- Analyte must favorably distribute one phase over the other
  → described by partition coefficient K

$$K = \frac{C_o}{C_w}$$

C_o = equilibrium concentration in immiscible organic solvent phase
C_w = equilibrium concentration in water phase



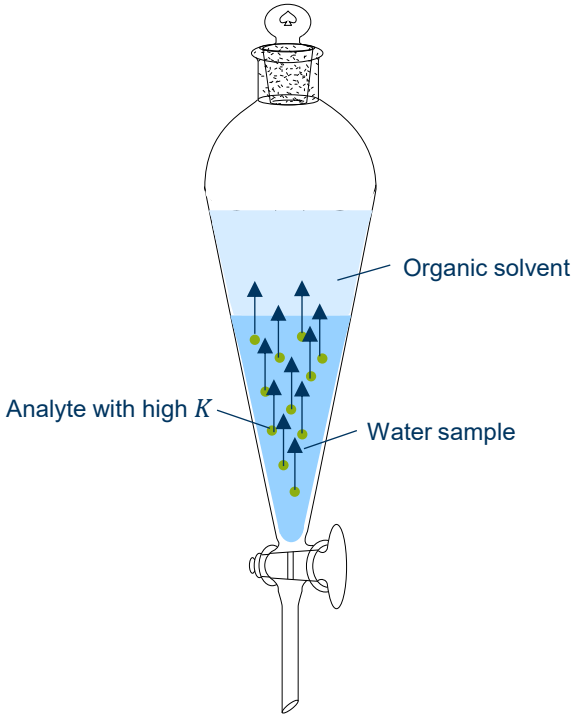
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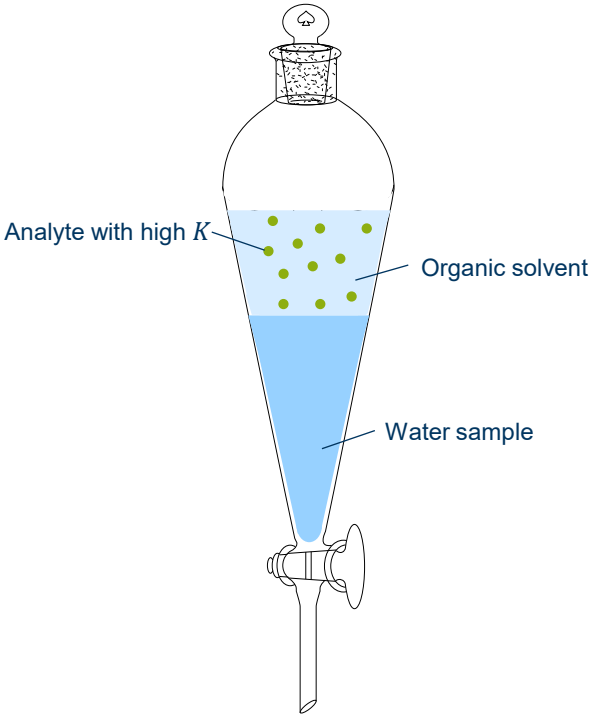
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- Relies on partitioning of analytes between two immiscible liquid phases with different densities
  - Normally a water phase and an organic solvent phase
- Analyte must favorably distribute one phase over the other → described by partition coefficient K

$$K = \frac{C_o}{C_w} \qquad E_w = \frac{1}{\left(1 + K\frac{V_o}{V_w}\right)^n} \qquad E_o = 1 - E_w$$

C_o = equilibrium concentration in immiscible organic solvent phase
C_w = equilibrium concentration in water phase
E = extraction efficiency
E_w = fraction of analyte remained in water phase
E_o = fraction of analyte remained in organic phase

V_o = volume of organic phase
V_w = volume of water phase
n = number of successive extractions with same solvent volume (V_o)



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Liquid-liquid extraction (LLE)

- Solvent miscibility chart

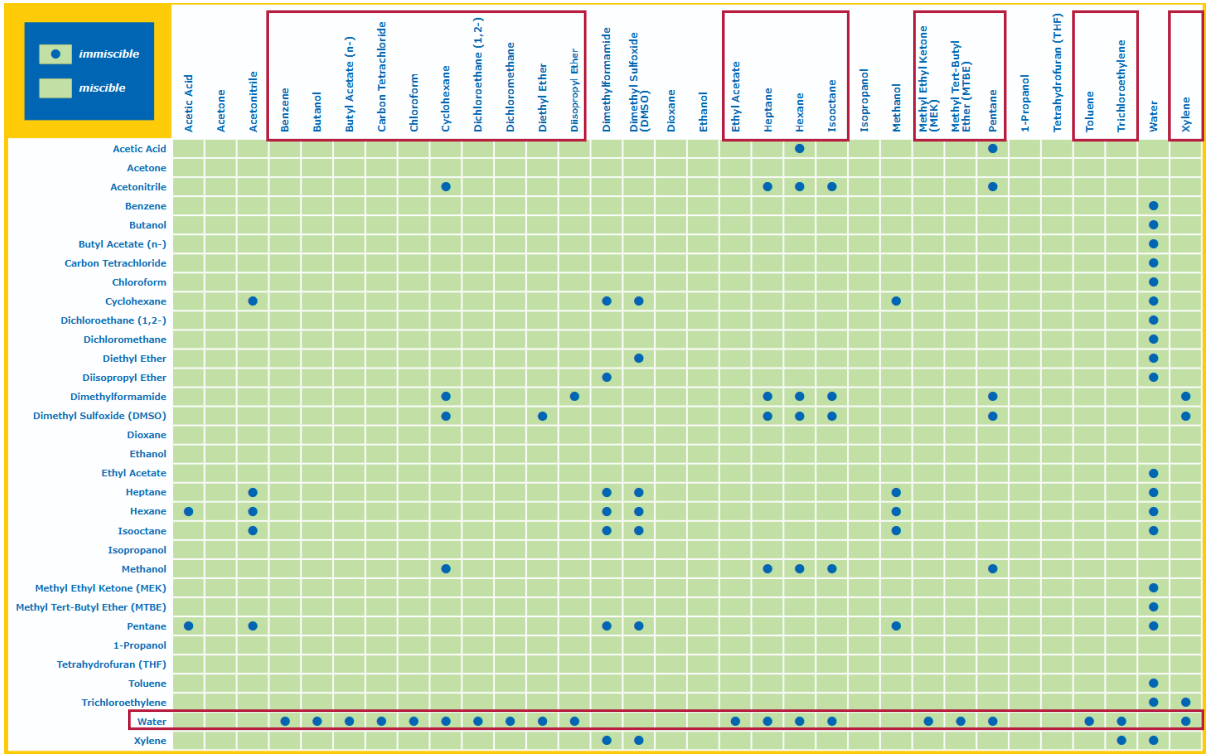
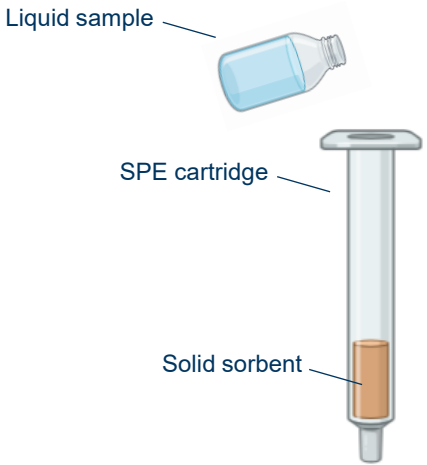


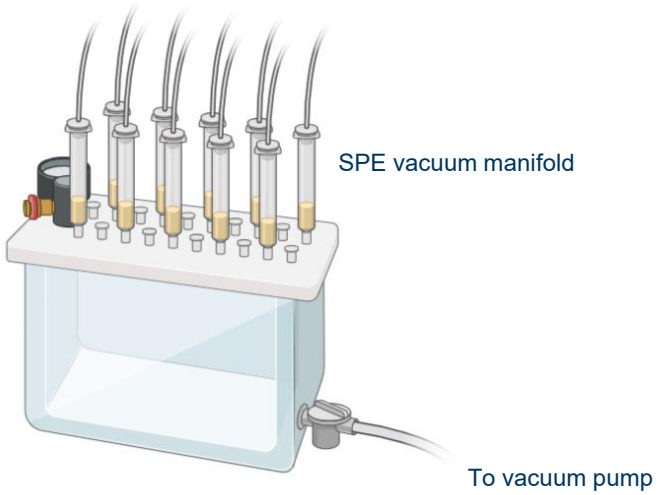
Figure taken from Merck at https://www.sigmaaldrich.com/DE/de/technical-documents/technical-article/analytical-chemistry/purification/solvent-miscibility-table?srsltid=AfmBOoqYdUB1kc_MMeyZ0LIxpsdir8RYx6j2satCCIW2ygDZy8qpVN4u

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Solid phase extraction (SPE)

- Retaining of analytes from a flowing liquid on a solid sorbent based on physicochemical interactions
- Extraction of semi-volatile organic compounds (NVOCs) and non-volatile organic compounds (NVOCs)





Illustrations from BioRender.com

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Solid phase extraction (SPE)

- Retaining of analytes from a flowing liquid on a solid sorbent based on physicochemical interactions
- Extraction of semi-volatile organic compounds (NVOCs) and non-volatile organic compounds (NVOCs)

Solid phase	Phase modification	Analytes
Reversed phase	C8, C18, …	Non-polar
Normal phase	Diol, cyanopropyl, …	Polar
Ion exchange phase	Anionic or cationic	Ionic

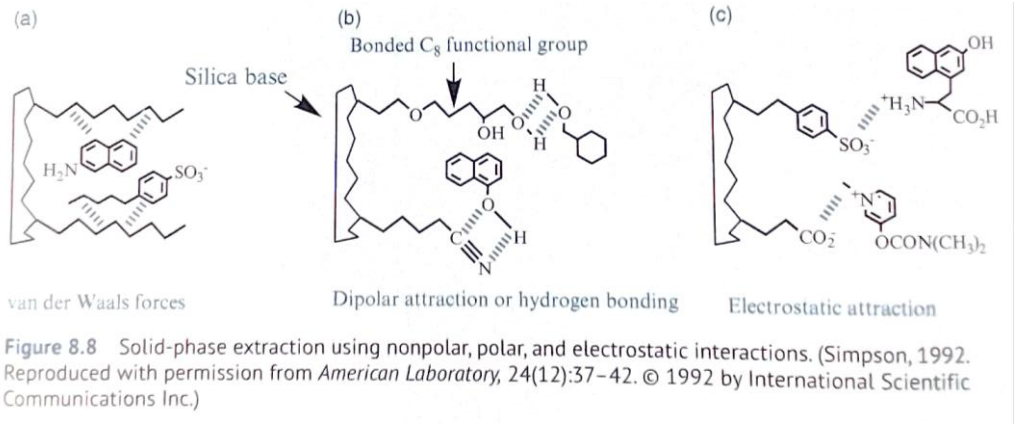


Figure 8.8 Solid-phase extraction using nonpolar, polar, and electrostatic interactions. (Simpson, 1992. Reproduced with permission from *American Laboratory*, 24(12):37–42. © 1992 by International Scientific Communications Inc.)

Cartridge from BioRender.com

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Solid phase extraction (SPE)

- SPE workflow

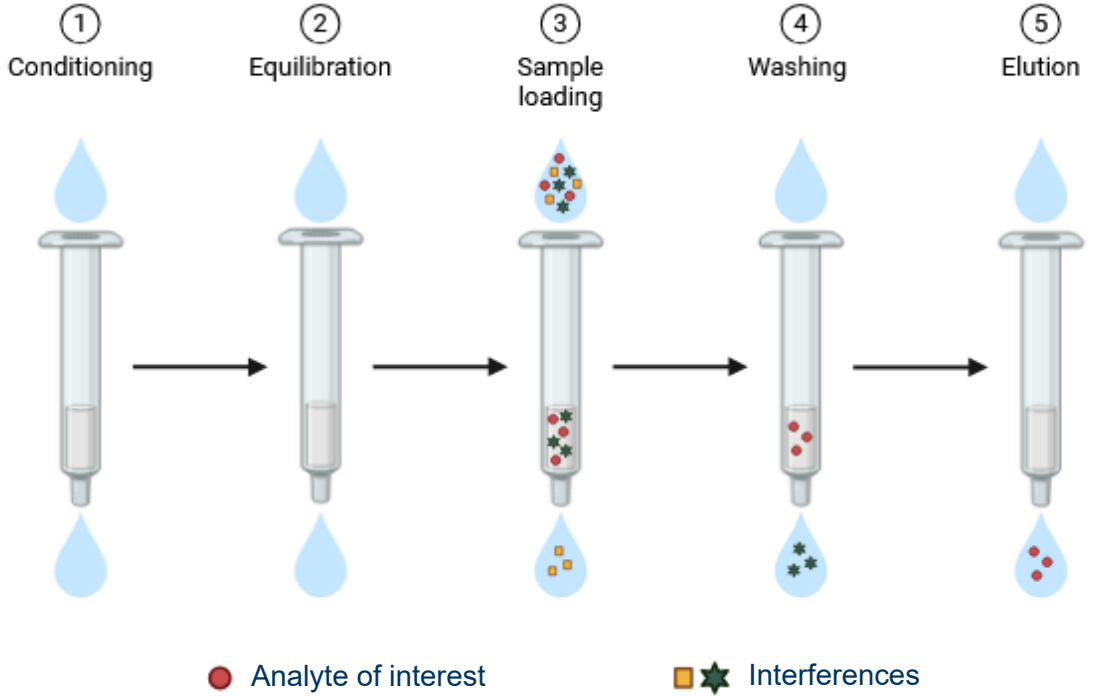


Illustration from BioRender.com

Static headspace extraction (SHE)

- Extraction of volatile organic compounds (VOCs) from a liquid or solid phase
- Sample is placed in a sealed vial
- VOCs diffuse into gas phase until headspace has reached state of equilibrium → described by partition coefficient K

$$K = \frac{C_g}{C_s} \qquad C_0 = C_g\left(\frac{1}{H} + \beta\right) \qquad \beta = \frac{V_g}{V_s}$$

C_g = equilibrium concentration in gas phase
C_s = equilibrium concentration in sample phase
C_0 = original concentration of volatile analyte in sample
H = Henry's law volatility constant of the analyte

β = phase ratio
V_g = Volume of gas phase
V_s = Volume of sample phase

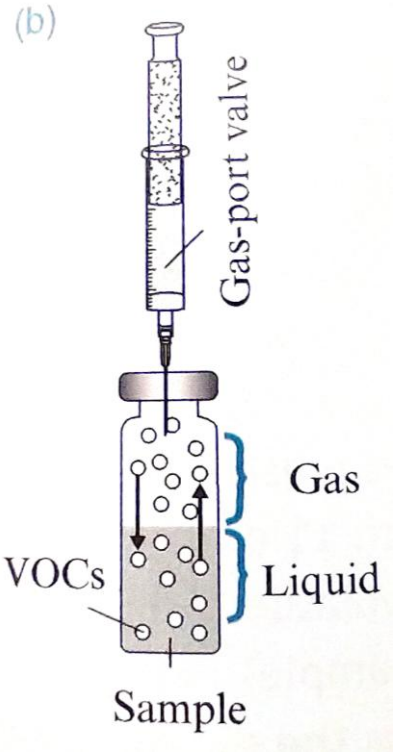


Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

Sample preparation of sediment samples

- Pretreatment workflow of sediments for NVOCs and SVOCs before extraction:
  - Drying
  - Homogenizing (e.g. mixing of several samples (composite sampling), grounding)
  - Sieving
    - Separating gravels (4.74-76.2 mm) and sands (75 µm-4.74 mm) from silt (2-75 µm) and clay (<2 µm)
- VOCs
  - Sediment → Extraction

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Sample preparation techniques for sediment samples

- Soxhlet extraction
- Pressurized liquid extraction (PLE)
- Dynamic head space extraction (purge-and-trap)
- Supercritical fluid extraction (SFE)
- Ultrasonic extraction (USE)
- Microwave-assisted extraction (MAE)
- Matrix solid-phase dispersion (MSPD)
- …

Soxhlet extraction

- Extraction of semi-volatile organic compounds (NVOCs) and non-volatile organic compounds (NVOCs)
- Solid sample placed in a permeable cellulose “thimble”
- Extraction solvent circulates in the closed system
  → repeated extraction with fresh solvent

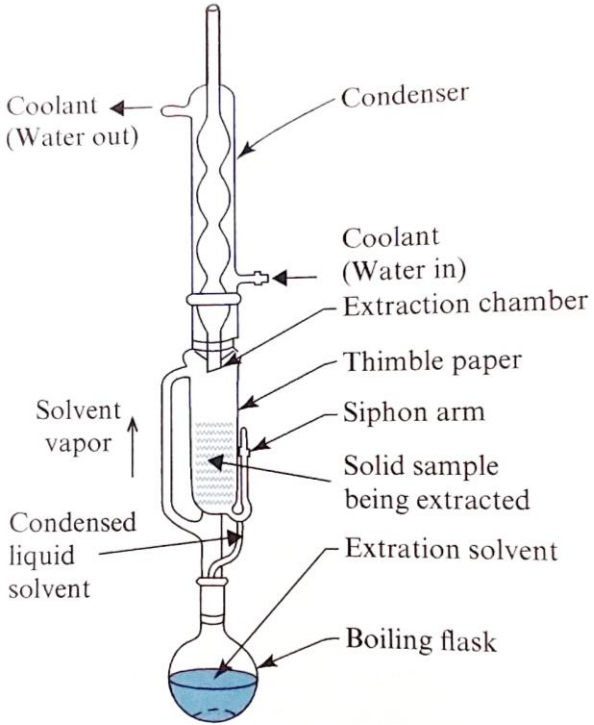
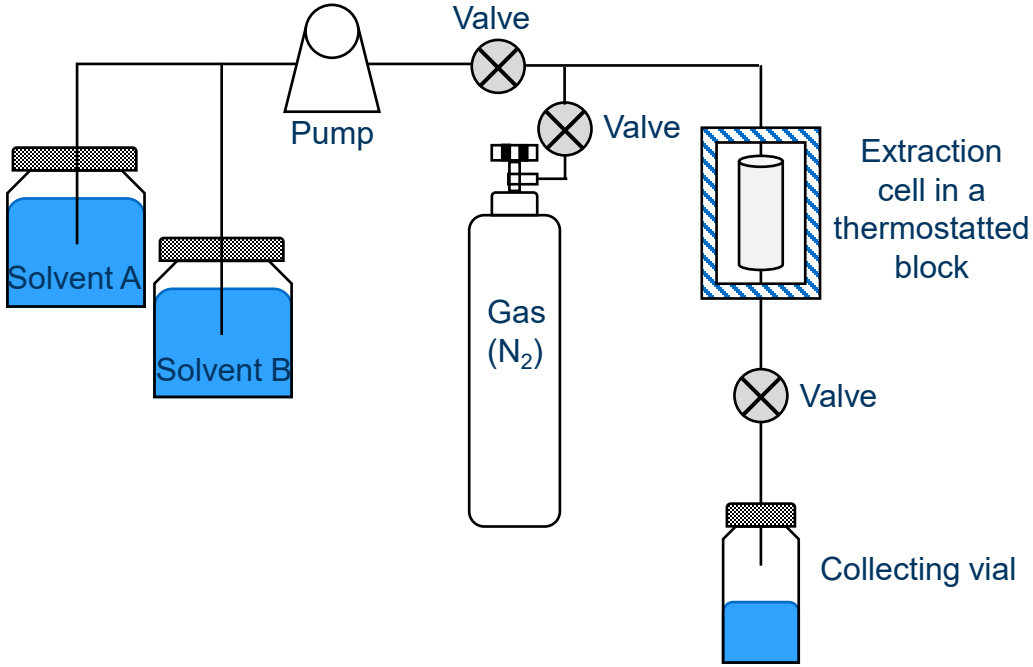


Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Pressurized liquid extraction (PLE)

- Fully automated extraction of semi-volatile organic compounds (NVOCs) and non-volatile organic compounds (NVOCs)
- Using of elevated temperatures (100-180 °C) and pressure (100-130 bar)
- Solubility parameters of extraction solvent or solvent mixtures can be optimized by solvent ratio in solvent mixtures, temperature and pressure



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Dynamic head space extraction (purge-and-trap)

- Extraction of volatile organic compounds (VOCs) from a liquid or solid phase
- Purging of sample with an inert gas and trapping of VOCs in a sorbent column
- Hydrophobic sorbent materials (e.g. graphitized carbon)
- Fast heating of the trap to desorb analytes from trap an inject into gas chromatography system

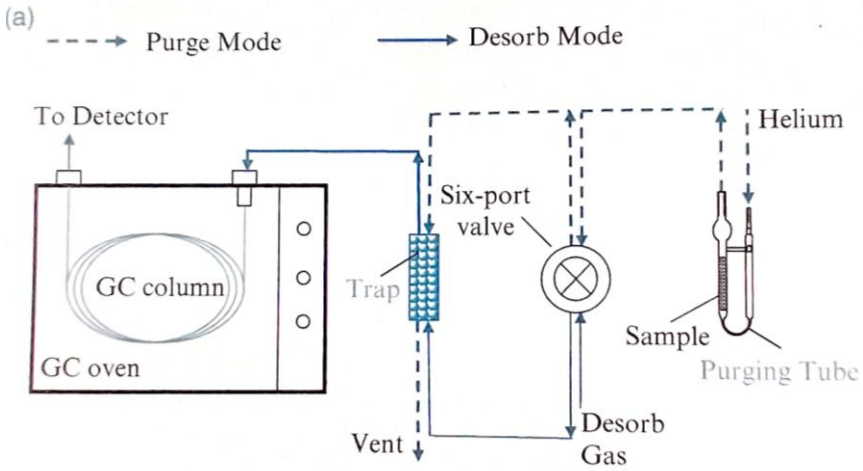
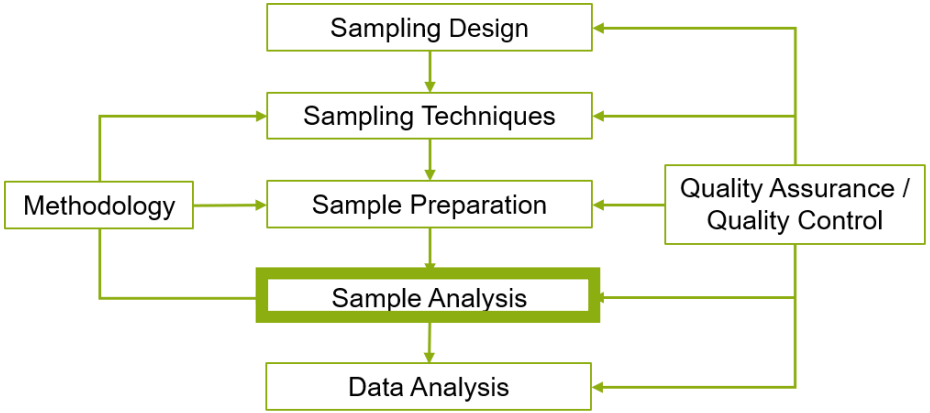


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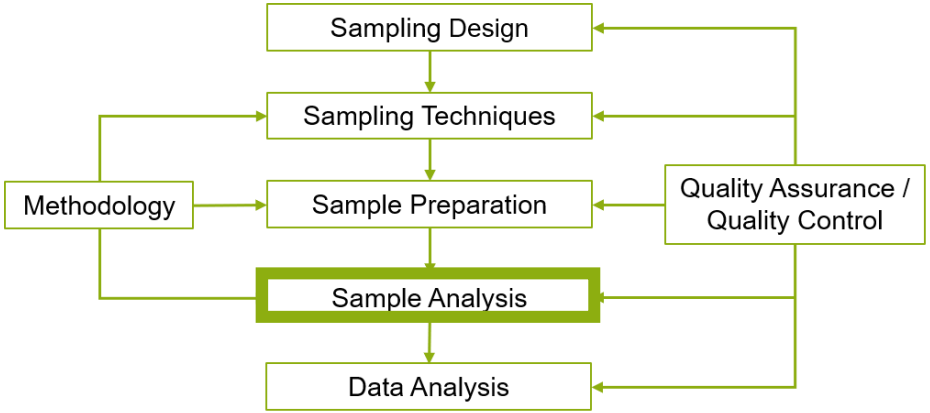
Sample analysis

- Which analytical technique to use?
- Qualitative or quantitative analysis?
- Is my method sensitive and selective enough?

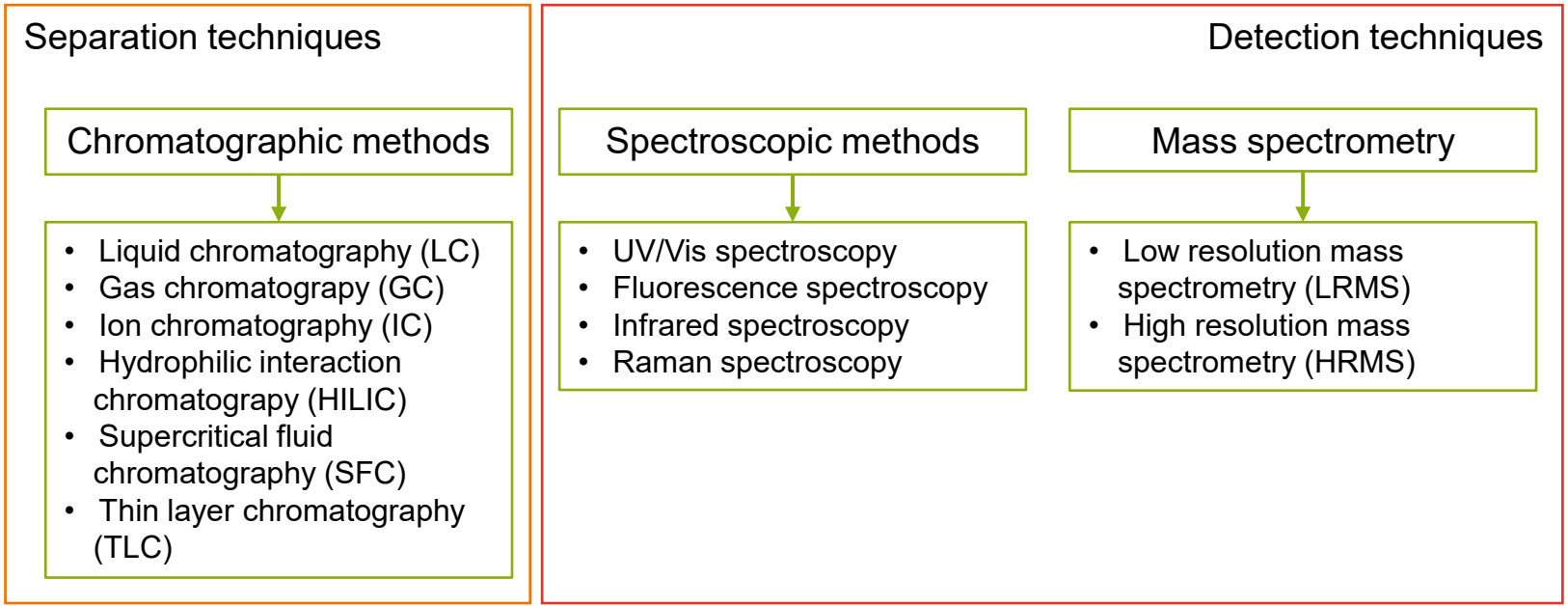


Sample analysis

- In environmental analysis often actually:
  - Sample separation → analyte detection



Analytical techniques for the analysis of organic compounds



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Chromatography

- Components of a mixture are separated based on differences in the rates at which they are carried through a stationary phase by a gaseous or liquid mobile phase
- Stationary phase is fixed in place in a column
- Mobile phase is forced through the column under pressure of gravity

General Classification	Specific method	Stationary phase	Type of Equilibrium
Gas chromatography (GC)	Gas-liquid (GLC)	Liquid adsorbed or bonded to a solid surface	Partitioning between gas and liquid
	Gas-solid	Solid	Adsorption
Liquid chromatography (LC)	Liquid-liquid, or partition	Liquid adsorbed or bonded to a solid surface	Partition between immiscible liquids
	Liquid-solid, or adsorption	Solid	Adsorption
	Ion exchange	Ion-exchange resin	Ion exchange
	Size exclusion	Liquid in interstices of a polymeric solid	Partition/sieving
	Affinity	Group-specific liquid bonded to a solid surface	Partition between surface liquid and mobile liquid

Elution in chromatography

- Sample solution containing to compounds A and B is injected onto the column
- Fresh mobile phase (Eluent) is added to force sample solution through the column
- Difference in distribution of A and B between mobile and stationary phase results in separation over time

$$K_c = \frac{C_s}{C_m}$$

K_c = Distribution constant
C_s = Concentration of analyte in stationary phase
C_m = Concentration of analyte in mobile phase

Chromatogram →

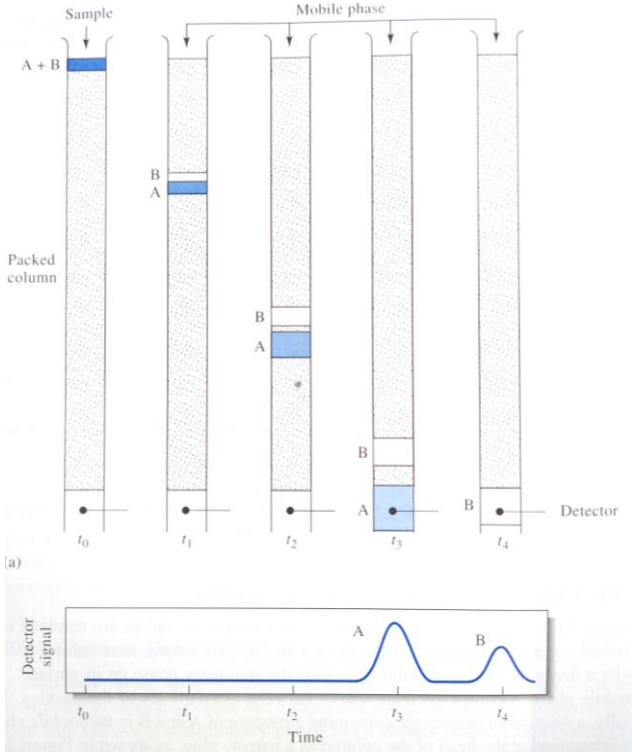
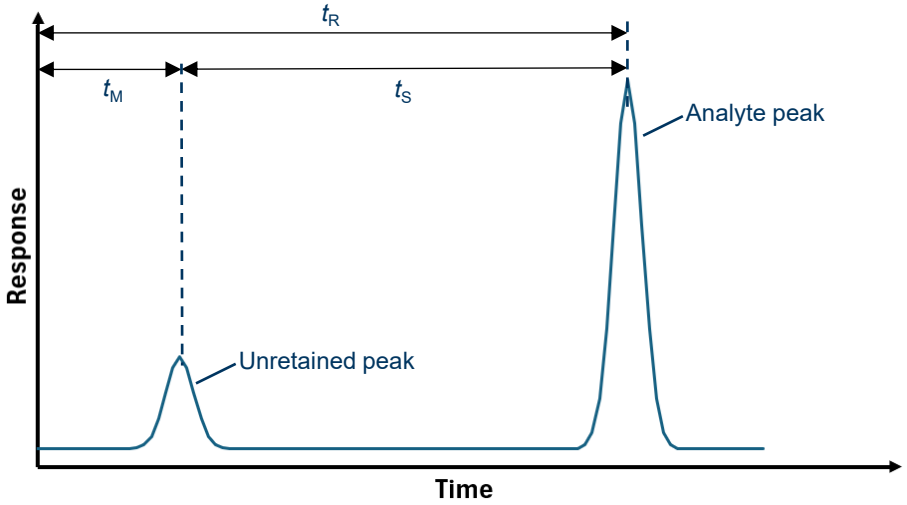


Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

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Retention times

- Compounds that do not interact with the stationary phase appear at the void time (t_M)
- Void time (t_M):
  - Time the mobile phase needs to pass through the column
- Retention time (t_R):
  - Time analyte needs from injection to reach the detector
- Time on stationary phase (t_S):
  - Time that the analyte spends on stationary phase

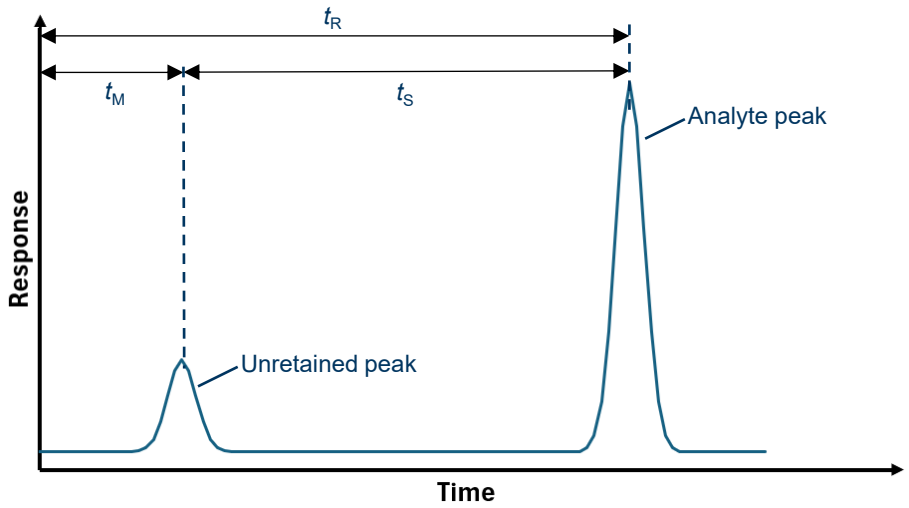


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Retention factor k'

- Describes the amount of time an analyte spends in the stationary phase relative to the time it spends in the mobile phase

$$k' = \frac{t_S}{t_M} = \frac{t_R - t_M}{t_M}$$

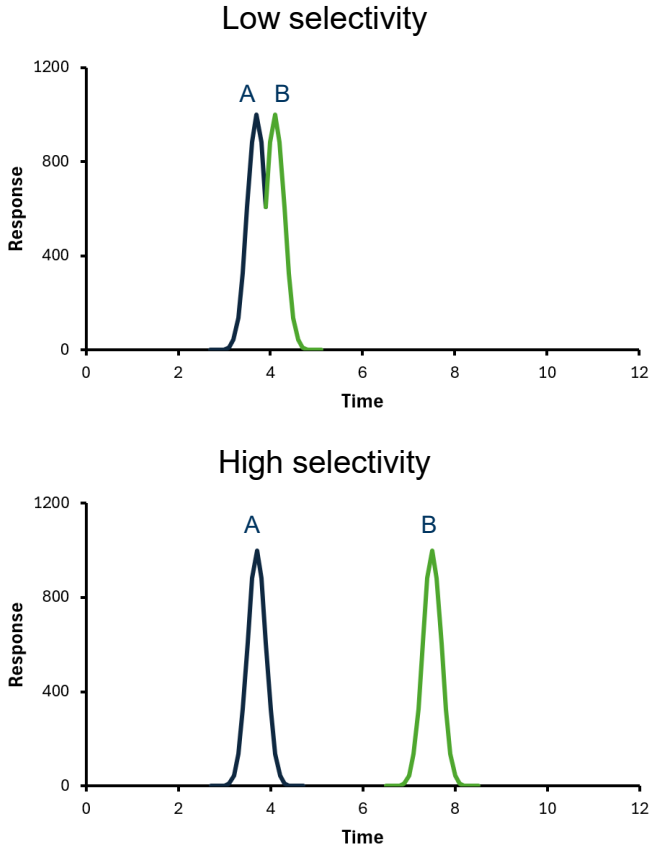


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Selectivity factor α

- Defined as the ratio of the retention factor of the more retained analyte B to the retention factor of the less retained analyte A
- Selectivity is influenced by type of stationary phase and type of mobile phase

$$\alpha = \frac{k'_B}{k'_A}$$



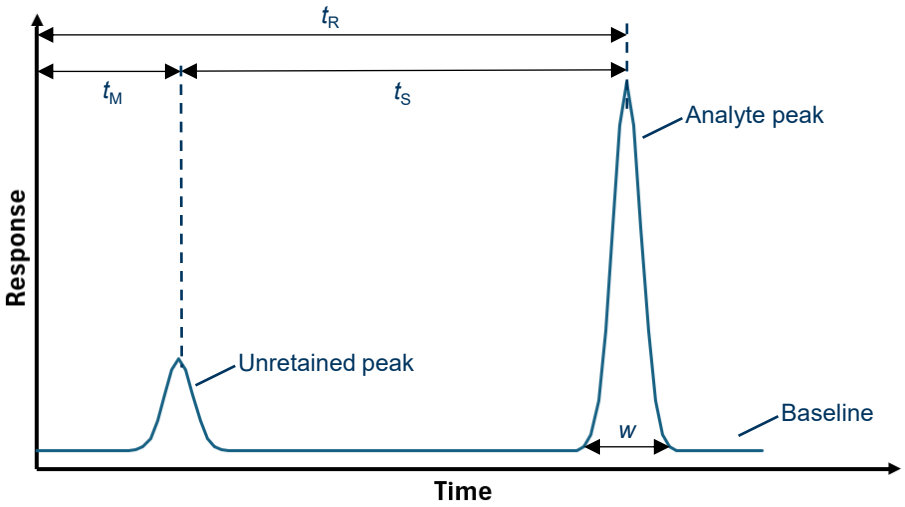
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Column efficiency

- Described by the number of theoretical plates N
- A theoretical plate represents a single equilibrium step of an analyte between mobile and stationary phase
- Column length L is divided into equal length N
- Length of N is called plate height H
- Columns with more N has greater resolving power

$$N = 16\left(\frac{t_R}{w}\right)^2 \qquad H = \frac{L}{N}$$

w = Peak width at baseline



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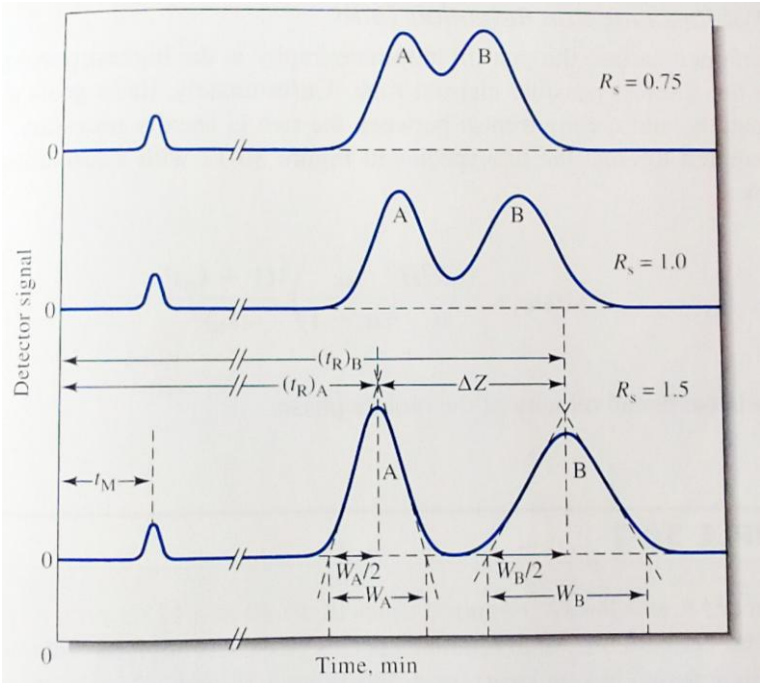
Resolution R

- Ability of stationary phase to separate two analytes

$$R = \left(\frac{k'}{1+k'}\right)\left(\frac{\alpha-1}{\alpha}\right)\left(\frac{\sqrt{N}}{4}\right) = \frac{t_{R,B} - t_{R,A}}{(w_A + w_B)/2}$$

Retention | Selectivity | Efficiency

Varied by:	Stationary phase Column length …	Stationary phase Eluent strength …	Stationary phase Mobile phase …



- Resolution ≥ 1.5 → baseline separation

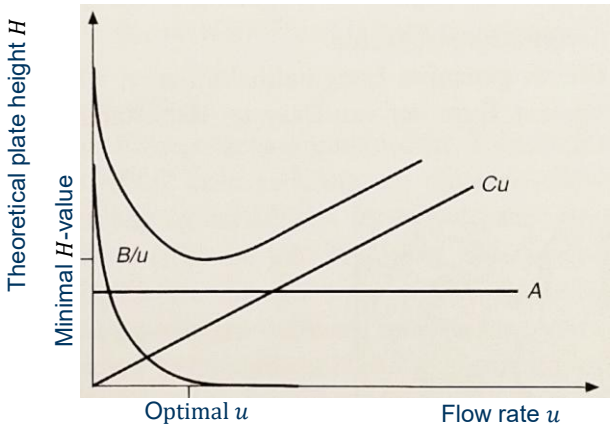
Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

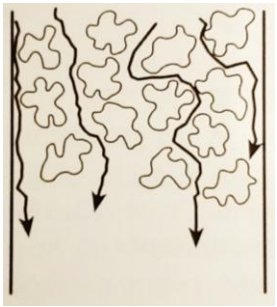
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Peak broadening effects and optimal flowrate

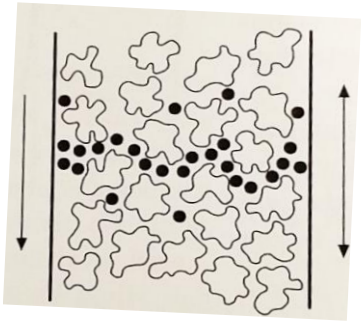
- Van-Deemter-Equation

$$H = \mathrm{A} + \frac{B}{u} + Cu$$

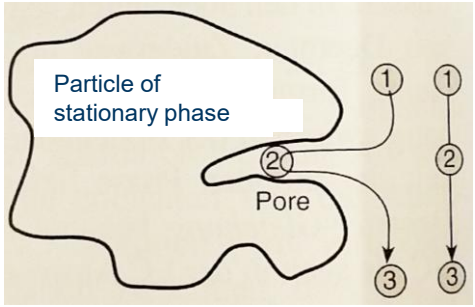




A-term: multiple path effect



B-term: longitudinal diffusion



C-term: mass transfer

Figures taken from Cammann, Instrumentelle Analytische Chemie, 1st Edition, **2001**.

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Liquid chromatography (LC)

- Partition LC
  - Liquid mobile phase
    - Water, methanol, acetonitrile, …
  - Solid stationary phase in packed columns
    - Silica
    - Bonded phase packings: C8, C18, …

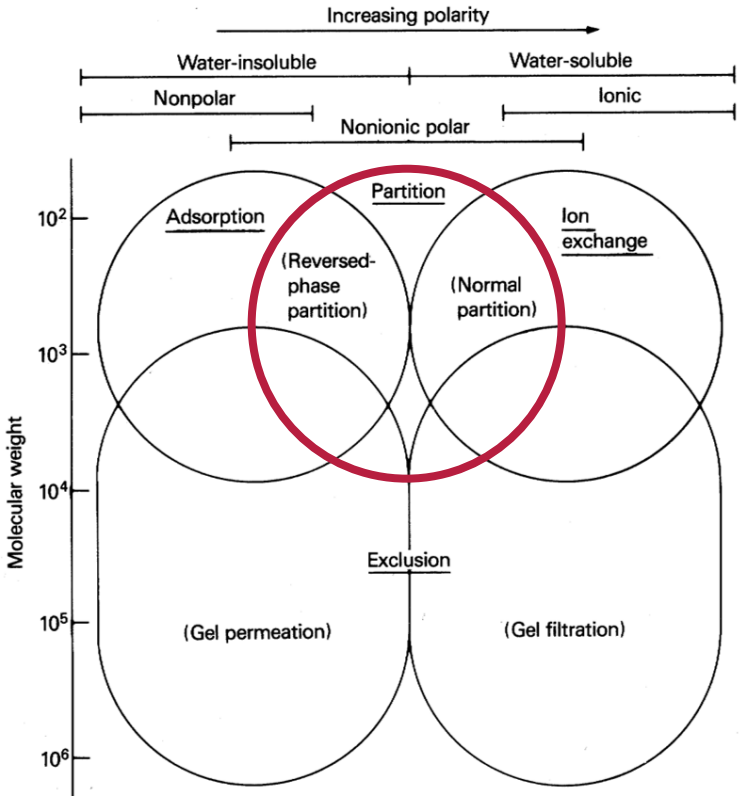


Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

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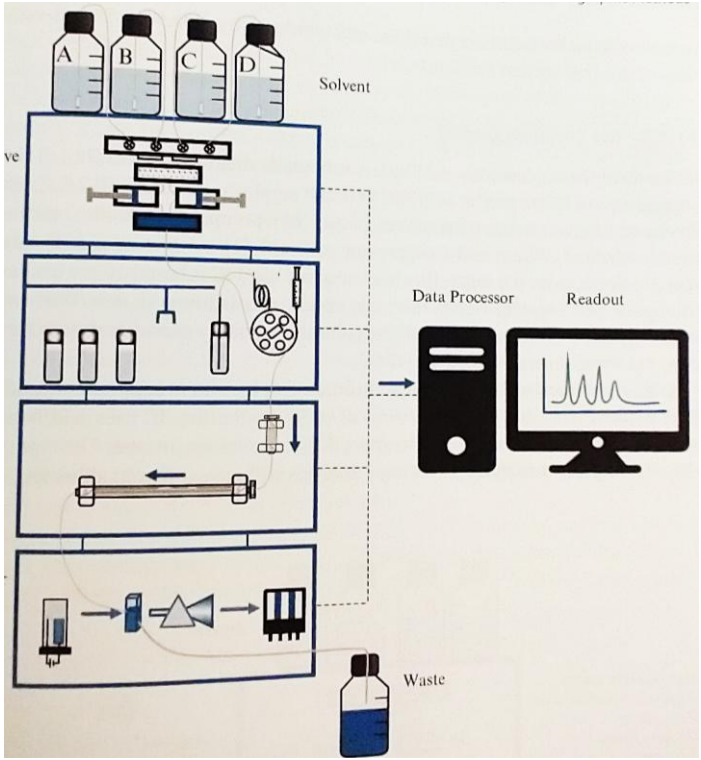
LC instrumentation

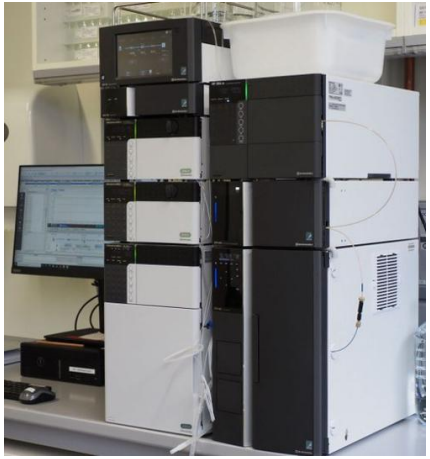
Pump

Autosampler

Column compartment

Detector







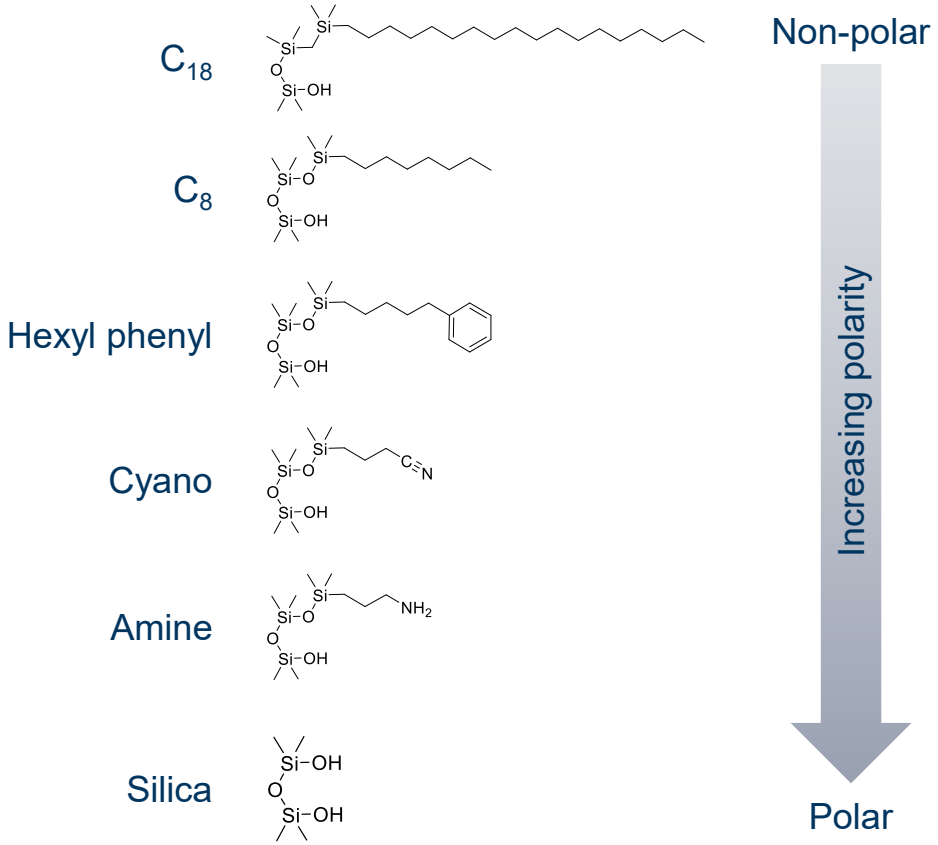
Typical dimensions in LC:
Column length: 50 to 250 mm
Column inner diameter: 2.1 to 4.6 mm
Flow rates: 0.3 to 1 mL/min

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

Picture of columns from lifesciences.danaher.com

LC stationary phases

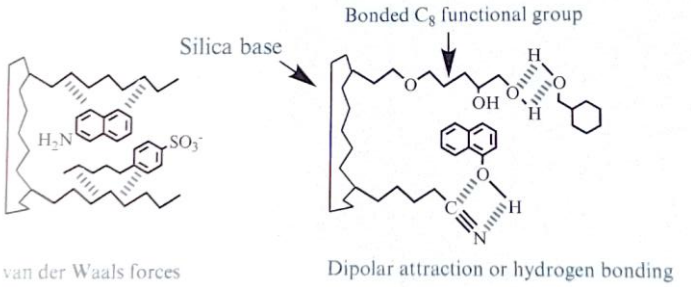
- Normal phase LC
  - Stationary phase: polar (e.g. silica)
  - Mobile phase: non-polar (e.g. hexane)
  - Separation of polar compounds
- Reversed phase LC
  - Stationary phase: non-polar (e.g. C_{18})
  - Mobile phase: polar (e.g. water)
  - Separation of medium to polar compounds



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LC separation mechanism

- Separation based on:
  - Van der Waals forces
  - Dipol/dipol interactions
  - Hydrogen bonding
  - π-π-interactions



C_{18}

C_8

Hexyl phenyl

Cyano

Amine

Silica

Non-polar

Increasing polarity

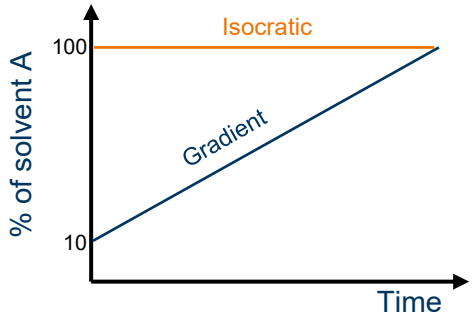
Polar

Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Isocratic vs. gradient elution

- Isocratic elution
  - Eluent has a constant flowrate and consists of one solvent
- Gradient elution
  - Eluent has a constant flowrate and consists of two or more solvents



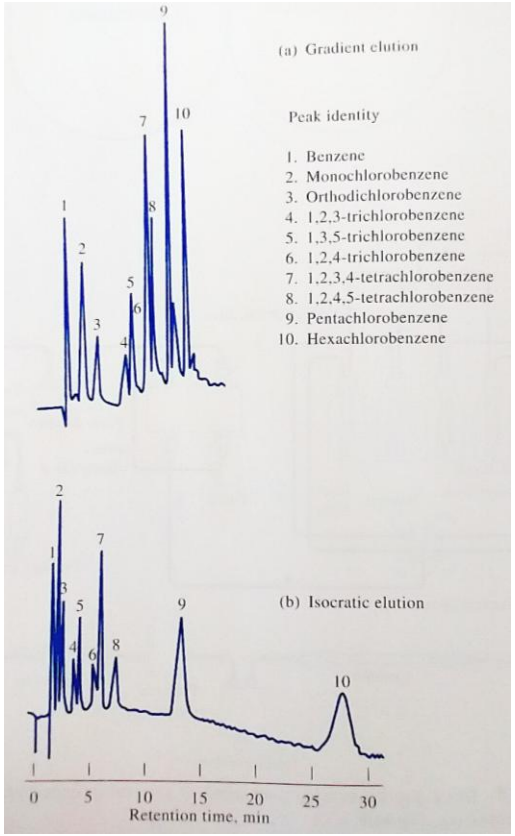


Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

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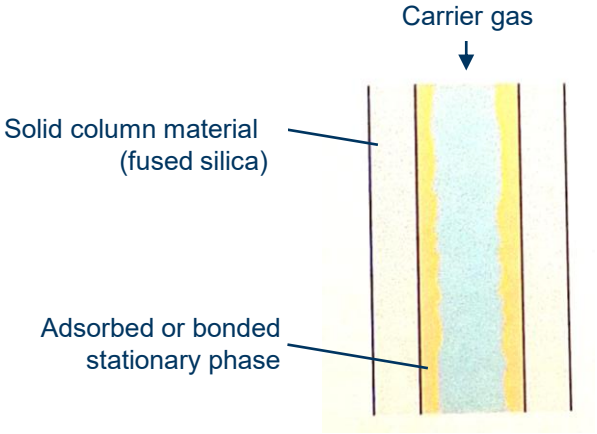
Gas chromatography (GC)

- Separation of volatile organic compounds (VOCs)
- Separation based on partitioning between gas and liquid adsorbed or bonded to a solid surface
- Derivatization of less volatile compounds to be able to analyze them with GC
  - E.g. Silylation of alcohols with trimethylsilyl (TMS)



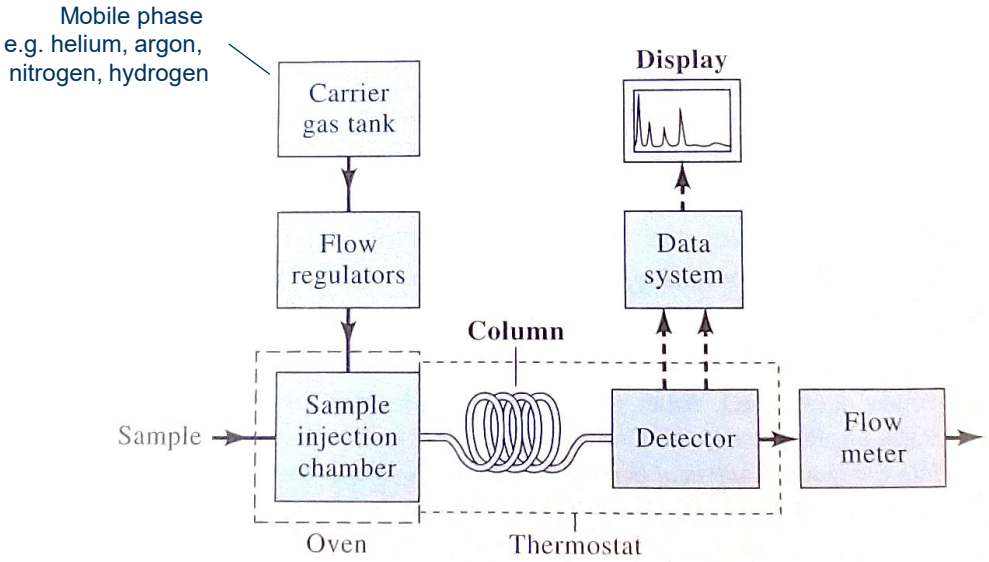
  - Produced silyl derivates are more volatile and more thermally stable

Open-tubular capillary columns



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GC instrumentation







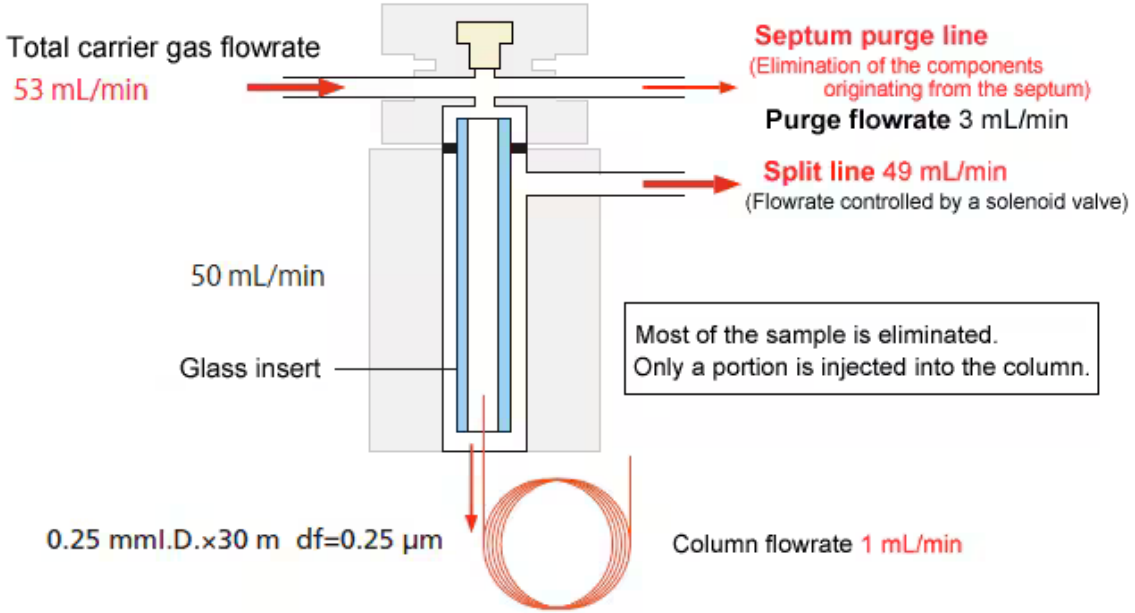
Typical dimensions for open-tubular capillary columns:
Column length: 15 to 60 m
Column inner diameter: 0.18 to 0.53 mm
Stationary phase film thickness: 0.1 to 5 µm
Flow rates: 0.3 to 8 mL/min

Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

Picture of GC-system and column from Agilent

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Injection of gas and liquids on open-tubular capillary columns





Injection temperature often 50 °C above the boiling point of the least volatile analyte in the sample

Spilt injection to avoid overloading of the column

Injection of 0.1 to 1 mL

Split ratios of e.g. 1:100 or 1:500

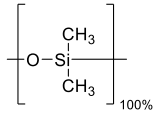
Figure of Split injector taken from Shimadzu.de

Figure of injection syringe taken from Fishersci.at

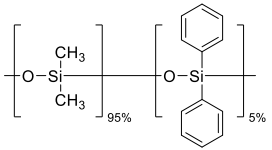
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Stationary phases for open-tubular capillary columns

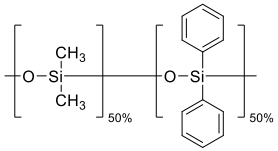
Polydimethylsiloxane



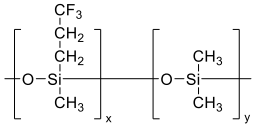
(5%-phenyl)-methylpolysiloxane



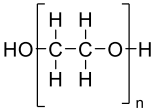
(50%-phenyl)-methylpolysiloxane



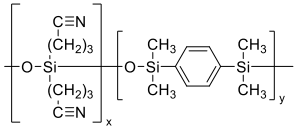
Trifluoropropyl dimethylpolysiloxane



Polyethylene glycol



Cyanopropyl dimethylpolysiloxane



Non-polar

Polar

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Temperature gradient

- Column temperature controlled by thermostatted oven
- A temperature equal to slightly above the average boiling point of the sample results in reasonable elution time (2 to 30 min)
- Temperature gradient for the analysis of samples with broad boiling range

Effect of temperature on gas chromatograms →

(a) Isothermal at 45 °C

(b) Isothermal at 145 °C

(c) Programmed at 30 °C to 180 °C

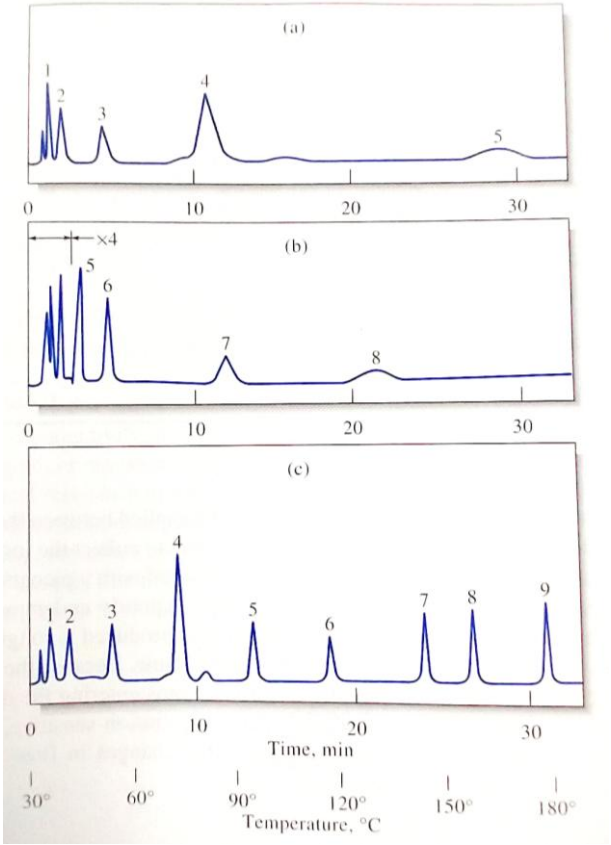


Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

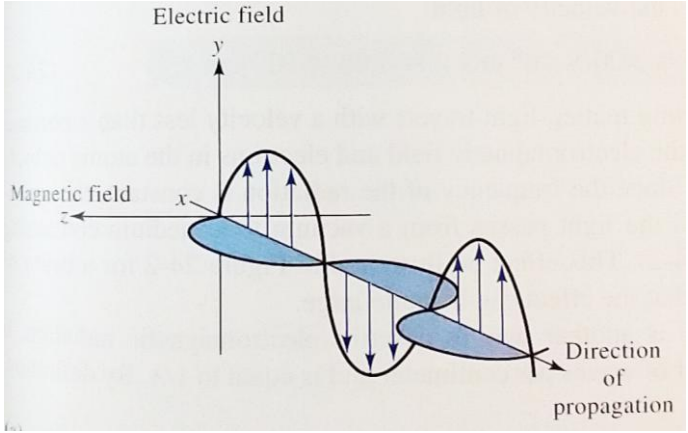
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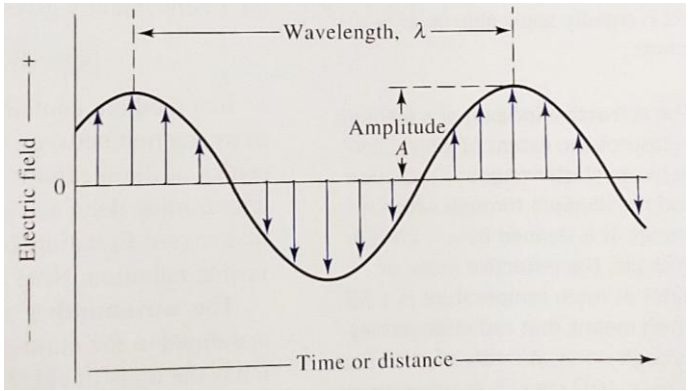
UV/Vis spectroscopy

- Spectroscopy → interaction between electromagnetic radiation and matter
- Energy of radiation E described by Planck's law:

$$E = hv = h\frac{c}{\lambda}$$

h = Planck constant (6.62 x 10^{-34} J s)
v = Frequency
c = Speed of light (3 x 10^{8} m/s)
λ = Wavelength





Figures taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

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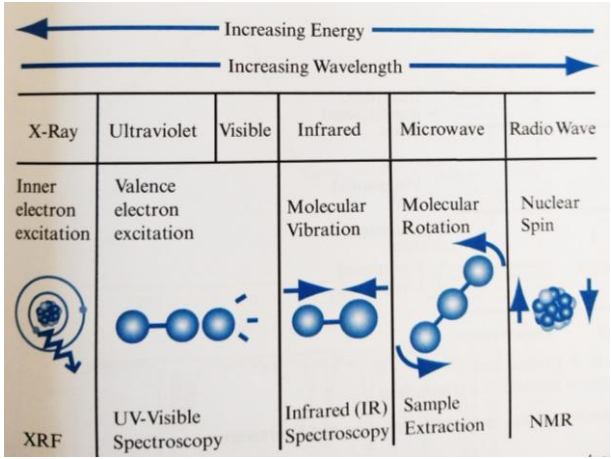
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$$E = hv = h\frac{c}{\lambda}$$

h = Planck constant (6.62 x 10^{-34} J s)
v = Frequency
c = Speed of light (3 x 10^{8} m/s)
λ = Wavelength

- E increases with increasing v and increases with decreasing λ
- Chemicals absorb electromagnetic radiation at certain energies (→wavelength)



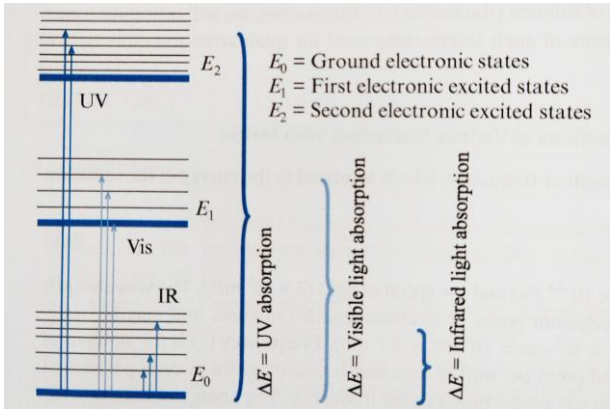
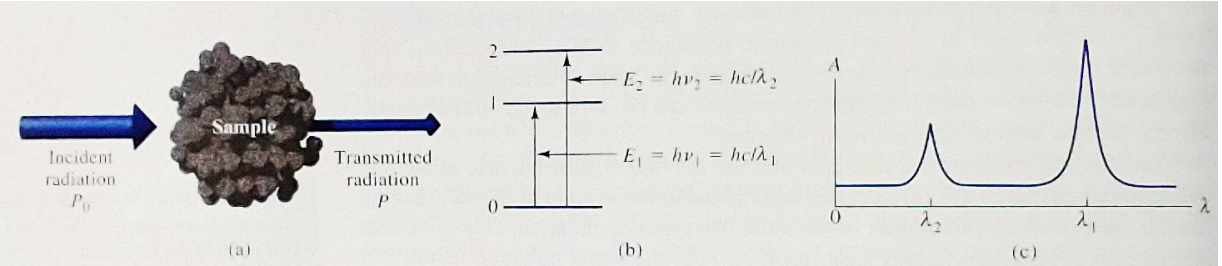


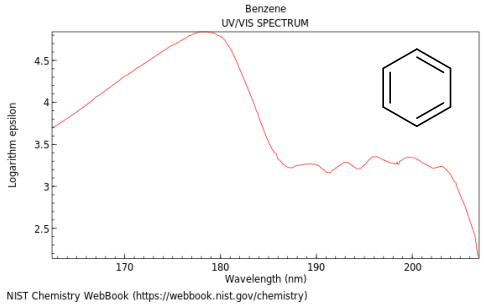
Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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UV/Vis molecular spectroscopy

- Absorption of electromagnetic radiation used for qualitative and quantitative analysis
  - Comparison of UV/Vis spectrum with spectra libraries for identification





Top figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

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UV/Vis spectroscopy

- Absorption of electromagnetic radiation used for qualitative and quantitative analysis
  - Comparison of UV/Vis spectrum with spectra libraries for identification
  - Using Beer's law for quantification
    - Absorbance A is proportional to the concentration C of the light-absorbing analyte
    - Molar absorptivity is an intrinsic property of the analyte species capacity to absorb light of a specific wavelength

$$A = \log\frac{P_0}{P} = \varepsilon l C$$

A = Absorbance
P_0 = Power of incident light
P = Power of transmitted light
ε = Molar absorptivity [L/(mol cm)]
l = Length of light path in sample cell [cm]
C = Concentration of analyte [mol/L]

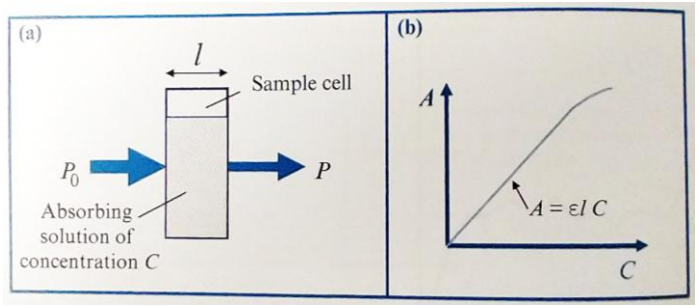
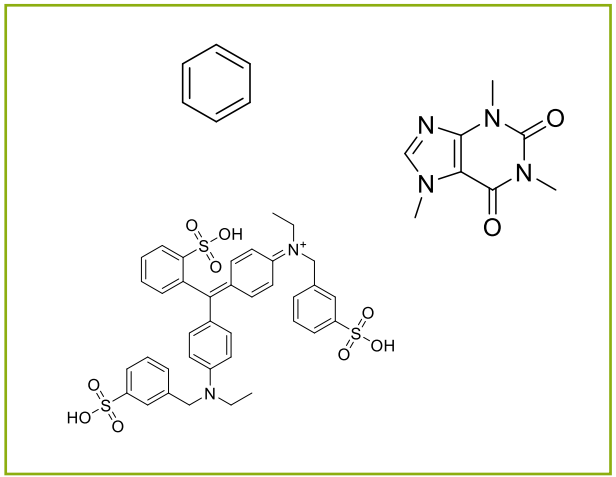


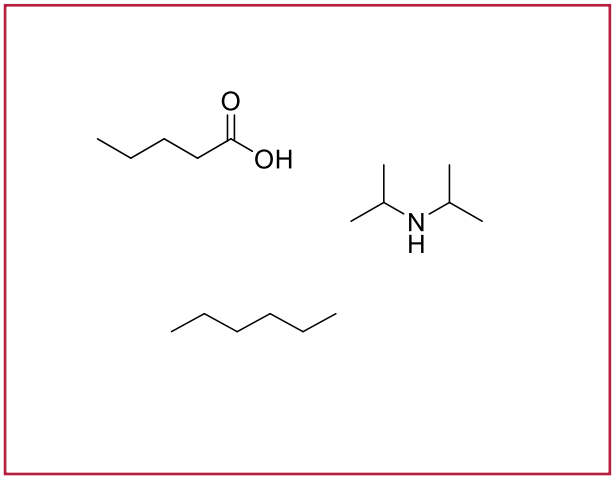
Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Organic compounds detectable with UV/Vis spectroscopy

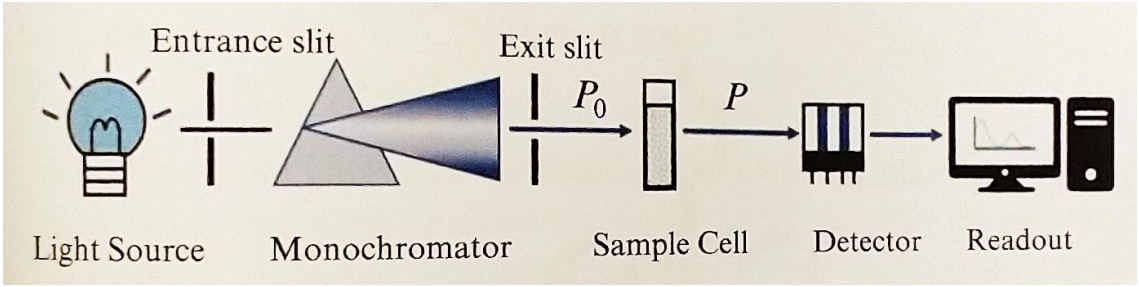
- Organic compounds with π-electron systems and conjugated double bonds work particularly well





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UV/Vis spectroscopy instrumentation



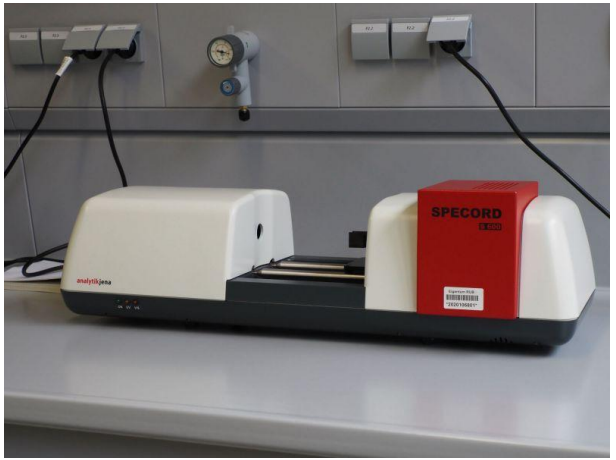
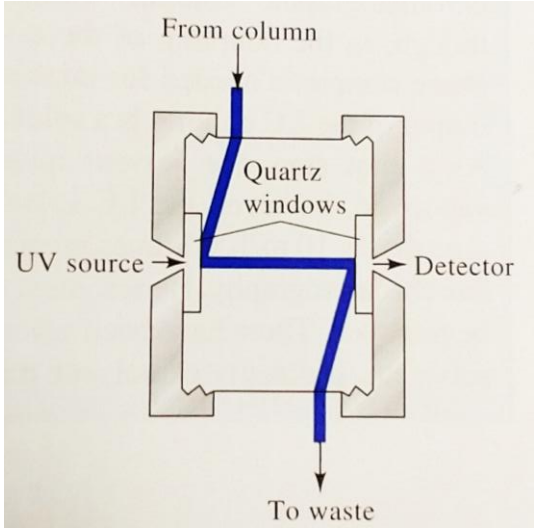


Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Coupling of LC-UV/Vis

- UV/Vis detectors can be easily coupled with LC
- Flow through cell



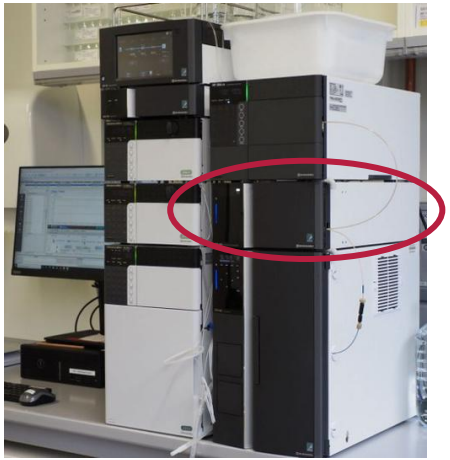
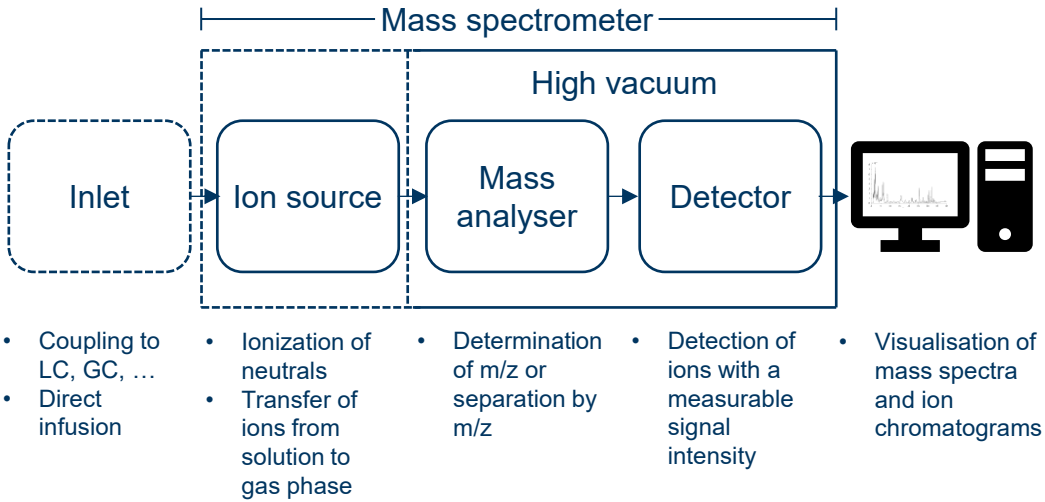


Figure taken from Skoog, West, Holler, Crouch, Fundamentals of Analytical Chemistry, 8th Edition, **2004**.

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Mass spectrometry (MS)

- Study of matter through the formation of gas-phase ions, their separation according to their mass-to-charge ratio (m/z) and their detection for qualitative and quantitative analysis

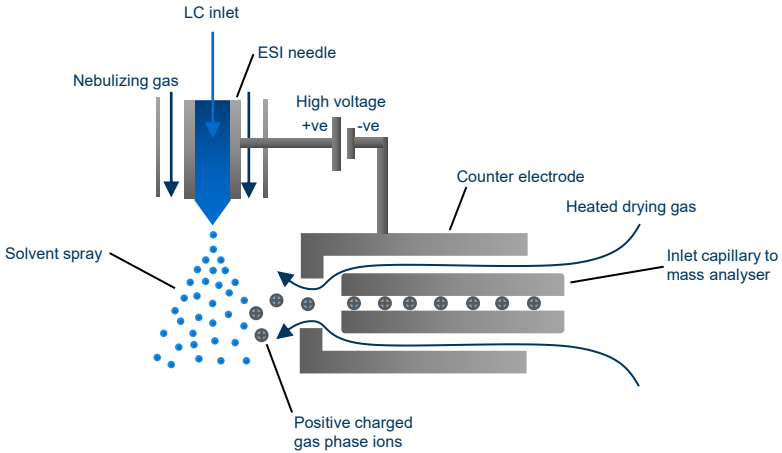


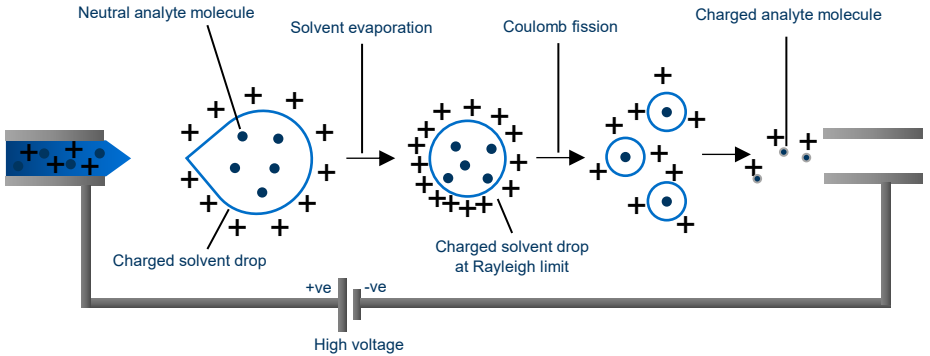


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Electrospray ionization (ESI)

- Ion source operating at ambient pressure used for liquid chromatography (LC)
- Generation of positive or negative charged ions by protonation or deprotonation of analyte molecules
  - ESI positive mode → $[M+H]^+$
  - ESI negative mode → $[M-H]^-$

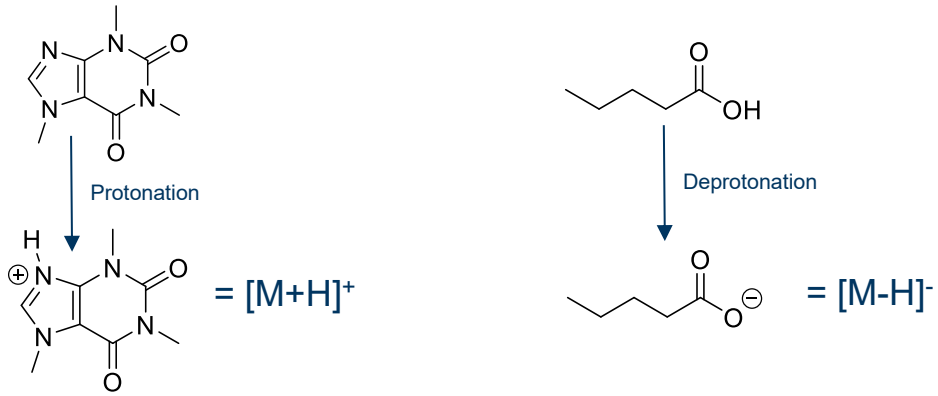




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Electrospray ionization (ESI)

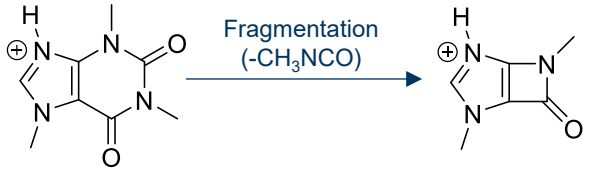
- Ion source at ambient pressure used for liquid chromatography (LC)
- Generation of positive or negative charged ions by protonation or deprotonation of analyte molecules
  - ESI positive mode → $[M+H]^+$
  - ESI negative mode → $[M-H]^-$
- Organic compounds need a functional group that can be protonated or deprotonated



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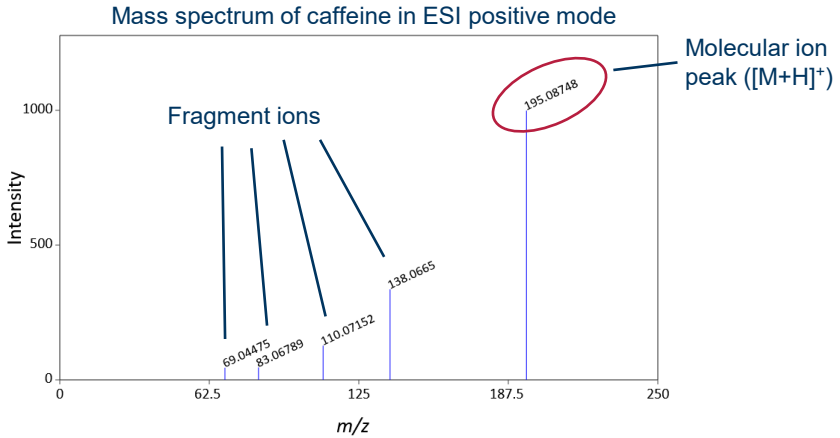
Electrospray ionization (ESI)

- Soft ionization technique → molecular ion peak in mass spectrum
- Fragmentation can occur during ionization process (in-source fragmentation) or controlled in the mass analyzer



Neutral molar mass of caffeine: 194 g/mol
Molar mass 1 x H: 1 g/mol
With one charge (z = 1) in ESI positive mode → m/z 195

m/z 138



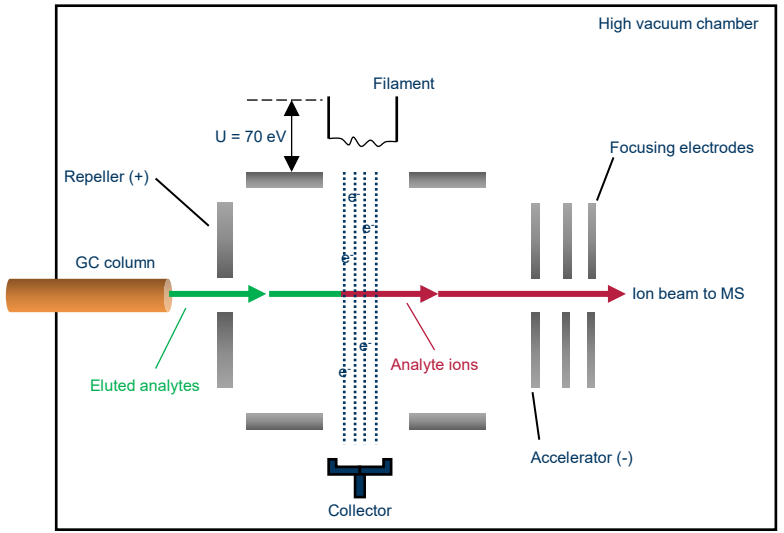
Mass spectrum taken from PubChem.

Electron ionization (EI)

- Ion source operating under vacuum used for gas chromatography (GC)
- Ionization in gas phase:

$$M + e^- \rightarrow M^{+\cdot} + 2e^-$$

- Only volatile organic compounds can be ionized using EI



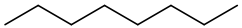
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Electron ionization (EI)

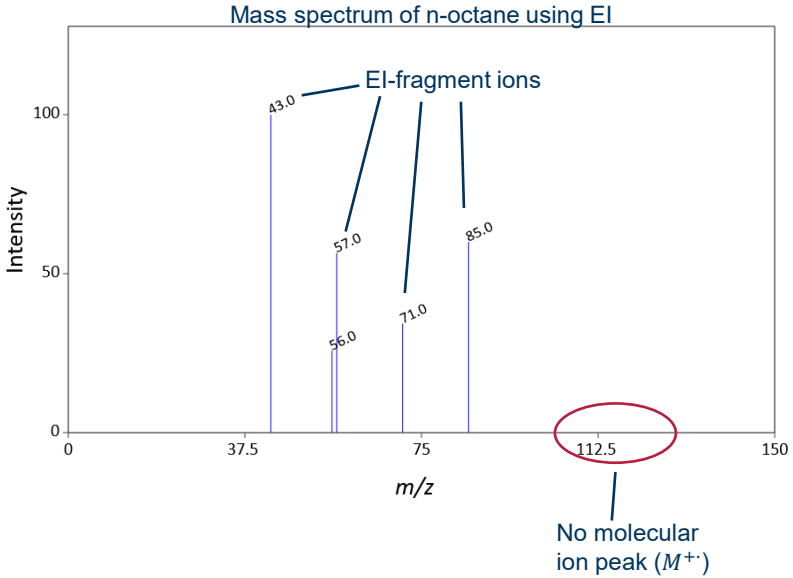
- Ion source operating under vacuum used for gas chromatography (GC)
- Ionization in gas phase:

$$M + e^- \rightarrow M^{+\cdot} + 2e^-$$

- Hard ionization technique → often no molecular ion peak



Neutral molar mass of n-octane: 114 g/mol



Mass spectrum taken from PubChem.

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Domains of organic compounds in relation to volatility and polarity within the region of LC-MS and GC-MS

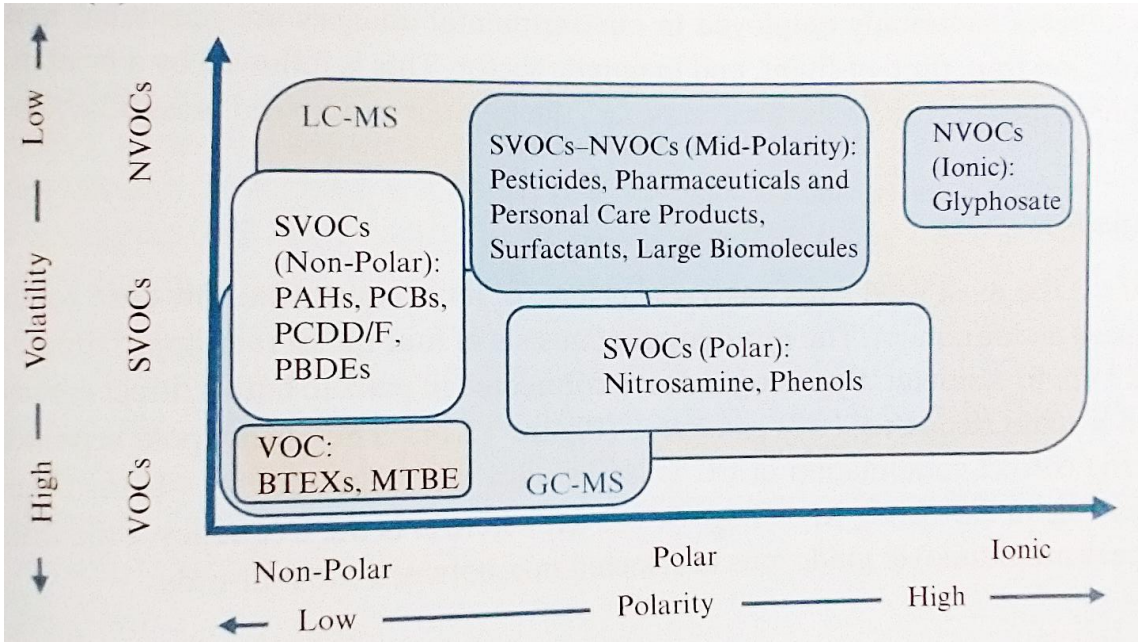


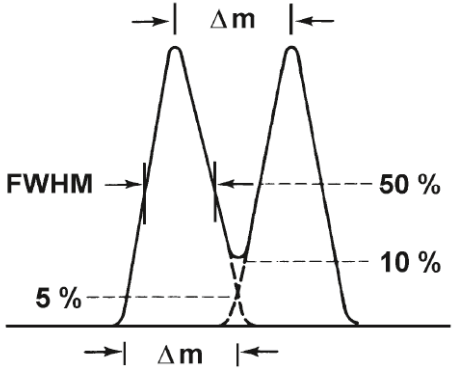
Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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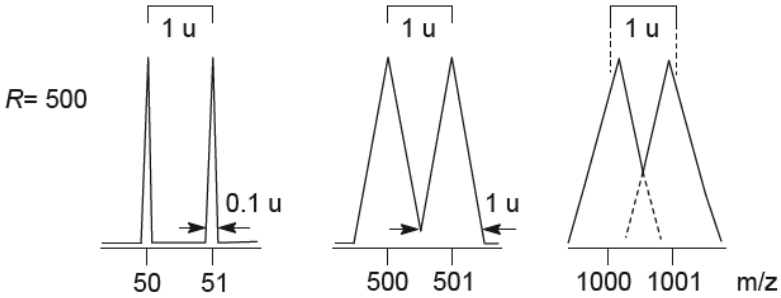
Mass analyser

- Low resolution mass spectrometry (LRMS)
  - Mass analyser: Quadrupole, triple quadrupole
- High resolution mass spectrometry (HRMS)
  - Mass analyser: Orbitrap, Time-of-flight, …
- Definition of LRMS and HRMS based on mass resolution R

$$R = \frac{m}{\Delta m} = \frac{m/z}{\Delta m/z}$$



FWHM = full width at half maximum



Figures taken from J. Gross, Massenspektrometrie, 2nd Edition, **2013**.

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Mass analyser

- Low resolution mass spectrometry (LRMS)
  - Mass analyser: Quadrupole, triple quadrupole
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  - Mass analyser: Orbitrap, Time-of-flight, …
- Definition of LRMS and HRMS based on mass resolution R

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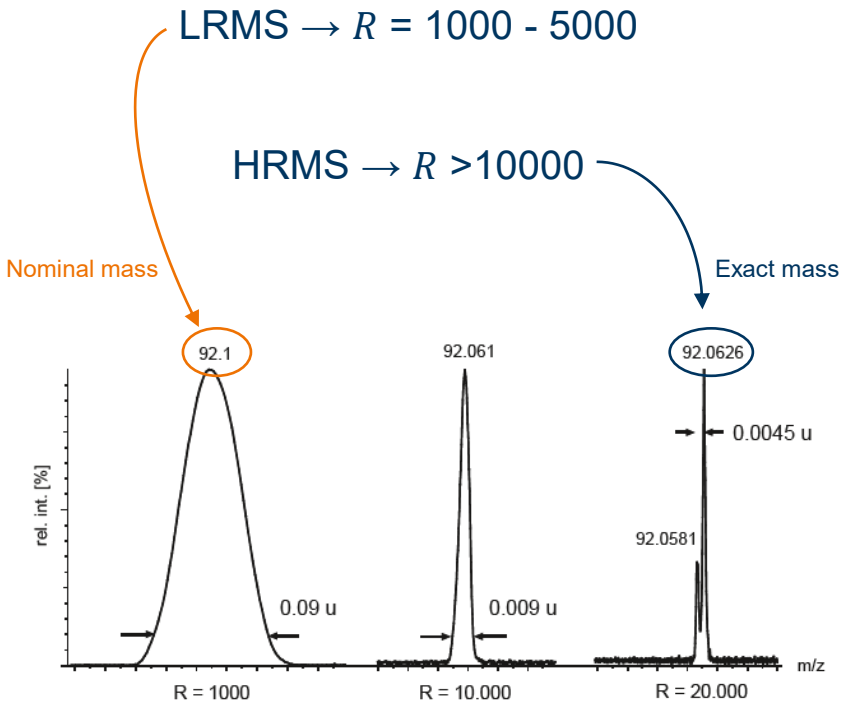
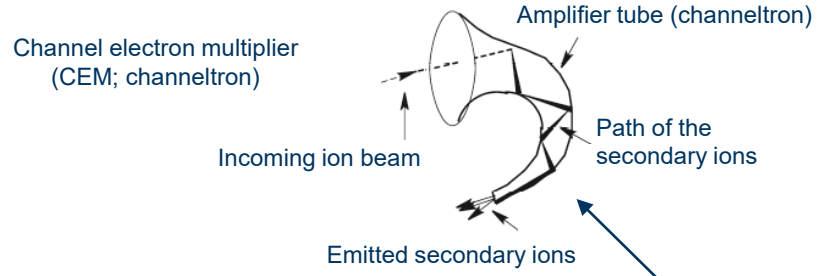


Figure taken from J. Gross, Massenspektrometrie, 2nd Edition, **2013**.

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Quadrupole mass analyser

- Low resolution mass analyser
- Resolving power R ~ 4000
- Opposing rods are connected in pairs to both a direct current (DC) generator and a radio frequency (RF) operator
- Separation of ions by setting up the correct combination of DC voltages and radio frequencies having an oscillating RF voltage
- Guiding of ions of selected m/z between the four metal rods → mass filter



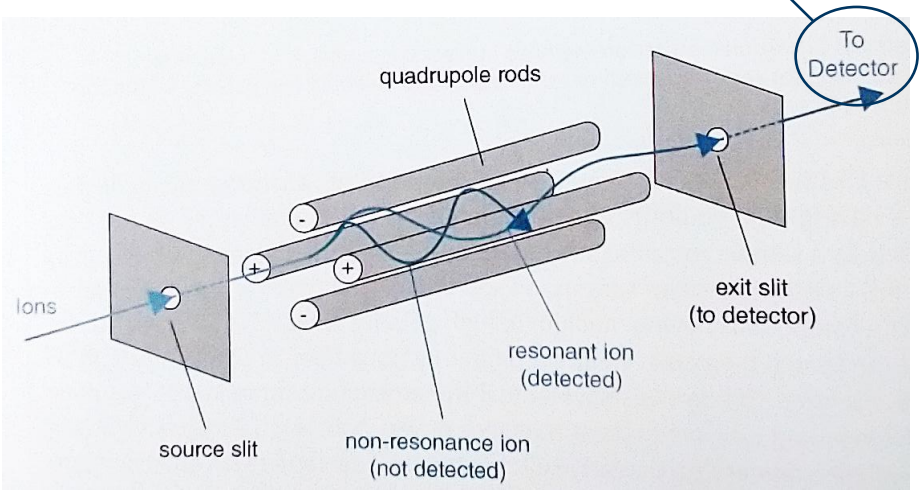
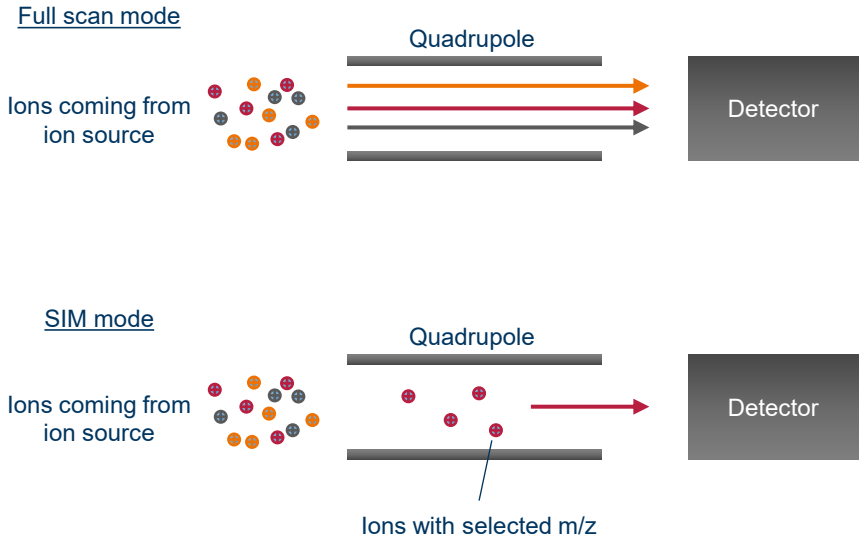


Figure of quadrupole taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.
Figure of channeltron taken from J. Gross, Massenspektrometrie, 2nd Edition, **2013**.

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Quadrupole (Q) mass analyser

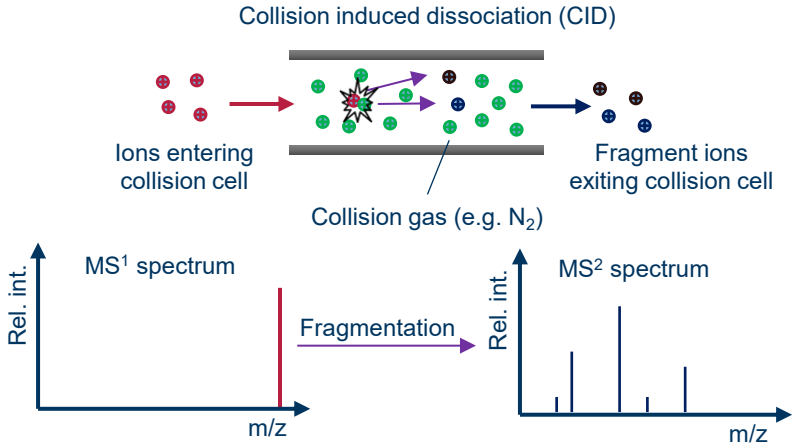
- Fullscan mode
  - All ions in a selected m/z range can pass the quadrupole
- Single ion monitoring (SIM) mode
  - Used for quantitative analysis
  - Quadrupole is used as a mass filter and only one selected m/z can pass through
  - Improved signal-to-noise ratios (S/N) and limit of quantifications (LOQ)

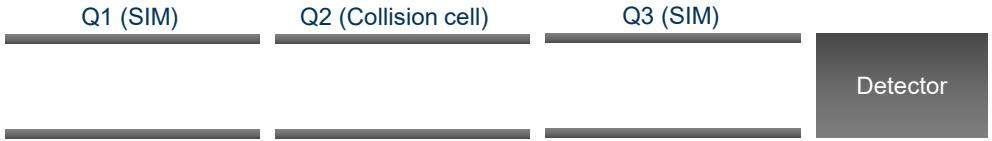


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Triple quadrupole (QQQ) mass analyser

- Three quadrupoles combined in a line
- Second quadrupole (Q2) is a collision cell using collision induced dissociation (CID) to produce fragment ions
  - Fragmentation under controlled conditions
  - Fragmentation pattern can be used for identification
  - Structural elucidation based on fragmentation pattern

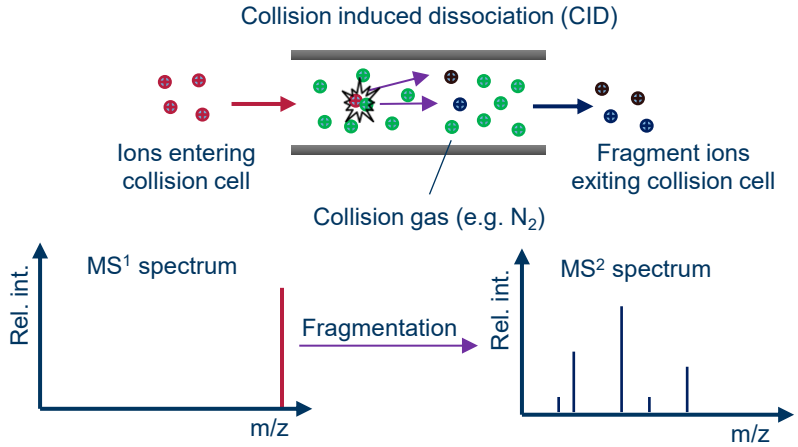


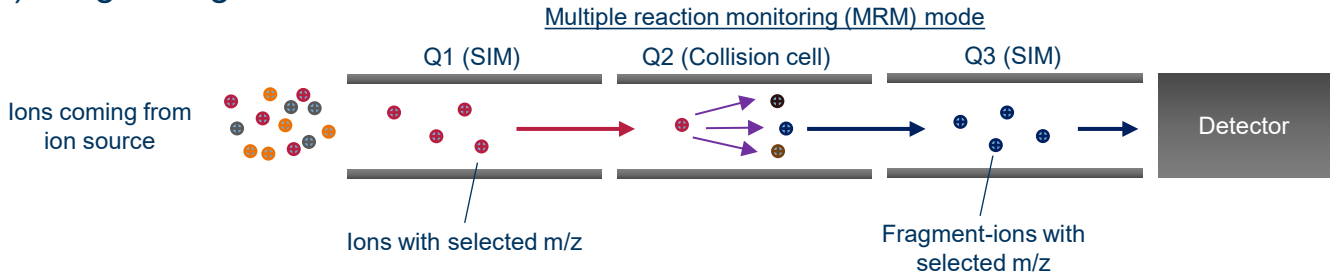


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Triple quadrupole (QQQ) mass analyser

- Three quadrupoles combined in a line
- Second quadrupole (Q2) is a collision cell using collision induced dissociation (CID) to produce fragment ions
- Multiple reaction monitoring (MRM)
  - Q1 and Q3 operated in SIM mode
  - High selectivity and high sensitivity
  - Limits of detection (LOD) and quantification (LOQ) in ng/L range





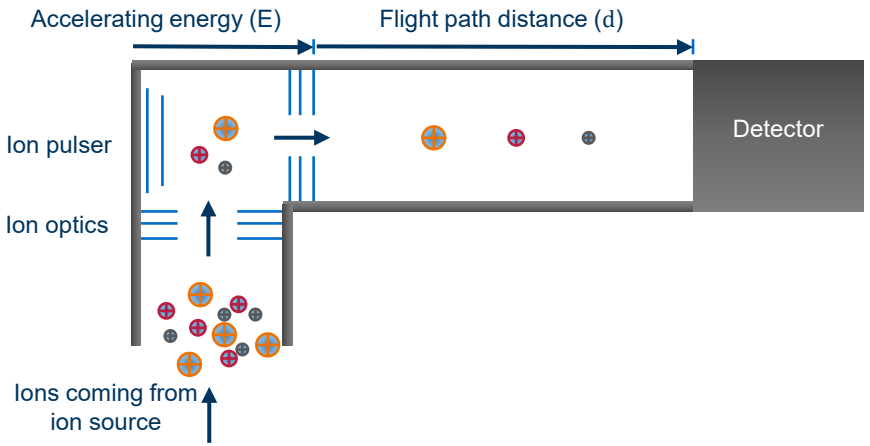
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Time of Flight (TOF) mass analyser

- High resolution mass analyser
- Resolving power R ~ 30000
- Separation on ions in a flight tube based on their m/z
- Larger ions will take longer time to reach the detector

$$t = d\sqrt{\frac{m}{2zeV}}$$

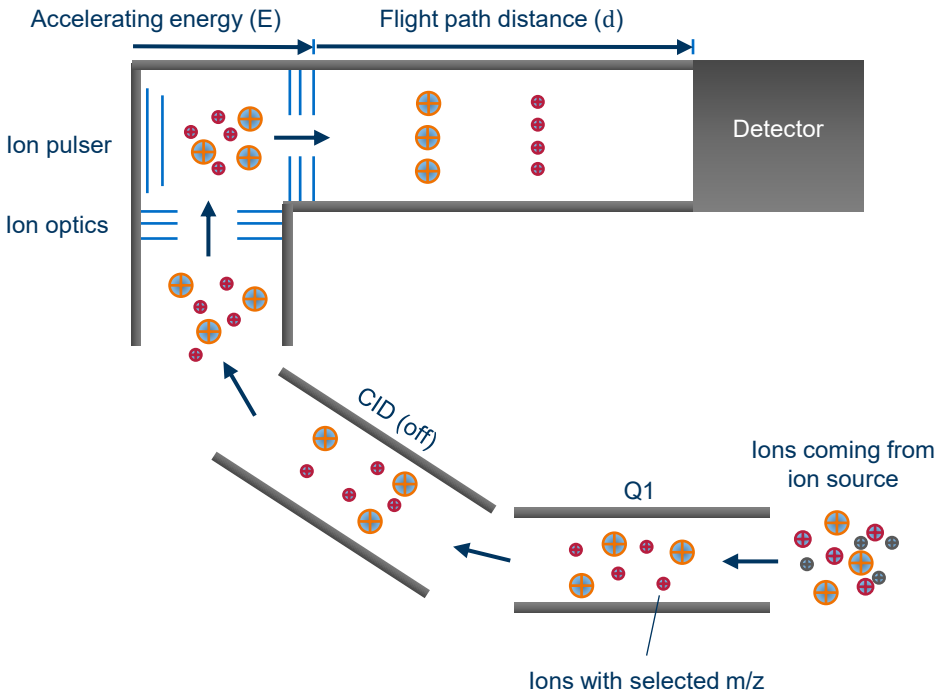
t = Flight time
d = Length of flight tube
m = Mass of the ion
z = Charge of the ion
e = Charge of one electron in coulomb (1 e^- = 1.6 x 10^{-19} C)
V = Positive electrical potential used for acceleration



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Quadrupole-Time of Flight (QTOF) mass analyser

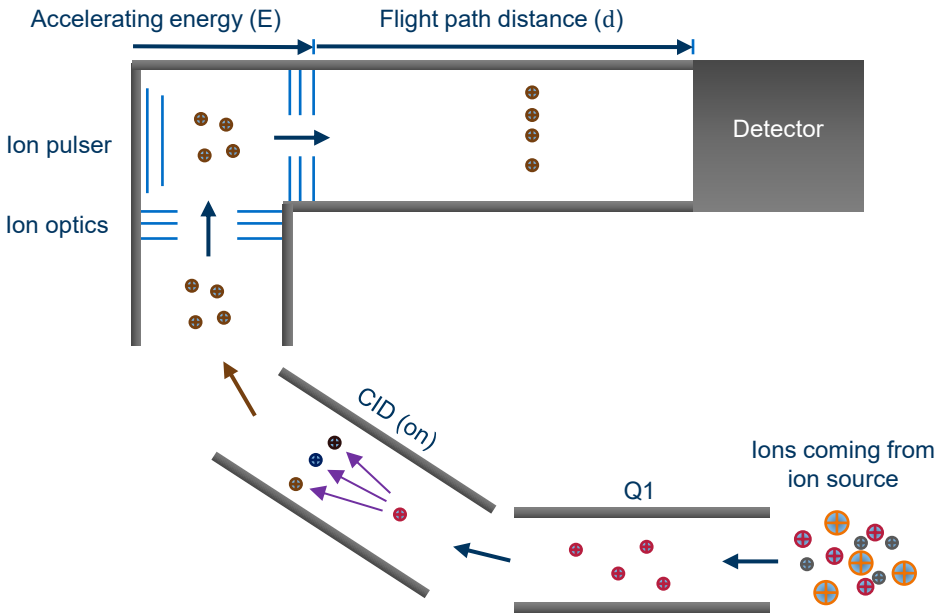
- Combination of quadrupole used as a mass filter and TOF for high mass resolution
- Ion ranges of interest can be selected by Q1



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Quadrupole-Time of Flight (QTOF) mass analyser

- Combination of quadrupole used as a mass filter and TOF for high mass resolution
- Ion ranges of interest can be selected by Q1
- Selected ion monitoring for high mass resolution also for fragment ions
- Lower sensitivity compared to triple quadrupole (QQQ), because TOF can not be used as a mass filter → disturbing ions may also reach the detector

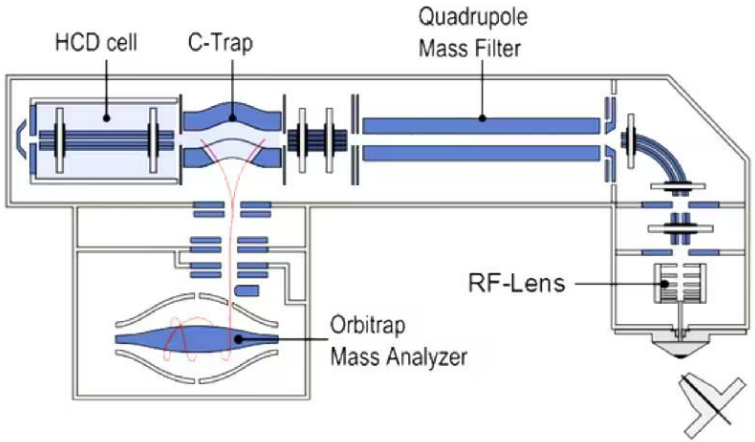


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Orbitrap mass analyser

- High resolution mass analyser
- Modern Orbitrap-systems reach resolving power up to R ~ 480000
- Axial oscillation frequency in Orbitrap combined with Fourier-Transformation for determination of m/z ratios
  - Low m/z → higher frequency
  - High m/z → lower frequency
- Can also be coupled with a quadrupole



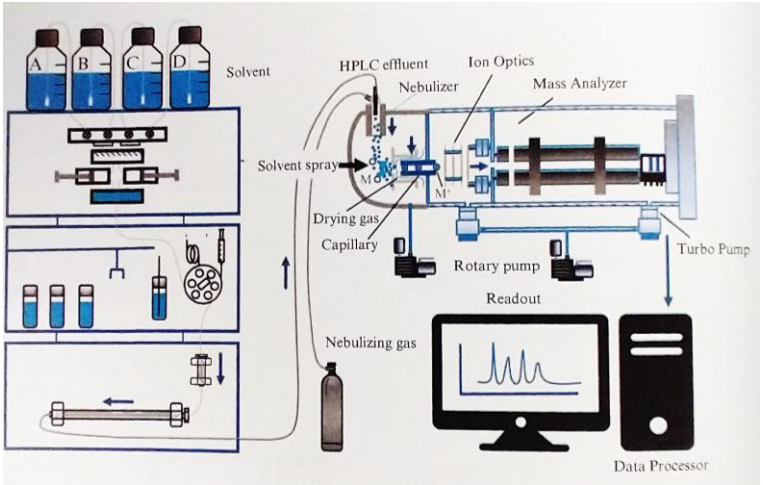


Figures taken from Thermofisher.com

Coupling of chromatography and mass spectrometry

- LC-MS and GC-MS are powerful hyphenated analytical techniques

LC-ESI-QMS



LC-ESI-QQQ

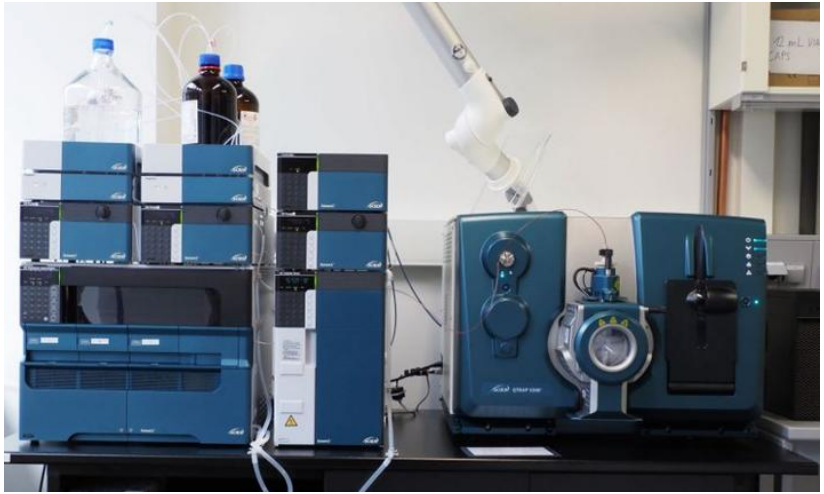


Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

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Coupling of chromatography and mass spectrometry

- LC-MS and GC-MS are powerful hyphenated analytical techniques

GC-EI-QMS

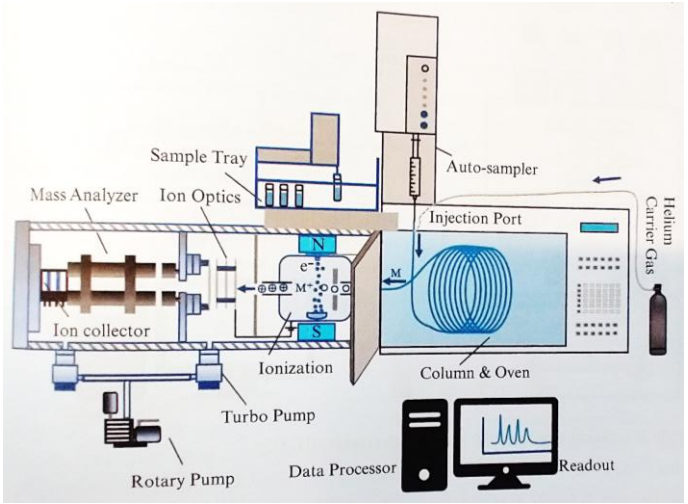




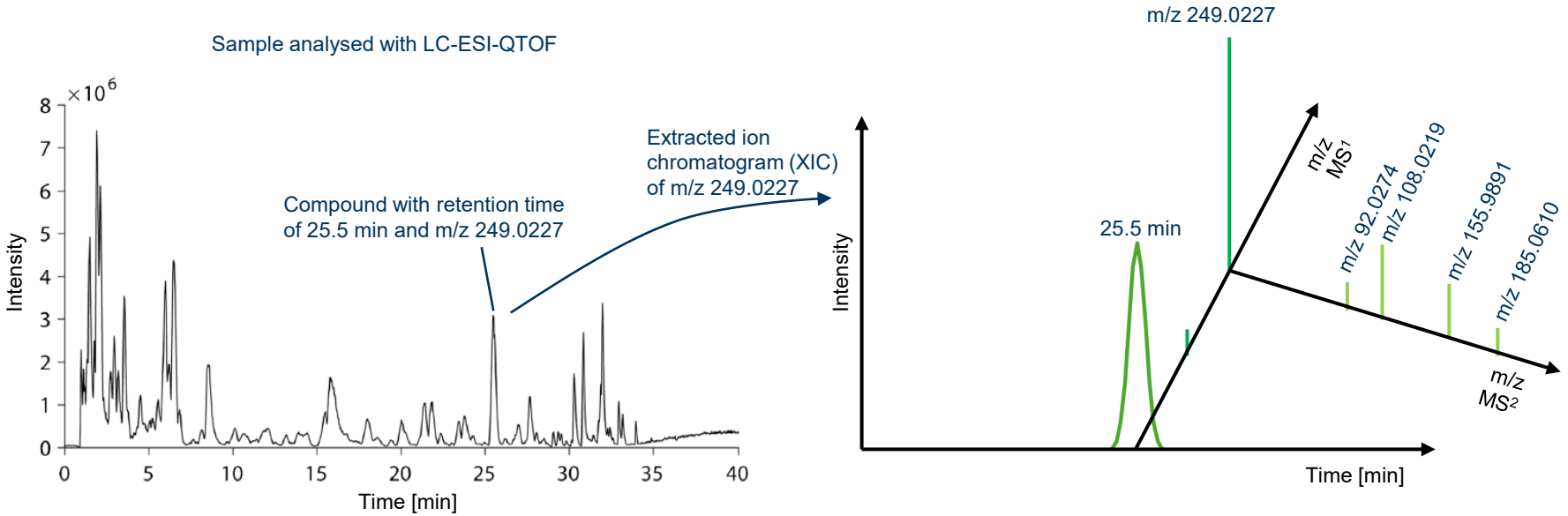
Figure taken from C. Zhang, Fundamentals of Environmental Sampling and Analysis, 2nd Edition, **2024**.

Picture of GC-system from Agilent

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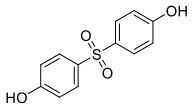
Ion chromatograms

- Chromatograms based on MS data



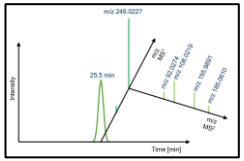
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Identification of compounds



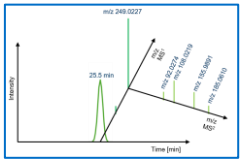


Injection of reference standard



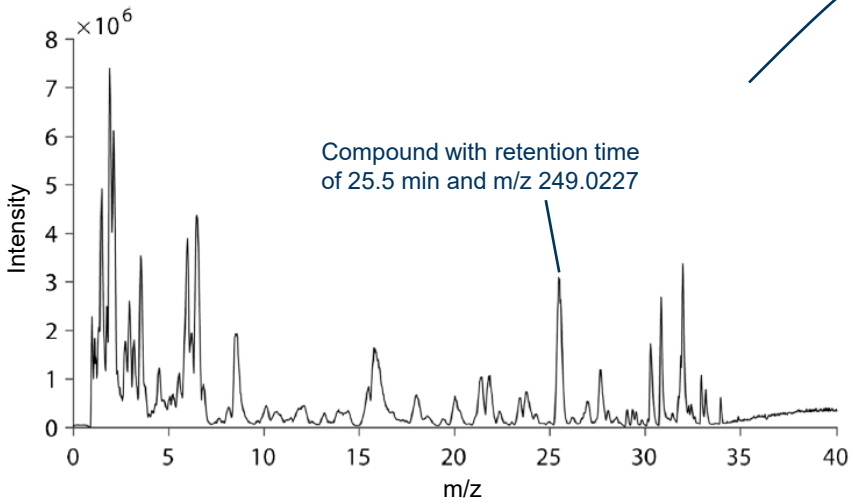
Sample data
(RT, MS^1, MS^2)

=

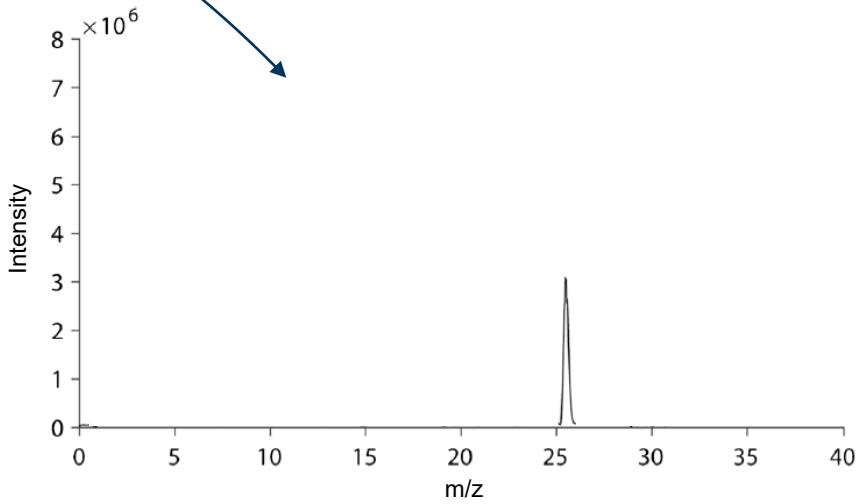


Reference data
(RT, MS^1, MS^2)

Sample analysed with LC-ESI-QTOF



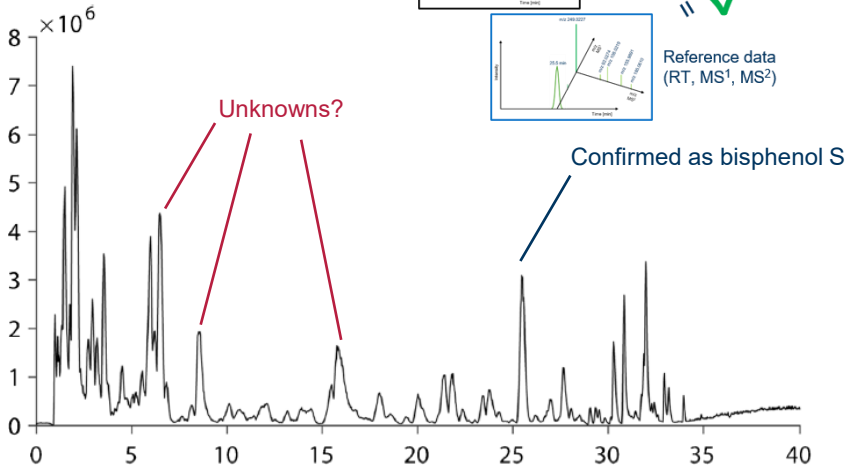
Compound with retention time
of 25.5 min and m/z 249.0227

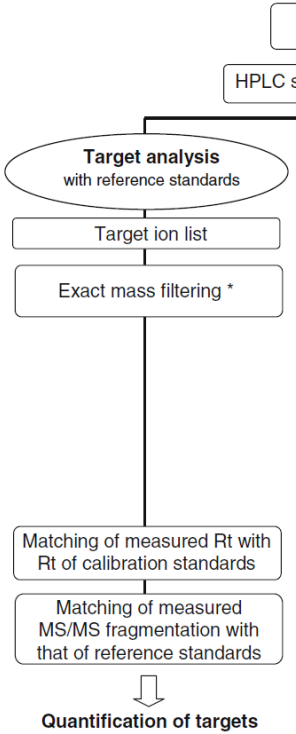


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Target analysis, suspect screening and Non-target screening

- Target analysis → quantification using external calibration





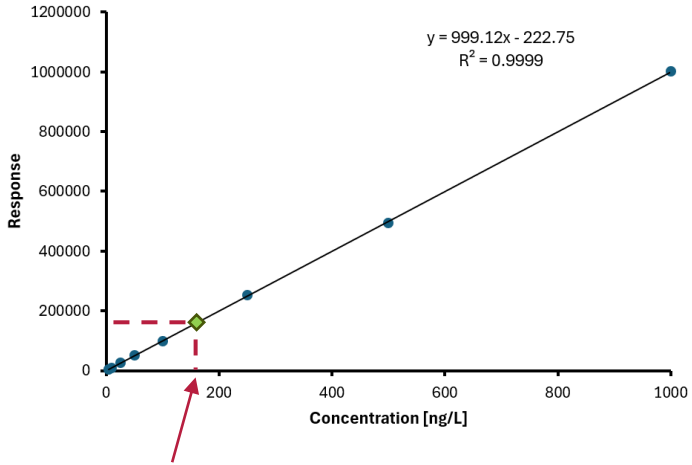
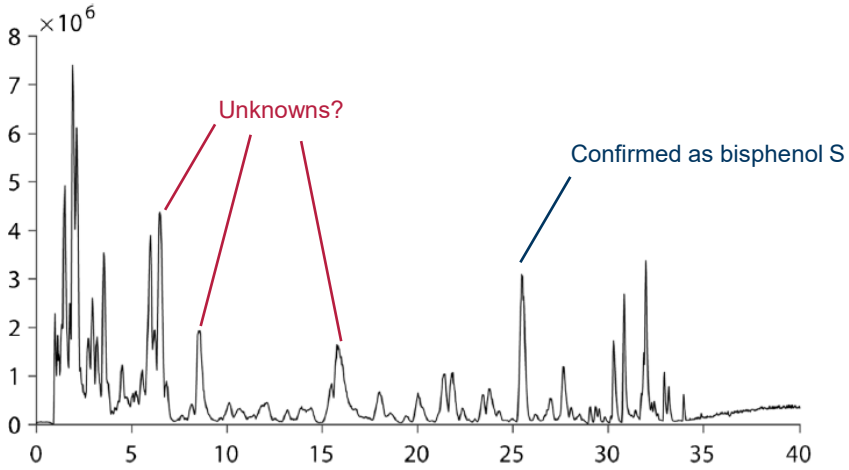


Figure taken from Krauss et al., Analytical Bioanalytical Chemistry, **2010**.

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Target analysis, suspect screening and Non-target screening

- Suspect screening → comparison of unknowns with suspect list including information about analytes
- Suspect list → compounds to be suspected in the sample (e.g. pesticides)



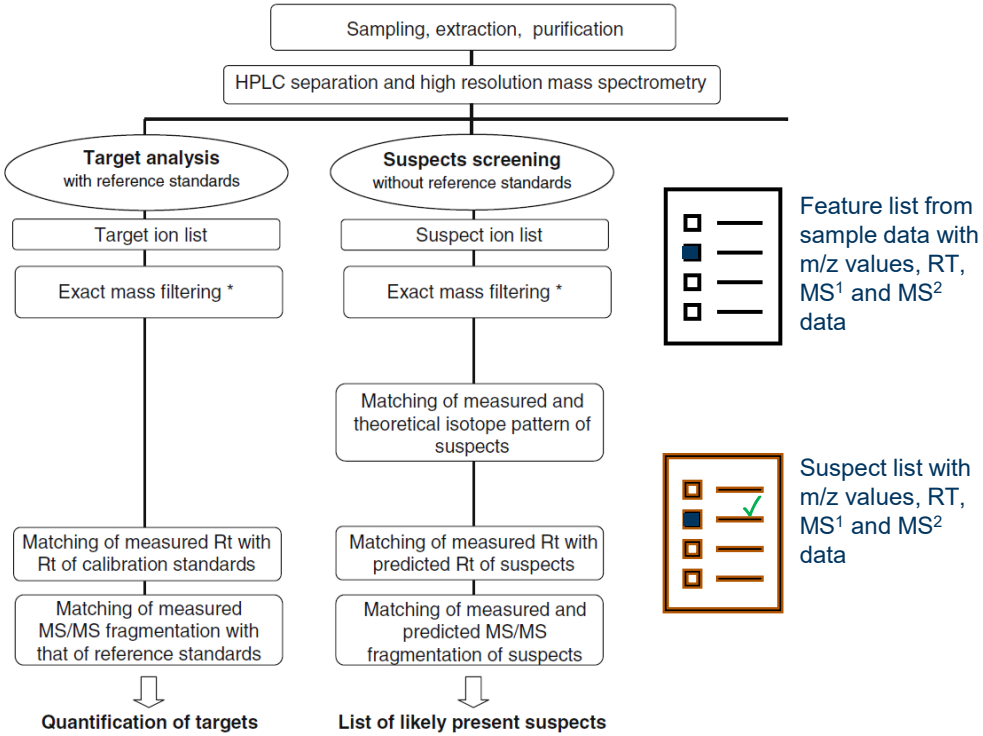
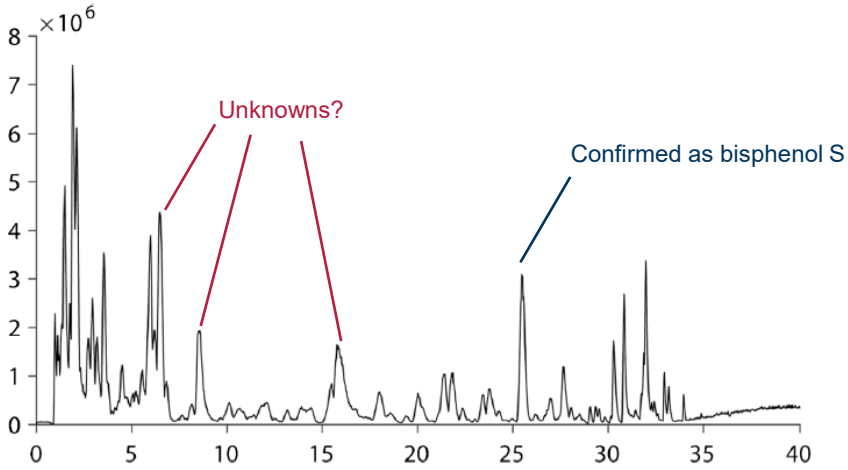


Figure taken from Krauss et al., Analytical Bioanalytical Chemistry, **2010**.

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Target analysis, suspect screening and Non-target screening

- Non-target screening → no information about unknowns beforehand
- Most important tool → high resolution mass spectrometry (HRMS)



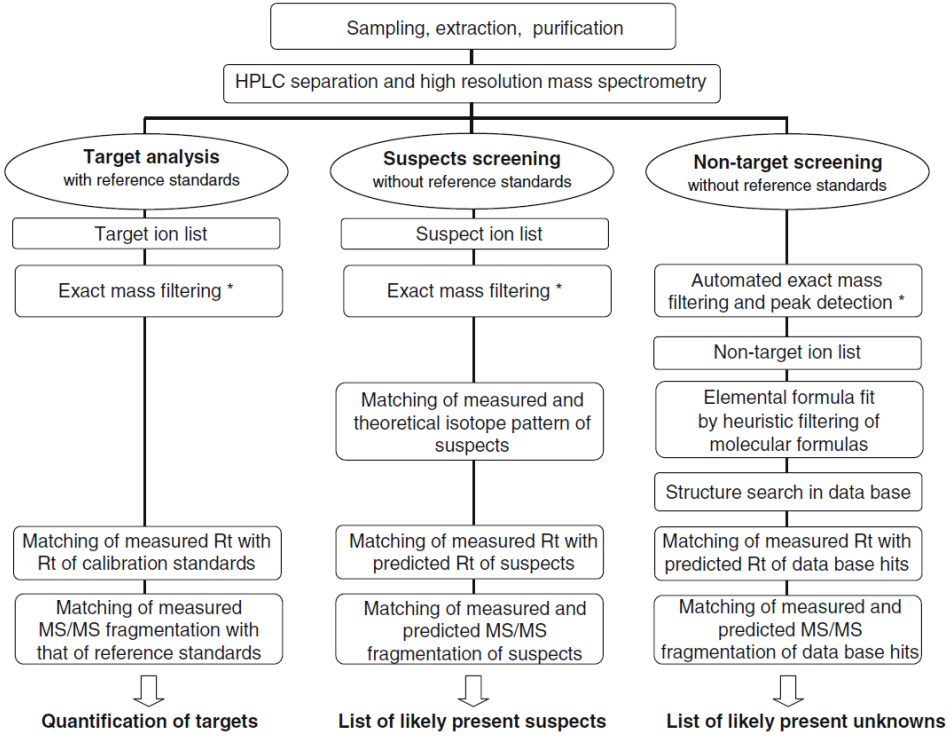
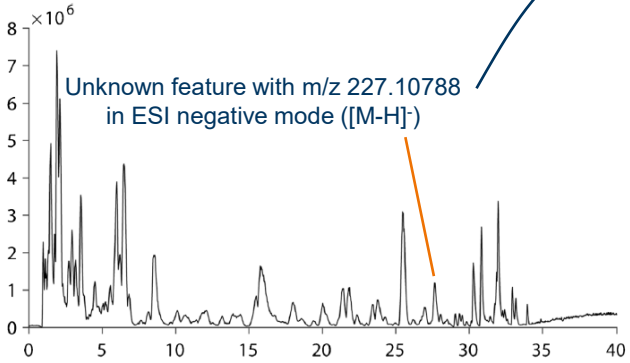


Figure taken from Krauss et al., Analytical Bioanalytical Chemistry, **2010**.

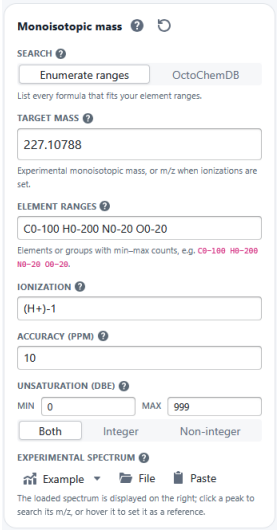
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Non-target screening

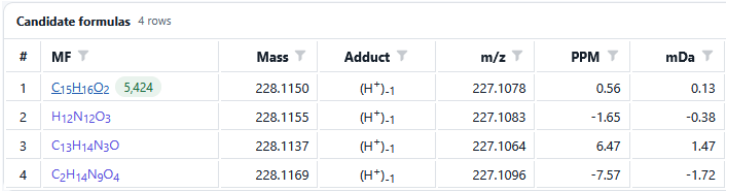
- Elemental formula calculated based on exact mass



Calculation of chemical formula
(e.g. on ChemCalc.org)

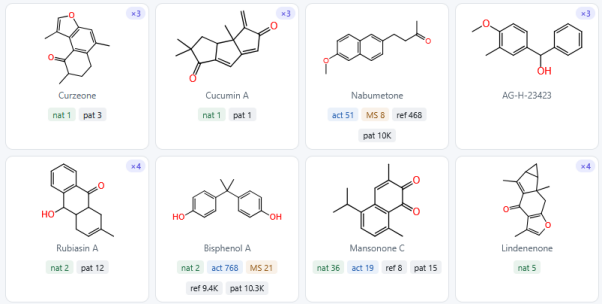


List of candidate formulas

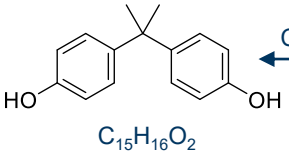


#	MF	Mass	Adduct	m/z	PPM	mDa
1	$C_{15}H_{16}O_2$ 5,424	228.1150	$(H^+)_{-1}$	227.1078	0.56	0.13
2	$H_{12}N_{12}O_3$	228.1155	$(H^+)_{-1}$	227.1083	-1.65	-0.38
3	$C_{13}H_{14}N_3O$	228.1137	$(H^+)_{-1}$	227.1064	6.47	1.47
4	$C_2H_{14}N_9O_4$	228.1169	$(H^+)_{-1}$	227.1096	-7.57	-1.72

Possible chemical structures (8 out of 4786)

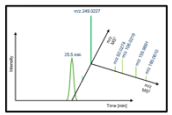


Bisphenol A



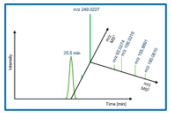
$C_{15}H_{16}O_2$

Confirmed



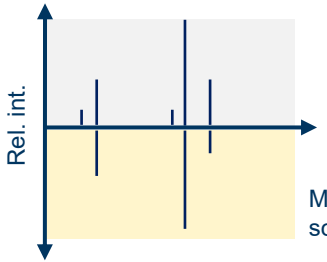
Sample data
(RT, MS^1, MS^2)

=



Reference data
(RT, MS^1, MS^2)

Buying reference standard
for confirmation



MS^2 spectra
Measured vs. data base

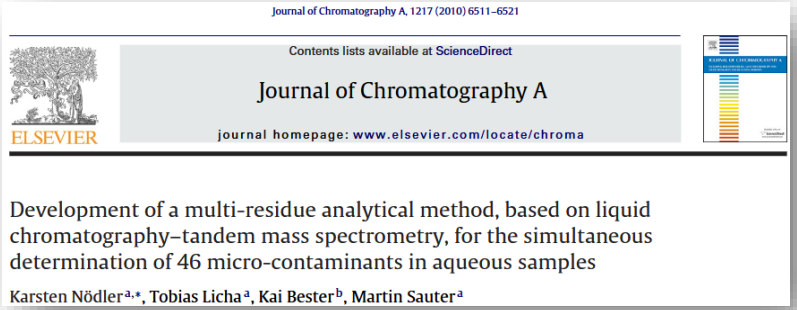
Matching
score: 96.4%

Calculation of chemical formula on Chemcalc.org.

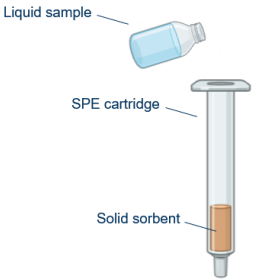
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Examples of research studies in environmental analysis

- Analysis of surface water samples
  - River water, seawater, treated wastewater
  - Target analysis of 46 organic contaminants



Solid phase extraction (SPE)

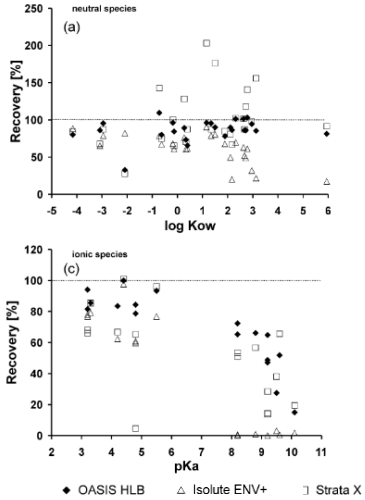


SPE sorbent:	HLB (= hydrophilic-lipophilic balance); often copolymer of N-vinylpyrrolidone and divinylbenzene
Conditioning:	4 mL methanol
Equilibration:	2 x 4 mL ultrapure water
Sample loading:	River water and seawater: 500 mL; wastewater: 100 mL
Washing:	2 x 1.5 mL ultrapure water
Elution:	2 x 2 mL methanol and 2 x 2 mL ethyl acetate

Examples of research studies in environmental analysis

- Analysis of surface water samples
  - River water, seawater, treated wastewater
  - Target analysis of 46 organic contaminants

Recovery tested on different SPE sorbents



Sample analysis:
Target analysis
LC-ESI-QQQ-MS in multiple reaction monitoring (MRM) mode
LC column: C18-Ether (150 x 2 mm, 3 µm particle size)
Mobile phase: Water with 0.015% formic acid + 5% methanol (solvent A) and methanol (solvent B)
Gradient elution, runtime 50 min
ESI in positive and negative mode with polarity switching
Method quantification limits (MQLs):

River and seawater:	1.2 to 28 ng/L
Wastewater:	5.0 to 160 ng/L

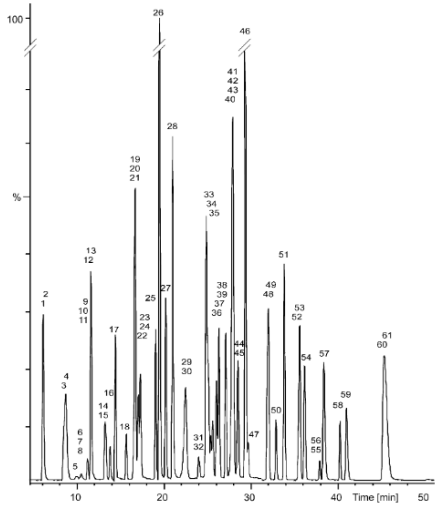


Fig. 2. Chromatogram (Total Ion Current, TIC) of a standard solution. Analytes and internal standards in order of elution: (1) sotalol, (2) iopamidol, (3) atenolol-D_7, (4) atenolol, (5) 3-methylxanthine, (6) iomeprol, (7) 1-methylxanthine, (8) iohexol, (9) desmethoxyiopromide, (10) paracetamol, (11) iopromide, (12) theobromine, (13) trimethoprim, (14) paraxanthine-D_6, (15) paraxanthine, (16) theophylline, (17) metoprolol, (18) caffeine, (19) desisopropylatrazine, (20) benzoylecgonine, (21) 1H-benzotriazole, (22) phenazone, (23) sulfamethoxazole-$^{13}C_6$, (24) sulfamethoxazole, (25) primidone, (26) citalopram, (27) desethylatrazine, (28) tolyltriazole, (29) erythromycin-N-methyl-^{13}C-D_3, (30) erythromycin, (31) fluoxetine-D_6, (32) fluoxetine, (33) pantoprazole, (34) sertraline, (35) clarithromycin, (36) ery-methyloxime, (37) roxithromycin, (38) carbamazepine-D_{10}, (39) carbamazepine, (40) lansoprazole, (41) atrazine-D_5, (42) metazachlor, (43) atrazine, (44) isoproturon-D_6, (45) isoproturon, (46) cetirizine, (47) diuron, (48) diazepam-D_5, (49) diazepam, (50) naproxen, (51) tetrazepam, (52) loratadine-D_4, (53) loratadine, (54) bezafibrate, (55) ibuprofen-D_3, (56) ibuprofen, (57) clofibric acid, (58) gemfibrozil, (59) diclofenac, (60) mecoprop-D_3 and (61) mecoprop.

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Examples of research studies in environmental analysis

- Analysis of surface water samples
  - River water, seawater, treated wastewater
  - Target analysis of 46 organic contaminants

36 out of 46 analytes detected in at least one sample

Concentrations of analytes in three different sample matrices expressed as mean result of duplicate extraction ± expanded uncertainty[a] in ng/L.

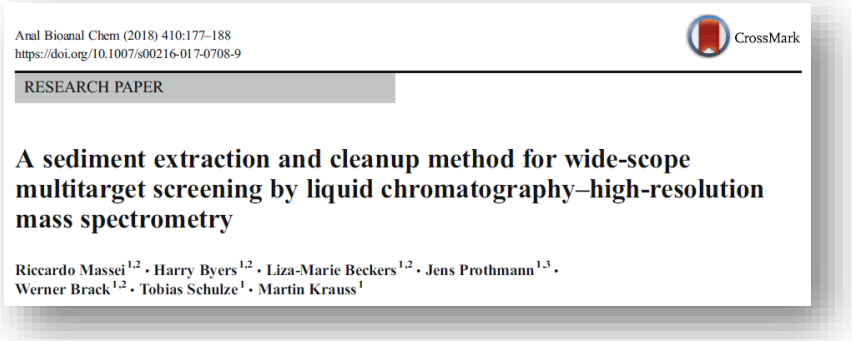
Compound	River sample	Seawater sample	Treated effluent sample
1H-benzotriazole	670 ± 41	29 ± 3.3	3846 ± 277
1-Methylxanthine	61 ± 14	nd[c]	–
3-Methylxanthine	–[b]	nd	–
Atenolol	66 ± 3.6	–	245 ± 16
Atrazine	2 ± 0.8	–	–
Benzoylecgonine	–	–	–
Bezafibrate	29 ± 2.5	–	168 ± 11
Caffeine	101 ± 8.7	58 ± 5.4	–
Carbamazepine	265 ± 10	26 ± 1.6	1429 ± 46
Cetirizine	20 ± 2.0	4 ± 1.3	105 ± 8.2
Citalopram	54 ± 4.7	–	431 ± 41
Clarithromycin	77 ± 12	14 ± 9.4	520 ± 35
Clofibric acid	21 ± 2.4	–	103 ± 9.3
Desethylatrazine	6 ± 1.0	–	–
Desisopropylatrazine	7 ± 3.4	–	–
Diazepam	10 ± 0.9	–	27 ± 6.1
Diclofenac	156 ± 6.0	–	1492 ± 89
Diuron	–	–	–
Erythromycin	22 ± 2.7	–	173 ± 23
Fluoxetine	–	nd	–
Gemfibrozil	–	–	–
Ibuprofen	–	–	–
Iohexol	1214 ± 88	–	5574 ± 718
Iomeprol	1258 ± 106	40 ± 11	6272 ± 572
Iopamidol	785 ± 100	–	5569 ± 619
Iopromide	268 ± 30	25 ± 11	2670 ± 208
Isoproturon	43 ± 2.0	–	35 ± 11
Loratadine	–	4 ± 4.0	–
Mecoprop	26 ± 1.2	–	159 ± 5.2
Metazachlor	–	–	–
Metoprolol	337 ± 22	–	2513 ± 161
Naproxen	–	–	–
Pantoprazole	13 ± 2.8	nd	149 ± 23
Paracetamol	1992 ± 198	–	–
Paraxanthine	36 ± 3.3	22 ± 2.4	–
Phenazone	27 ± 1.7	–	56 ± 23
Primidone	60 ± 4.3	–	216 ± 20
Roxithromycin	16 ± 6.4	–	78 ± 11
Sertraline	–	nd	–
Sotalol	195 ± 9.0	–	1314 ± 130
Sulfamethoxazole	93 ± 4.4	7 ± 1.5	509 ± 35
Tetrazepam	–	–	–
Theobromine	–	23 ± 4.2	–
Theophylline	21 ± 3.0	–	–
Tolyltriazole	741 ± 45	37 ± 3.6	4803 ± 404
Trimethoprim	95 ± 4.2	–	681 ± 28

[a] expanded uncertainties calculated according to Konieczka and Namieśnik [49].
[b] Not detected.
[c] Not determined.

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Examples of research studies in environmental analysis

- Analysis of river sediments



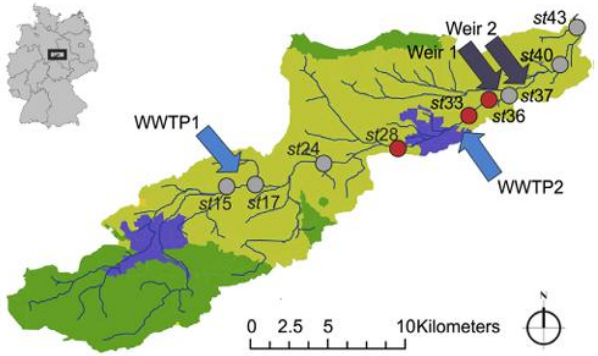
Anal Bioanal Chem (2018) 410:177–188
https://doi.org/10.1007/s00216-017-0708-9

CrossMark

RESEARCH PAPER

A sediment extraction and cleanup method for wide-scope multitarget screening by liquid chromatography–high-resolution mass spectrometry

Riccardo Massei[1,2] · Harry Byers[1,2] · Liza-Marie Beckers[1,2] · Jens Prothmann[1,3] · Werner Brack[1,2] · Tobias Schulze[1] · Martin Krauss[1]

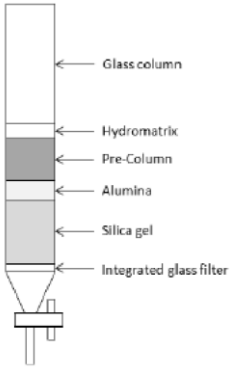


● Sediments collected at red spots

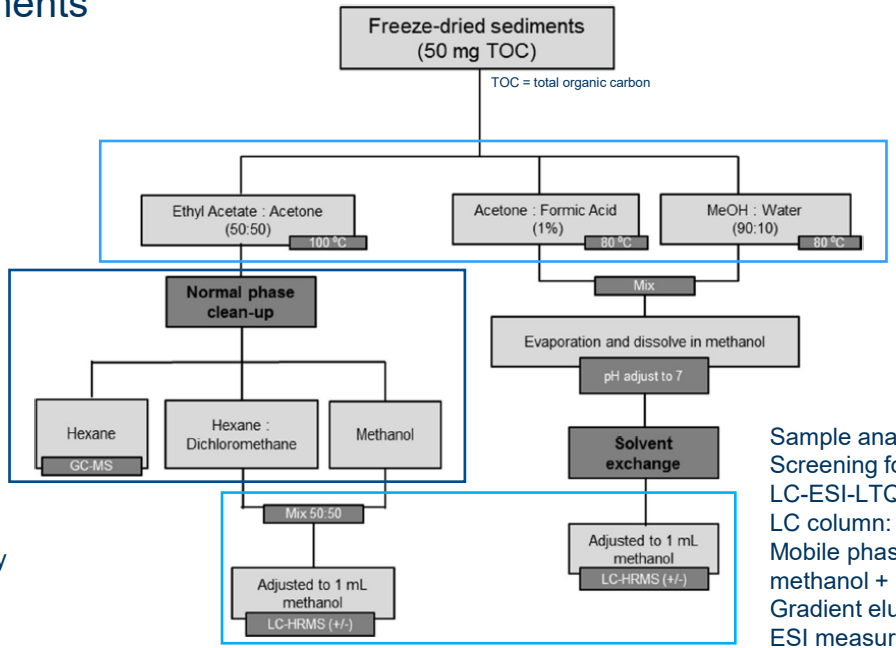
Sampling tool:	stainless steel scoop
Sample bottles:	stainless steel bowls
Sampling approach:	composite sampling (aliquots of the top 5 cm from 5 to 10 points within an area of 5-25 m^2)
Sample pretreatment:	homogenized, freeze-dried, sieved to ≤63 µm
Sample storage:	-20 °C

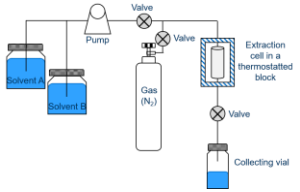
Examples of research studies in environmental analysis

- Analysis of river sediments

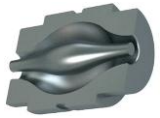


Fractionation/clean-up:
Normal phase column chromatography





Extraction:
Sequential extraction of 50 mg of sediment (TOC equivalent, grain size ≤63 µm) with pressurized liquid extraction (PLE)



Sample analysis:
Screening for 310 target compounds
LC-ESI-LTQ/Orbitrap-MS
LC column: C18 (100 x 3 mm, 2.6 µm particle size)
Mobile phase: Water + 0.1% formic acid (solvent A) and methanol + 0.1% formic acid (solvent B)
Gradient elution, runtime 50 min
ESI measured separately in positive and negative mode
MS resolving power in MS^1: 70000 and in MS^2: 35000
Method detection limits (MDLs): 0.016 to 12.8 µg/g_{TOC}

Examples of research studies in environmental analysis

- Analysis of river sediments

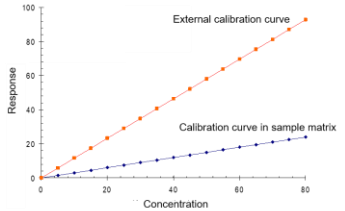
Investigation of matrix effects with spiked samples

TOC (mg mL^{-1})	Compound (ng mL^{-1})	Average recovery ± standard deviation (%)	
		Pesticides (n = 38)	Pharmaceuticals (n = 51)
20	40	72 ± 2	73 ± 3
50	100	61 ± 2	63 ± 5
100	200	54 ± 2	59 ± 6
200	500	44 ± 4	53 ± 10
500	1000	35 ± 1	39 ± 4

Recovery decreases with increasing amount of sample matrix

Table 1 Effect of sample matrix (given as the concentration of TOC in the vial) on recoveries of pharmaceuticals and pesticides in spiked neutral and basic/acidic extracts mimicking their dilution of organic carbon (n = 2)

$$\textit{Matrix effect on recovery} = \frac{\textit{Peak area of analyte with matrix}}{\textit{Peak area of analyte without matrix}}$$



Concentrations in ng/g_{TOC} of 18 out of 69 detected compounds

Compound	Site 14 (Silstedter Bach)	Site 36 (Weir Gross Quenstedt)	Site 41 (Salzgraben)	Compound class
N-Acetyl-4-aminoantipyrine		35		Pharmaceutical TP
7-Amino-4-methylcoumarin	68	195	223	Industrial
Amiodarone		33		Pharmaceutical
Bendiocarb		32	76	Insecticide
Benzocaine	60	44	104	Pharmaceutical
Bifonazole		45	7.6	Pharmaceutical
Bisoprolol		63	74	Pharmaceutical
Boscalid			8.7	Fungicide
N-Butylbenzenesulfonamide		802		Industrial
Carbamazepine	20	146	122	Pharmaceutical
Carbendazim		7.2	38	Biocide/fungicide
N-Cyclohexyl-2-benzothiazolamine	66	10		Industrial
Crotamiton	31		7.2	Pharmaceutical
DEET	45		41	Insect repellent
Diazinon		150		Insecticide
7-Diethylamino-4-methylcoumarin	19	1085	1327	Industrial
Diphenhydramine	59			Pharmaceutical
Diphenyl phosphate		226	100	Flame retardant TP

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Examples of research studies in environmental analysis

- Suspect and non-target analysis of river samples

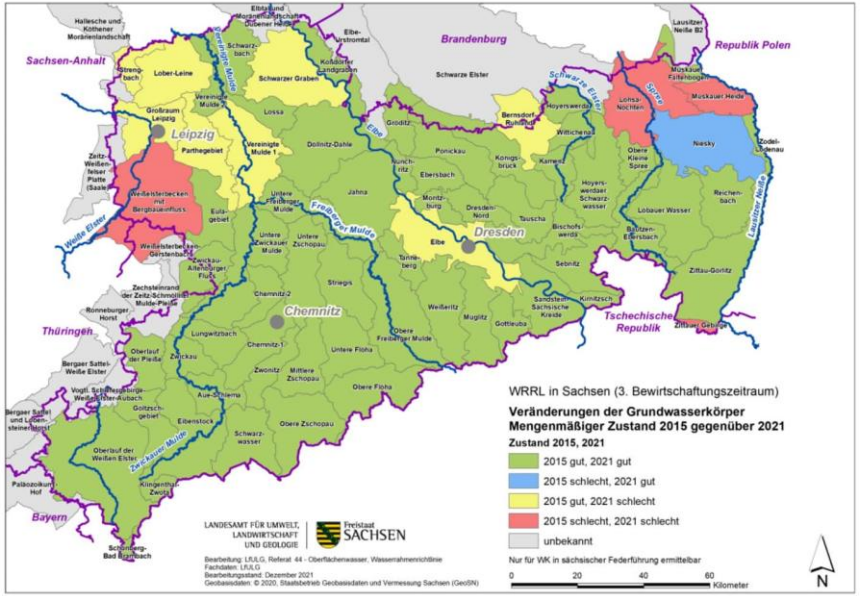
Analytical and Bioanalytical Chemistry (2025) 417:6121–6137
https://doi.org/10.1007/s00216-025-06124-3

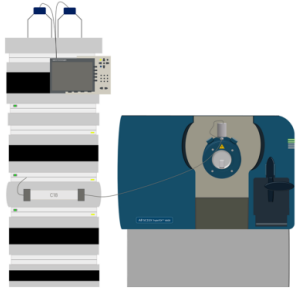
RESEARCH PAPER

A category approach to characterize river sites based on non-target screening and target analysis

Toni Köppe[1] · Nina Hermes[1] · Kevin S. Jewell[1] · Arne Wick[1] · Sylvia Rohde[2] · Thomas A. Ternes[1]

Sampling area for river water (Saxony, Germany)





Analysis of 520 river water samples from 79 locations

Non-target analysis using LC-ESI-QTOF-MS
LC column: C18 (150 x 2.1 mm, 3.5 µm particle size)
Mobile phase: Water + 0.1% formic acid (solvent A) and acetonitrile + 0.1% formic acid (solvent B)
Gradient elution, runtime 40 min
ESI measured separately in positive and negative mode
MS resolving power in MS^1: 35000 and in MS^2: 30000

Filtration of samples (0.45 µm cellulose) → no further sample preparation → direct injection into LC-MS system

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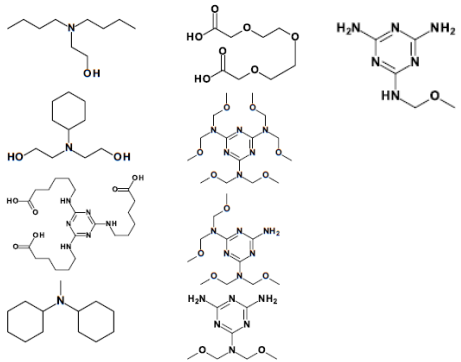
Examples of research studies in environmental analysis

- Suspect and non-target analysis of river samples
  - Identification of 376 compounds with suspect analysis using a spectral-library containing about 1400 compounds
  - Two sampling sites identified with high industrial pollution
  - Identification of unknown compounds at industrial polluted sites with non-target analysis

5400 unknown features detected at two sampling sites with industrial pollution
Filtering by
- Spatially specific
- Accurate MS2 spectrum
- Detected an a specific minimum intensity (50 cps)

Dataset reduced to ~1500 unknown features
- Selection of 150 most intense features for further identification:

Identification of 9 compounds



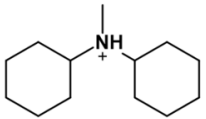
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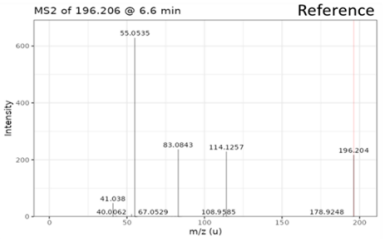
Examples of research studies in environmental analysis

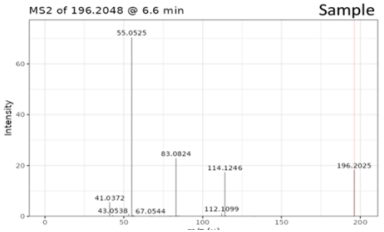
- Analysis of river sediments

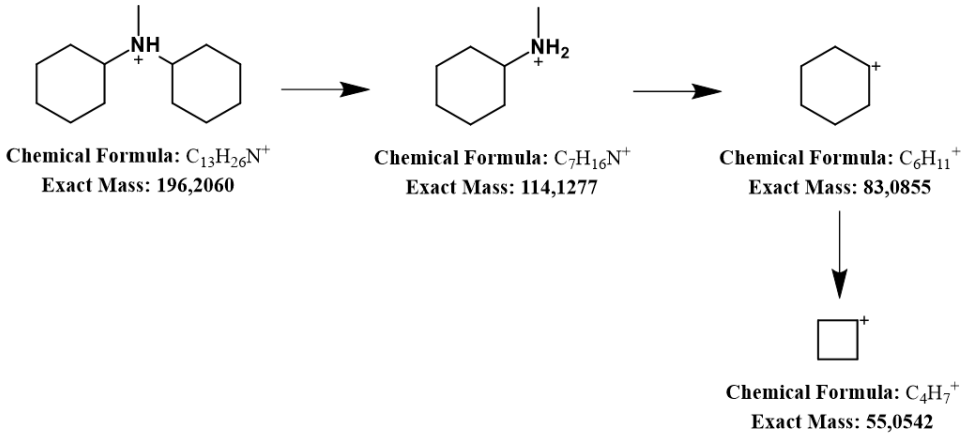
N,N-Dicyclohexylmethylamine (NDM)

m/z reference [Da]:	196.2060
mean m/z sample [Da]:	196.2047
Δ m/z [Da]:	0.0013
t_R refrence [min]:	6.6
mean t_R sample [min]:	6.5
Δ t_R [min]:	0.1









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